



Dissertation

On the Principled Design of Echo State Networks for Chaotic Forecasting

Daniel Estevez-Moya

Born on: 12th July 1993 in Havana, Cuba

23rd April 2026

First referee

Dr. Hölger Kantz

Second referee

Dr. Ernesto Estevez-Rams

Third referee

Dr. Francesco Sorrentino

Supervisors

Dr. Hölger Kantz

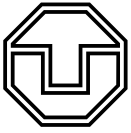
Dr. Ernesto Estevez-Rams

Statement of authorship

I hereby certify that I have authored this document entitled *On the Principled Design of Echo State Networks for Chaotic Forecasting* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. There were no additional persons involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, 23rd April 2026

Daniel Estevez-Moya



Abstract

Forecasting the future evolution of chaotic dynamical systems from data is fundamentally limited by the exponential divergence of nearby trajectories, yet within this limit the question of how closely a data-driven model can approach the theoretical predictability horizon remains open. Echo State Networks (ESNs)—a reservoir computing paradigm in which a large, fixed recurrent network is coupled with a trainable linear readout—offer a computationally efficient framework for this task, but three methodological gaps have hindered reproducible and comparable benchmarking: the lack of a principled treatment of the synchronization transient, the absence of a geometry-aware temporal discretization protocol, and the reliance on brittle, discontinuous evaluation metrics.

This thesis addresses all three gaps and integrates the resulting tools into a unified, automated pipeline for the principled design of Echo State Networks for chaotic forecasting. First, a quantitative framework for analysing transient times is developed, combining an analytical upper bound for contracting reservoirs with ensemble statistics that reveal heavy-tailed transient-time distributions near and above the spectral unit circle. A graph-theoretic decomposition of the reservoir into core, dendritic, and quiescent nodes establishes the design principle that the fraction of dynamically active nodes should approach unity ($c + d \rightarrow 1$), a condition realised by the proposed *dendrocycle* topology. Second, a derivative-free discretization method based on the *sagitta*—the perpendicular deviation of a trajectory sample from the chord connecting its temporal neighbours—is introduced. The sagitta scales quadratically with the time step, yields idempotent selection rules, and enables fair cross-system comparison by normalising trajectories to a common level of geometric difficulty. Third, the *Expected Forecast Horizon* (EFH), a smooth, bounded, and differentiable surrogate of the classical valid prediction time, is constructed through a survival-process analogy; it recovers the hard hitting time in the appropriate limit, concentrates its gradients at the failure onset, and exhibits substantially lower sensitivity to noise than the conventional metric. Finally, a systematic hyperparameter optimization campaign using the tree-structured Parzen estimator, conducted on a benchmark suite of four chaotic systems spanning nearly two orders of magnitude in Lyapunov exponent, demonstrates that the three tools compose naturally into a single pipeline. The optimizer, searching freely over an eight-dimensional configuration space, independently rediscovers the $c + d \approx 1$ design principle in all systems, confirming that eliminating quiescent reservoir nodes is a universal prerequisite for high-performance chaotic forecasting. All methods are implemented in the open-source library ResDAG.

Contents

Abstract	V
Introduction	1
1 Echo State Networks	7
1.1 Reservoir Computing	7
1.1.1 Liquid State Machines	9
1.1.2 Echo State Networks	9
1.2 The Echo State Property	10
1.2.1 Formal Definition	10
1.2.2 The Spectral Radius Misconception	11
1.2.3 Input-Dependence of the Echo State Property	12
1.2.4 Connection to Generalized Synchronization	13
1.3 ESN Variants	14
1.3.1 Leaky Integrator ESN	14
1.3.2 Deep Echo State Networks	14
1.3.3 Parallel and Ensemble ESNs	15
1.4 Reservoir Topologies	15
1.4.1 Random Sparse Reservoirs	15
1.4.2 Minimal Complexity Reservoirs	16
1.5 Training Echo State Networks	17
1.5.1 Open-Loop Phase: State Collection	18
1.5.2 Closed-Loop Phase: Autonomous Prediction	18
1.6 Critical Issues in the ESN Framework	19
1.6.1 Hyperparameter Sensitivity	20
1.6.2 Time Discretization	20
1.6.3 Benchmarking Practices	21
1.7 Outlook	21
2 Transient Times in Echo State Networks	23
2.1 Motivation	23
2.2 Quantification	25
2.3 Analysis	26

2.4	Transient Time Distributions	30
2.5	Parameter Dependency	32
2.6	Zero-Input Dynamics and Graph Structure	36
2.7	Topology and Forecast Performance	38
2.8	Summary and Design Principles	41
3	Geometric Time Discretization	43
3.1	The discretization problem in forecasting	43
3.2	Linearization error and curvature	44
3.3	The sagitta as a derivative-free curvature proxy	47
3.4	Selection rules and idempotence	52
3.5	Practical considerations	55
3.5.1	Possible caveats	57
3.6	Conclusions	58
4	Chaotic forecasting evaluation	59
4.1	Expected forecast horizon: definition and interpretation	60
4.1.1	Forecasting setting, closed-loop prediction, and normalization	60
4.1.2	Pointwise error signal	61
4.1.3	Single benchmark curve	62
4.1.4	Soft acceptability gate	63
4.1.5	Soft survival process and definition of EFH	64
4.1.6	EFH as a metric and as a differentiable training objective	65
4.2	Theoretical properties	67
4.2.1	Bounds and monotonicity	67
4.2.2	Limiting cases and the connection to a hitting time	68
4.2.3	Survival and hazard representation	70
4.2.4	Gradient concentration at the failure onset	71
4.2.5	Stability under the normalization convention	73
4.2.6	Robustness to noise	74
4.2.7	Behaviour across different error profiles	75
4.3	Conclusions	75
5	Hyperparameter Optimization	77
5.1	Experimental setup	77
5.2	Optimization landscape and convergence	80
5.3	Hyperparameter importance	81
5.4	Marginal effects of individual hyperparameters	83
5.5	Influence of core and dendritic fractions	89
5.6	Conclusions	90
6	Conclusions	93
A	Appendix: Transient Times	111
A.1	Bounds on Initial Separation	111
B	Appendix: Geometric time discretization	113
B.1	Derivation of the sagitta-curvature relation	113

B.2	Proof of Proposition 3.1	116
B.3	Proofs of Theorems 3.3 and 3.4	119
B.3.1	Idempotence of the absolute rule	119
B.3.2	Idempotence of the relative rule	120
B.3.3	Remarks	121
B.4	Extension to scalar time series	121
B.4.1	Embedding lag via average mutual information	122
B.4.2	Embedding dimension via false nearest neighbors	122
B.4.3	Integration with the sagitta rule	123

Introduction

Forecasting the future state of a dynamical system from its past observations is one of the oldest and most consequential problems in the physical sciences. When the system is chaotic, the task acquires a fundamental difficulty: trajectories that start from nearby initial conditions diverge exponentially, so that any model— however faithful— will eventually lose track of the true state. The rate of this divergence, quantified by the maximal Lyapunov exponent $\lambda_{\max}$, sets a hard ceiling on the forecast horizon measured in physical time [1, 2]. Within that ceiling, though, the question of *how well* a data-driven model can approach the theoretical limit is wide open, and it is precisely this question that motivates the present work.

The data-driven modelling of chaotic systems has a substantial history. Takens' embedding theorem [3] provided the theoretical foundation by showing that a scalar time series, appropriately delay-embedded, can faithfully reconstruct the topology of the underlying attractor. Casdagli [4] demonstrated that nonlinear deterministic models can outperform linear stochastic ones on chaotic data, and subsequent decades saw steady progress through local methods, radial basis functions, and neural networks [2]. The modern era, however, is dominated by recurrent neural networks, which learn the dynamics directly from sequences without explicit embedding construction.

Recurrent neural networks have been the workhorse of sequence modelling since the backpropagation-through-time algorithm made their training feasible [5]. Yet the practical difficulties of gradient-based training— vanishing and exploding gradients [6, 7], sensitivity to initialization [8], and substantial computational cost— have long motivated the search for alternatives. Reservoir computing, introduced independently by Jaeger under the name Echo State Networks (ESNs) [9] and by Maass, Natschläger and Markram as Liquid State Machines [10], offers a radically different paradigm: a large, fixed, randomly connected recurrent network is driven by the input signal, and only a linear readout from the reservoir states is trained. This decoupling of the recurrent dynamics from the learning problem eliminates the need for gradient propagation through the reservoir and reduces training to a single ridge-regression solve [11]. The theoretical underpinnings of this approach— chiefly the *echo state property*, which ensures that the reservoir state is a unique function of the input history [12, 13]— place reservoir computing on firm ground as a nonlinear functional approximation scheme [14], while its practical appeal lies in the fast, closed-form training

that avoids the pitfalls of gradient-based optimization [15]. For comprehensive reviews of the reservoir computing framework and its physical implementations, we refer to Tanaka *et al.* [16] and Cucchi *et al.* [17].

The simplicity of the ESN framework has made it a remarkably effective tool for chaotic forecasting. Jaeger and Haas demonstrated early on that ESNs can predict chaotic signals several Lyapunov times into the future [18], and Pathak *et al.* showed that reservoir computers can reconstruct the full attractor of spatiotemporally chaotic systems from data alone [19]. Lu *et al.* [20] extended this to observer problems where unmeasured variables must be inferred from partial observations. Comparative studies have confirmed that, for chaotic dynamics, reservoir computing is often competitive with or superior to deep recurrent architectures such as LSTMs and GRUs, at a fraction of the training cost [21, 22]. More recently, Gauthier *et al.* [23] proposed a next-generation reservoir computing framework that replaces the random recurrent network with a polynomial feature map, further broadening the design space.

Despite these successes, three methodological gaps have limited the reproducibility and comparability of ESN-based forecasting studies, and addressing them is the central goal of this thesis.

Transient dynamics and reservoir topology. Before the echo state property can be exploited, an initial transient must elapse during which the reservoir synchronises with the driving signal. This *washout* period is routinely handled by discarding a fixed number of initial steps, but the duration required for synchronization depends in a non-trivial way on the spectral radius, the leak rate, the network size, and—crucially—the topology of the reservoir graph [9, 24]. Underestimating the transient corrupts the training data; overestimating it wastes signal. A systematic, quantitative understanding of transient times and their dependence on architectural choices is therefore a prerequisite for reliable ESN deployment, yet it has been largely absent from the literature.

Time discretization. Chaotic systems modelled by ordinary or delay differential equations must be discretised before they can be fed to a reservoir. The choice of sampling interval Δt has a profound effect on the difficulty of the forecasting task—too large a step introduces aliasing, too small a step wastes reservoir capacity on near-trivial one-step predictions—yet it is rarely discussed in systematic terms. Most studies adopt a convenient round number or follow a precedent set by earlier work, making cross-system comparisons unreliable. What is needed is a principled, data-driven rule that selects Δt based on the geometric complexity of the trajectory, so that different dynamical systems are sampled at comparable levels of difficulty.

Forecast evaluation. The standard metric for assessing chaotic forecasts is the *valid prediction time* (VPT): the first instant at which the normalised error exceeds a fixed threshold. Because the VPT is defined through a hard indicator function, it is non-differentiable, discontinuous in the model parameters, and sensitive to noise. Minor fluctuations in the error signal around the threshold can produce large jumps in the reported forecast horizon, obscuring meaningful differences between models and making the VPT ill-suited as an

optimization objective. A smooth, differentiable alternative that preserves the physical interpretation of a forecast horizon while admitting stable black-box optimization has, to our knowledge, not been proposed prior to this work.

The natural culmination of these three contributions is a fourth: once we have a principled discretization, a reliable transient estimate, and a smooth objective function, we can pose the ESN design problem as a proper hyperparameter optimization (HPO) problem and attack it with modern sequential model-based methods. Hyperparameter optimization has become a central concern in machine learning, driven by the recognition that manual tuning is unreliable and that even random search vastly outperforms grid search [25, 26]. Model-based sequential strategies such as Bayesian optimization [27] and tree-structured Parzen estimators (TPE) [26] can find superior configurations with far fewer evaluations [28]. In this thesis we deploy the Optuna framework [29], which implements the TPE sampler, to systematically optimise ESNs for chaotic forecasting— using the tools developed in the preceding chapters as integral parts of the pipeline.

The remainder of this document is organised as follows.

Chapter 1: Echo State Networks. We provide a self-contained introduction to the reservoir computing paradigm, with emphasis on the mathematical foundations relevant to chaotic forecasting. After reviewing the general framework, we give a detailed treatment of the echo state property, including its formal definition in the language of nonautonomous dynamical systems [30], the well-known misconception that a spectral radius below unity is sufficient (or necessary) for the ESP to hold [12, 31], and the connection to generalised synchronization [32, 33]. We survey the main ESN variants— leaky integrators [34], deep reservoirs [35], and ensemble architectures— as well as the principal reservoir topologies: random sparse graphs, minimal-complexity structures such as simple cycle reservoirs [36] and cycle reservoirs with jumps [37], and the dendrocycle topology that will play a central role in our exploratory chapters. The chapter closes by identifying the three open problems— transient control, time discretization, and forecast evaluation— that motivate the rest of the thesis.

Chapter 2: Transient Times in Echo State Networks. We introduce a quantitative framework for analysing the synchronization transient of a driven reservoir. For networks with spectral radius $\rho < 1$, we derive an analytical upper bound on the transient time and verify it numerically. For $\rho \geq 1$, where the bound does not apply, we characterise the transient-time distribution and show that it develops a heavy tail with rare realizations that take an order of magnitude longer than the median. We study how transient times depend on the spectral radius, network size, leak rate, and link density. A graph-theoretic analysis decomposes the reservoir into core nodes (lying on cycles), dendritic nodes (downstream of cycles), and quiescent nodes (on acyclic subgraphs), and reveals that the fraction of core and dendritic nodes is a strong predictor of forecasting performance. This leads to the design principle that $c + d \rightarrow 1$ — maximising the fraction of dynamically active nodes— which the dendrocycle topology achieves by construction. The chapter establishes concrete guidelines for setting the washout period and motivates the use of transient pre-screening as a pruning criterion in hyperparameter searches.

Chapter 3: Geometric Time Discretization. We propose a derivative-free method for selecting the sampling step Δt of a continuous dynamical system based on the geometry of its trajectory. The key quantity is the *sagitta*— the perpendicular distance from the midpoint of a chord to the curve— which we show scales as $s(t; h) = \frac{1}{2}|\mathbf{a}_\perp(t)| h^2 + \mathcal{O}(h^3)$, providing a curvature proxy that requires no differentiation of the data. We define both absolute and relative selection rules, prove their idempotence (resampling at the selected step and reapplying the rule yields the same step), and establish a standard protocol with default tolerance $\varepsilon = 0.1$ and 95th-percentile aggregation. Applied to a suite of chaotic systems including the Lorenz [38], Rössler [39], and Mackey-Glass [40] attractors, the sagitta method produces sampling intervals that are geometrically comparable across systems with widely different natural time scales, thereby enabling fair cross-system benchmarking.

Chapter 4: Chaotic Forecasting Evaluation. We introduce the Expected Forecast Horizon (EFH), a smooth, bounded, and differentiable summary of forecast quality that replaces the binary valid prediction time. The EFH is constructed through a survival-process analogy: a soft sigmoidal gate converts the pointwise error signal into a probability of continued acceptable prediction, and the EFH is defined as the sum of the resulting survival curve. We prove that the EFH is bounded, monotone in the error threshold τ , and recovers the hard hitting time in the limit of vanishing softness parameter $\delta \rightarrow 0^+$. A gradient-concentration result shows that the EFH derivative is localised at the failure onset, making it an effective training signal. The EFH is scale equivariant, robust to noise— exhibiting a substantially lower coefficient of variation than the hard VPT— and admits a natural interpretation in terms of survival analysis. We propose its use both as an evaluation metric and as a differentiable optimization objective for hyperparameter tuning.

Chapter 5: Hyperparameter Optimization. The final chapter brings the preceding contributions together in a systematic hyperparameter optimization campaign. Using the Optuna framework [29] with the tree-structured Parzen estimator [26], we search over the full ESN configuration space— spectral radius, network size, input scaling, leak rate, regularization, and the topological parameters of the dendrocycle— on a benchmark suite comprising one chaotic discrete map (Hénon [41]) and three continuous flows (Lorenz [38], Rössler [39], and Chen [42]), with continuous systems preprocessed via the sagitta rule from Chapter 3. We first validate the optimization machinery on the Hénon map, showing how the landscape saturates rapidly to a well-defined stripe and demonstrating that this saturation reflects genuine model behaviour rather than optimization noise. We then show that the TPE sampler [29], searching freely over the dendrocycle parameters, converges organically to topologies with $c + d \approx 1$, thereby rediscovering the design principle of Chapter 2 without imposing it as a constraint. The chapter concludes with optimised forecasting results across all five systems and practical guidelines for the end-to-end pipeline: sagitta preprocessing, dendrocycle topology, EFH-driven hyperparameter optimization, and transient pre-screening.

The software developed in the course of this work is available as the open-source library¹ **ResDAG** [43], which provides GPU-accelerated ESN training, the sagitta discretization method,

¹The software is available at <https://github.com/El3ssar/ResDAG>.

the EFH metric, and Optuna integration for hyperparameter optimization. All experiments reported in this thesis are fully reproducible through this library.

1 Echo State Networks

1.1 Reservoir Computing

Recurrent neural networks (RNNs) constitute a broad family of connectionist models whose defining feature is the presence of cycles in their computational graph. Unlike feedforward architectures, which map a fixed-dimensional input to an output through a directed acyclic composition of transformations, RNNs maintain an internal state that evolves over time, allowing them to process sequential data of arbitrary length. The state update at discrete time n for a generic RNN takes the form

$$\mathbf{x}(n) = F(\mathbf{x}(n-1), \mathbf{u}(n)), \quad (1.1)$$

where $\mathbf{x}(n) \in \mathbb{R}^{N_x}$ denotes the internal state vector, $\mathbf{u}(n) \in \mathbb{R}^{N_u}$ is the external input, and F is a nonlinear map determined by the network's weights and activation functions. The recurrent connections endow RNNs with a form of dynamical memory: the current state $\mathbf{x}(n)$ implicitly encodes information about the entire history of past inputs $\{\mathbf{u}(k)\}_{k \leq n}$, filtered through the nonlinear dynamics of the network.

Training an RNN to perform a desired input-output mapping requires adjusting all connection weights—input-to-hidden, hidden-to-hidden, and hidden-to-output—so that the network's trajectory through state space produces the correct output sequence. The two classical algorithms for this purpose are Backpropagation Through Time (BPTT) [5] and Real-Time Recurrent Learning (RTRL) [44]. BPTT unfolds the recurrent computation into a deep feedforward graph and applies the chain rule backward through T time steps, yielding a computational cost of $\mathcal{O}(T \cdot N_x^2)$ per weight update and a memory requirement proportional to the unfolding depth. RTRL computes exact gradients online by propagating a sensitivity matrix forward in time, but at a prohibitive cost of $\mathcal{O}(N_x^4)$ per step.

Beyond the computational burden, both algorithms suffer from a fundamental pathology: the *vanishing and exploding gradient problem*. As gradients are propagated through the repeated application of the recurrent Jacobian $\partial F / \partial \mathbf{x}$, they either shrink exponentially toward zero or grow without bound [6–8]. The vanishing regime makes it effectively impossible to learn dependencies over long time horizons, since the error signal from distant time steps is attenuated below the noise floor of finite-precision arithmetic. The exploding regime

causes training instability, with gradient magnitudes oscillating wildly across updates. These difficulties are not merely practical inconveniences but reflect structural properties of gradient flow in iterated nonlinear maps, and they have historically limited the applicability of fully trained RNNs to tasks involving long-range temporal dependencies.

A crucial empirical observation that paved the way for a radically different approach was reported by Schiller and Steil [45], who analyzed the dynamics of weight modifications during standard RNN training. They demonstrated that, across a variety of tasks, the dominant changes occurred in the output layer weights, while the recurrent weights remained comparatively stable after the initial phase of learning. This finding suggested that the internal recurrent dynamics, once initialized within a suitable operating regime, already provided a sufficiently rich nonlinear transformation of the input— and that the primary role of training was merely to learn an appropriate linear readout from this pre-existing dynamical representation. The recurrent kernel, in other words, could be viewed as a fixed nonlinear feature extractor whose outputs need only be linearly combined to approximate the target function.

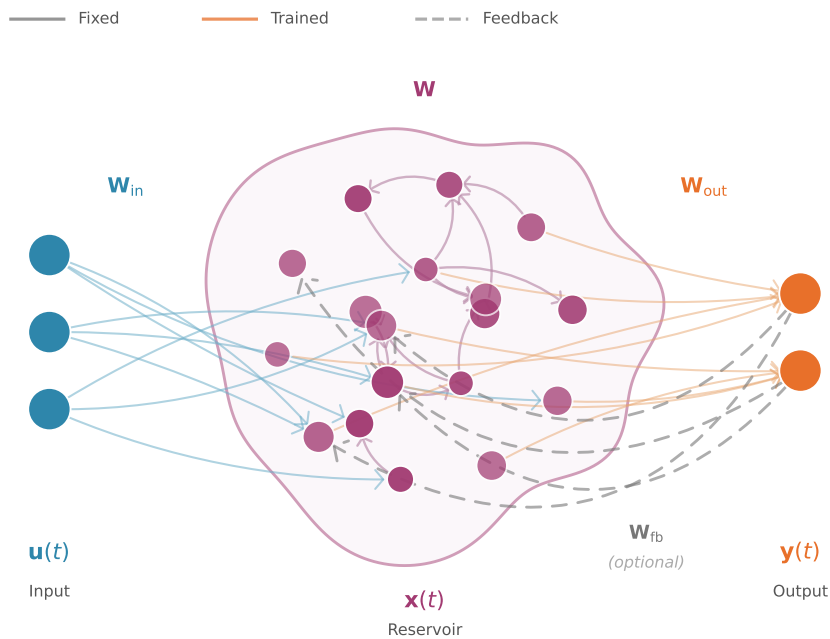


Figure 1.1: Schematic of the reservoir computing framework. The input signal $\mathbf{u}(n)$ is fed into a reservoir with fixed input weights $\mathbf{W}_{\text{in}}$ and recurrent weights $\mathbf{W}$. The high-dimensional state $\mathbf{x}(n)$ is read out by a trainable linear layer $\mathbf{W}_{\text{out}}$.

This insight crystallized into the *reservoir computing* (RC) paradigm, which was formalized independently and almost simultaneously through two complementary approaches: the *Echo State Network* (ESN), introduced by Jaeger [9], and the *Liquid State Machine* (LSM), proposed by Maass, Natschläger, and Markram [46]. The term “reservoir computing” itself was coined later by Verstraeten et al. [47] to unify these frameworks under a common conceptual umbrella, which also encompasses the Backpropagation-Decorrelation (BPDC) learning rule for RNNs developed by Steil [48].

The elegance of RC lies in this separation of concerns (Figure 1.1). The reservoir’s role is to provide a sufficiently rich, nonlinear, high-dimensional representation of the input

history. No gradient computation through the recurrent dynamics is required, and the vanishing/exploding gradient problem is entirely circumvented. The readout's role is to extract task-relevant information from this representation. Since linear regression is a convex optimization problem, the training is guaranteed to find the global optimum, is computationally inexpensive ($\mathcal{O}(N_x^2)$ for offline regression), and is numerically stable when properly regularized.

The RC paradigm decomposes the computational architecture into two functionally distinct components. The first is a large, fixed, recurrent dynamical system— the *reservoir*— that receives external input and transforms it into a high-dimensional state trajectory. The reservoir's internal weights are drawn from some random distribution and remain frozen; they are never modified by the learning algorithm. The second component is a trainable *readout layer*, typically a simple linear map, that projects the reservoir's state onto the desired output. Because only the readout weights are learned, the training problem reduces to linear regression, which is solved in closed form by methods such as ridge regression (Tikhonov regularization [11]) or, in the online setting, by recursive least squares [49].

1.1.1 Liquid State Machines

The Liquid State Machine, introduced by Maass et al. [46], was motivated by computational neuroscience and framed the reservoir as a model of cortical microcircuits. In an LSM, the reservoir consists of a randomly connected network of spiking neurons— typically integrate-and-fire or more biologically detailed models— that operate in continuous time. The input is encoded as a spatiotemporal pattern of spikes, and the reservoir's "liquid" state evolves according to the network's intrinsic dynamics. The metaphor of a liquid is apt: just as the surface of a body of water retains a transient, fading record of recent perturbations, the reservoir's state encodes a decaying trace of the input history.

Maass et al. established the theoretical foundation for this framework through two key properties: the *separation property*, which requires that distinct input streams produce distinguishable reservoir states, and the *approximation property*, which requires that a sufficiently rich readout can approximate arbitrary continuous functionals of the input [46]. Together, these ensure that the LSM constitutes a universal approximator for time-varying input-output mappings, under appropriate conditions on the reservoir and readout.

While LSMs remain influential in computational neuroscience and neuromorphic computing, their reliance on spiking neuron models and continuous-time dynamics introduces substantial complexity in both simulation and analysis. For the purposes of this thesis, whose focus is on the mathematical analysis and principled design of reservoir architectures for time series forecasting, we adopt the Echo State Network formulation, which operates in discrete time with sigmoidal neurons and is considerably more tractable both analytically and computationally.

1.1.2 Echo State Networks

The Echo State Network, introduced by Jaeger [9], is the discrete-time, rate-coded counterpart of the LSM. An ESN consists of a reservoir of N_x nonlinear units with sigmoidal activation

functions, connected by a fixed recurrent weight matrix $\mathbf{W} \in \mathbb{R}^{N_x \times N_x}$, driven by an external input through a fixed input weight matrix $\mathbf{W}_{\text{in}} \in \mathbb{R}^{N_x \times N_u}$, whose non-zero entries are drawn independently from a uniform distribution on $[-a, a]$; the *input scaling* $a > 0$ controls the amplitude of the driving signal as seen by the reservoir. The standard ESN state update equation is

$$\mathbf{x}(n) = f(\mathbf{W}_{\text{in}} \mathbf{u}(n) + \mathbf{W} \mathbf{x}(n-1)), \quad (1.2)$$

where $f : \mathbb{R}^{N_x} \rightarrow \mathbb{R}^{N_x}$ is a component-wise activation function, typically $\tanh$. The output is computed as

$$\mathbf{y}(n) = \mathbf{W}_{\text{out}} \mathbf{z}(n), \quad (1.3)$$

where $\mathbf{z}(n)$ is a feature vector that may include the reservoir state $\mathbf{x}(n)$, the input $\mathbf{u}(n)$, and optionally a bias term, and $\mathbf{W}_{\text{out}} \in \mathbb{R}^{N_y \times \dim(\mathbf{z})}$ is the trainable readout matrix.

From the perspective of dynamical systems theory, an ESN is an *input-driven* (nonautonomous) discrete-time dynamical system [30]. The reservoir state $\mathbf{x}(n)$ evolves under the combined influence of the driving input $\mathbf{u}(n)$ and the internal recurrent dynamics governed by $\mathbf{W}$. This viewpoint is fundamental to the theoretical analysis of ESNs, as it allows us to deploy the full machinery of nonautonomous dynamical systems theory— including Lyapunov exponents, conditional Lyapunov exponents, invariant measures, and generalized synchronization— to characterize the reservoir’s behavior.

Since the internal states $\mathbf{x}(n)$ and the output $\mathbf{y}(n)$ are time series, the tools of time series analysis are equally applicable. In particular, for forecasting tasks where the target output is the future evolution of a dynamical system observed through a scalar or vector measurement function, the ESN can be understood as implementing a nonlinear version of delay embedding, analogous to Takens’ embedding theorem [3]. The reservoir state provides a high-dimensional representation of the input history that, under suitable conditions, reconstructs the essential features of the underlying attractor.

1.2 The Echo State Property

The central theoretical requirement for the reservoir computing paradigm to function is the *echo state property* (ESP). Informally, the ESP states that the reservoir’s response should be uniquely determined by the input signal, regardless of the initial state. That is, after a sufficiently long driving period, two copies of the same reservoir initialized with different states but driven by the same input sequence should converge to the same trajectory. If this property fails, the reservoir’s output depends on its initialization, and the readout cannot learn a well-defined input-output mapping.

1.2.1 Formal Definition

Let us formalize the ESP following the framework of Jaeger [9] with the corrections from [12] and the input-dependent formulation of Manjunath and Jaeger [13]. Consider the state update map $T_{\mathbf{u}} : \mathbb{R}^{N_x} \rightarrow \mathbb{R}^{N_x}$ parametrized by an input $\mathbf{u}$:

$$T_{\mathbf{u}}(\mathbf{x}) = f(\mathbf{W}_{\text{in}} \mathbf{u} + \mathbf{W} \mathbf{x}). \quad (1.4)$$

Given a bi-infinite input sequence $\dots, \mathbf{u}(-1), \mathbf{u}(0), \mathbf{u}(1), \dots$, the ESN state at time n starting from initial condition $\mathbf{x}_0$ at time n_0 is obtained by iterated application:

$$\mathbf{x}(n; n_0, \mathbf{x}_0) = T_{\mathbf{u}(n)} \circ T_{\mathbf{u}(n-1)} \circ \dots \circ T_{\mathbf{u}(n_0+1)}(\mathbf{x}_0). \quad (1.5)$$

Definition 1.1 (Echo State Property). *An ESN with state update (1.2) has the echo state property with respect to a class of input sequences $\mathcal{U}$ if, for every input sequence $(\mathbf{u}(n))_{n \in \mathbb{Z}} \in \mathcal{U}$ and for every pair of initial conditions $\mathbf{x}_0, \mathbf{x}'_0 \in \mathcal{A}$ (where $\mathcal{A}$ is a compact admissible set),*

$$\lim_{n \rightarrow \infty} \|\mathbf{x}(n; 0, \mathbf{x}_0) - \mathbf{x}(n; 0, \mathbf{x}'_0)\| = 0. \quad (1.6)$$

When the ESP holds, the reservoir state $\mathbf{x}(n)$ is asymptotically a function only of the input history $\{\mathbf{u}(k)\}_{k \leq n}$. In other words, there exists a unique mapping— an *echo function*— from the space of left-infinite input sequences to the reservoir state space:

$$\mathbf{x}(n) = h(\dots, \mathbf{u}(n-1), \mathbf{u}(n)), \quad (1.7)$$

where h is a causal, time-invariant functional that does not depend on the initial condition. This is the “echo” that gives the property its name: the reservoir’s current state is an echo of the driving input history.

We can establish the equivalence between Definition 1.1 and the existence of a unique compatible state sequence by a contraction argument. Since the activation function $f = \tanh$ satisfies $\|f(\mathbf{a}) - f(\mathbf{b})\| \leq \|\mathbf{a} - \mathbf{b}\|$ (being a contraction in each component), we have for any input $\mathbf{u}$:

$$\|T_{\mathbf{u}}(\mathbf{x}) - T_{\mathbf{u}}(\mathbf{x}')\| = \|f(\mathbf{W}_{\text{in}}\mathbf{u} + \mathbf{W}\mathbf{x}) - f(\mathbf{W}_{\text{in}}\mathbf{u} + \mathbf{W}\mathbf{x}')\| \leq \|\mathbf{W}(\mathbf{x} - \mathbf{x}')\| \leq \|\mathbf{W}\| \cdot \|\mathbf{x} - \mathbf{x}'\|, \quad (1.8)$$

where $\|\mathbf{W}\|$ denotes the operator norm. If $\|\mathbf{W}\| < 1$, then $T_{\mathbf{u}}$ is a contraction for every input $\mathbf{u}$, and by the Banach fixed-point theorem applied to the composed sequence of maps, any two trajectories converge exponentially:

$$\|\mathbf{x}(n; 0, \mathbf{x}_0) - \mathbf{x}(n; 0, \mathbf{x}'_0)\| \leq \|\mathbf{W}\|^n \cdot \|\mathbf{x}_0 - \mathbf{x}'_0\| \xrightarrow{n \rightarrow \infty} 0. \quad (1.9)$$

The condition $\|\mathbf{W}\| < 1$ is thus *sufficient* for the ESP to hold for any input. However, this is a conservative condition involving the operator norm (largest singular value), which is in general much larger than the spectral radius $\rho(\mathbf{W})$ (largest absolute eigenvalue).

1.2.2 The Spectral Radius Misconception

It is here that one of the most persistent and damaging misconceptions in the reservoir computing literature must be addressed. Since Jaeger’s original report [9], a widespread heuristic has been to set the spectral radius $\rho(\mathbf{W}) < 1$ as the design criterion for ensuring the ESP. This heuristic has propagated through the literature to such an extent that many papers treat $\rho(\mathbf{W}) < 1$ as both a necessary and sufficient condition for the ESP. As Jaeger

himself has noted in no uncertain terms: “This has led in the literature to a far-spread but erroneous identification of the ESP with a spectral radius below 1” [50].

The situation is, in fact, considerably more subtle.

$\rho(\mathbf{W}) < 1$ is not sufficient for the ESP. Yildiz, Jaeger, and Kiebel [12] provided definitive analytical examples demonstrating that the ESP can be violated even when $\rho(\mathbf{W}) < 1$. Through a local bifurcation analysis of the autonomous system ($\mathbf{u} = \mathbf{0}$), they showed that specific weight matrix structures with spectral radius strictly below unity can exhibit multistability—the coexistence of multiple attractors—which directly contradicts the ESP. The key insight is that the spectral radius governs the linearized dynamics around fixed points, but the global nonlinear dynamics can exhibit qualitatively different behavior. A reservoir with $\rho(\mathbf{W}) = 0.95$, for instance, can possess multiple stable fixed points in the zero-input regime [12], meaning that different initializations lead to permanently distinct trajectories.

$\rho(\mathbf{W}) < 1$ is not necessary for the ESP. Conversely, when the input is sufficiently large, the driving signal can “wash out” the reservoir’s internal dynamics and force convergence even when $\rho(\mathbf{W}) > 1$. Jaeger observed this empirically and noted that “the larger the input amplitude, the further above unity the spectral radius may be while still obtaining the ESP” [50]. The input acts as a synchronizing force: strong driving can suppress the reservoir’s autonomous tendency to diverge.

The resolution of this apparent paradox lies in recognizing that the ESP is fundamentally a property of the *driven* system, not of the autonomous reservoir. It depends jointly on the reservoir architecture *and* the input statistics.

1.2.3 Input-Dependence of the Echo State Property

The input-dependent nature of the ESP was given a rigorous mathematical treatment by Manjunath and Jaeger [13], who established a remarkable *zero-one law*: for ergodic input sources, the ESP holds with probability either zero or one. That is, for a given reservoir and a given stationary ergodic input process, the ESP is not a “sometimes” property—it either holds almost surely or fails almost surely.

This result places the ESP firmly within the framework of nonautonomous dynamical systems theory [30]. The reservoir driven by a random input constitutes a random dynamical system, and the ESP corresponds to the existence of a unique random attractor that attracts all initial conditions almost surely. The relevant quantity is not the spectral radius of $\mathbf{W}$ alone, but the *Lyapunov exponents* of the driven system, which measure the average rate of exponential divergence or convergence of nearby trajectories under the combined action of the recurrent dynamics and the driving input.

The practical implications are significant. A reservoir that satisfies the ESP for one class of inputs (e.g., Gaussian white noise of a given variance) may fail it for another (e.g., a slowly varying sinusoid or a constant input). The design and validation of ESN architectures must

therefore account for the specific input regime in which the network will operate. This observation will be central to the analysis developed in Chapter 2, where we study how the transient behavior of the ESP convergence depends on both the reservoir topology and the input characteristics.

Further algebraic conditions for the ESP have been developed by Buehner and Young [31], who provided tighter bounds than the original operator-norm condition, and by Ozturk et al. [51], who analyzed the relationship between spectral radius and reservoir dynamics in more detail. The universal approximation capacity of ESNs with the ESP has been established by Grigoryeva and Ortega [52], who proved that ESNs with the fading memory property (a consequence of the ESP) can approximate any causal, time-invariant, continuous filter with fading memory to arbitrary precision. This result was further refined by Gonon and Ortega [53].

1.2.4 Connection to Generalized Synchronization

The ESP admits a natural interpretation in the language of *generalized synchronization* from the theory of coupled dynamical systems [32, 54]. In the classical drive-response framework, a *drive* system $\mathbf{u}(n)$ unidirectionally forces a *response* system $\mathbf{x}(n)$. Generalized synchronization is said to occur when the response state becomes a deterministic function of the drive state:

$$\mathbf{x}(n) = \phi(\mathbf{u}(n), \mathbf{u}(n-1), \dots), \quad (1.10)$$

for some (possibly very complex) functional ϕ . Comparing with Eq. (1.7), we see that the ESP is precisely the statement that the reservoir achieves generalized synchronization with the input signal. The echo function h is the synchronization functional ϕ .

This connection is not merely a terminological convenience. It provides a set of quantitative tools for analyzing the ESP and the quality of the reservoir's representation. In particular, the *conditional Lyapunov exponents* (CLEs) of the driven reservoir—the Lyapunov exponents computed for the response system conditioned on the driving signal—characterize the rate at which different initial conditions converge under the same input. When all CLEs are negative, the ESP holds and the convergence is exponential. The magnitude of the largest (least negative) CLE quantifies how rapidly the initial transient decays, which is directly relevant to the choice of the transient discard length in practice.

Hart [33] recently demonstrated that the CLEs play a critical role in attractor reconstruction tasks. Specifically, for a reservoir computer trained to autonomously reproduce the chaotic attractor of the target system, the maximal CLE of the driven reservoir must be significantly more negative than the most negative Lyapunov exponent of the target system. This provides a quantitative link between the synchronization dynamics during training and the fidelity of the learned autonomous dynamics. Lymburn et al. [55] further explored the reservoir's perspective on generalized synchronization, relating the smoothness of the synchronization functional to the reservoir's ability to generalize.

1.3 ESN Variants

The basic ESN architecture defined by Eq. (1.2) has been extended in several directions to address specific limitations or to exploit particular structural priors. In this section, we survey the principal variants.

1.3.1 Leaky Integrator ESN

The standard ESN update (1.2) has an implicit time constant determined entirely by the reservoir weights. In many applications, the input signal operates on a different time scale from the natural dynamics of the $\tanh$ neurons, creating a mismatch that degrades performance. The *leaky integrator* ESN, introduced by Jaeger et al. [56], addresses this by incorporating an explicit leak rate $\alpha \in (0, 1]$ into the state update:

$$\mathbf{x}(n) = (1 - \alpha) \mathbf{x}(n-1) + \alpha f(\mathbf{W}_{\text{in}} \mathbf{u}(n) + \mathbf{W} \mathbf{x}(n-1)). \quad (1.11)$$

The leak rate α controls the speed of the reservoir dynamics. When $\alpha = 1$, Eq. (1.11) reduces to the standard ESN. As $\alpha \rightarrow 0$, the state update becomes increasingly sluggish, effectively implementing a low-pass filter that smooths the reservoir response and extends the effective memory. This can be interpreted as a discrete-time approximation to a continuous-time reservoir with time constant $\tau = 1/\alpha$. The leaky integrator formulation thus provides a tunable mechanism for matching the reservoir's intrinsic time scale to the temporal structure of the input, which is especially valuable when the input contains both fast and slow components.

The ESP conditions for leaky integrator ESNs were analyzed in [56] and [12]. The key observation is that the leak rate modifies the effective weight matrix: the autonomous dynamics are governed by $(1 - \alpha)\mathbf{I} + \alpha\mathbf{W}'$ (where $\mathbf{W}'$ accounts for the linearized activation), so the stability properties depend on α in a nontrivial way.

1.3.2 Deep Echo State Networks

A natural extension of the single-layer ESN is the *deep* ESN, which stacks multiple reservoir layers in a hierarchical architecture [57–59]. In a deep ESN with L layers, the state update for layer ℓ is:

$$\mathbf{x}^{(\ell)}(n) = f(\mathbf{W}_{\text{in}}^{(\ell)} \mathbf{x}^{(\ell-1)}(n) + \mathbf{W}^{(\ell)} \mathbf{x}^{(\ell-1)}(n-1)), \quad \ell = 1, \dots, L, \quad (1.12)$$

where $\mathbf{x}^{(0)}(n) = \mathbf{u}(n)$ is the external input. Each layer receives the output of the previous layer as its input, creating a hierarchy of temporal abstractions.

The motivation is analogous to that of deep feedforward networks: by composing multiple nonlinear transformations, the network can learn increasingly abstract representations. In the reservoir computing context, deeper layers are expected to capture longer-range temporal dependencies and more complex temporal features than a single layer of equivalent total size [57]. Gallicchio and Micheli [58] extended the ESP analysis to the deep setting, showing

that the ESP for the full deep architecture requires analyzing the contraction properties across all layers, characterized by a *global* condition involving the spectral properties of all layer weight matrices jointly.

1.3.3 Parallel and Ensemble ESNs

An alternative to depth is breadth: *parallel* or *ensemble* ESN architectures employ multiple independent reservoirs operating simultaneously on the same input. Each reservoir may differ in size, topology, spectral radius, or leak rate, providing diverse dynamical representations of the input signal. The readout layer then combines the states from all reservoirs:

$$\mathbf{y}(n) = \mathbf{W}_{\text{out}} \begin{bmatrix} \mathbf{x}_1(n) \\ \vdots \\ \mathbf{x}_K(n) \end{bmatrix}, \quad (1.13)$$

where $\mathbf{x}_k(n)$ is the state of the k -th reservoir at time n .

The ensemble approach exploits the diversity of the individual reservoirs to improve robustness and generalization [60]. Different reservoirs may specialize in capturing different temporal scales or different aspects of the input dynamics, and the linear readout can optimally weight their contributions. This is conceptually related to ensemble methods in machine learning (e.g., bagging), but with the important difference that the diversity arises from the random initialization of the reservoir parameters rather than from resampling the training data.

1.4 Reservoir Topologies

The topology of the recurrent weight matrix $\mathbf{W}$ — that is, the graph structure of the connections among reservoir neurons— has a significant influence on the ESN’s dynamical properties, memory capacity, and computational performance. While the original ESN formulation employed sparse random matrices [9], subsequent work has explored a variety of structured topologies [61–66].

1.4.1 Random Sparse Reservoirs

The default ESN construction generates $\mathbf{W}$ as a sparse random matrix: each entry is nonzero with probability p (the *connection density* or *sparsity* parameter) and, when nonzero, is drawn from some distribution (e.g., uniform on $[-1, 1]$). The matrix is then rescaled so that $\rho(\mathbf{W})$ equals a prescribed target spectral radius. This construction is inspired by the Erdős-Rényi random graph model [67], where each possible edge exists independently with a fixed probability.

Random sparse reservoirs have the advantage of simplicity and generally provide good performance across a range of tasks. The sparsity serves both a computational purpose (reducing the cost of the matrix-vector products in the state update) and a dynamical one

(creating a heterogeneous network with diverse local connectivity patterns). However, the random construction introduces variability: different random seeds produce reservoirs with different performance, and there is no guarantee that any particular realization is optimal or even adequate for a given task.

1.4.2 Minimal Complexity Reservoirs

A striking line of work by Rodan and Tiño [36, 68] demonstrated that highly structured, deterministic reservoir topologies can match or exceed the performance of random sparse reservoirs while being far simpler to construct and analyze (Figure 1.2).

Simple Cycle Reservoir (SCR). The simplest nontrivial topology is the *cycle* (ring) reservoir, where the N_x neurons are arranged in a single directed cycle:

$$W_{i,i+1} = r, \quad i = 1, \dots, N_x - 1, \quad W_{N_x,1} = r, \quad (1.14)$$

with all other entries zero. The single parameter r controls both the connectivity and the spectral radius ($\rho(\mathbf{W}) = |r|$). Despite its extreme simplicity—no randomness, no tunable sparsity, a single free parameter—the SCR was shown to perform competitively with standard random reservoirs on a variety of benchmarks [36].

Cycle Reservoir with Jumps (CRJ). An extension of the SCR adds “jump” connections that skip a fixed number of neurons along the cycle [68]. These long-range connections introduce shortcuts in the graph, enhancing the mixing of information across the reservoir while preserving the deterministic, parameter-efficient construction.

The theoretical significance of these results was established recently: Li, Fong, and Tiño [69] proved that simple cycle reservoirs are *universal*—they can approximate any fading-memory filter to arbitrary precision, given a sufficiently large cycle. This universality result was further extended by Fong et al. [70], who established the universality of real minimal complexity reservoirs more broadly. These results imply that the complex random topologies traditionally used in ESNs are not necessary for computational universality; the essential ingredient is the interplay between nonlinear activation and recurrent memory, which is present even in the simplest cyclic structure.

Delay Line Reservoir (DLR). An even simpler structure is the pure delay line, where each neuron connects only to its successor: $W_{i,i+1} = r$. This is a cycle with the closing edge removed. The DLR has a finite memory capacity equal to its length N_x , since information propagates along the chain without any recurrence. While limited, the DLR serves as a useful theoretical baseline and connects reservoir computing to the classical tapped-delay-line models from signal processing.

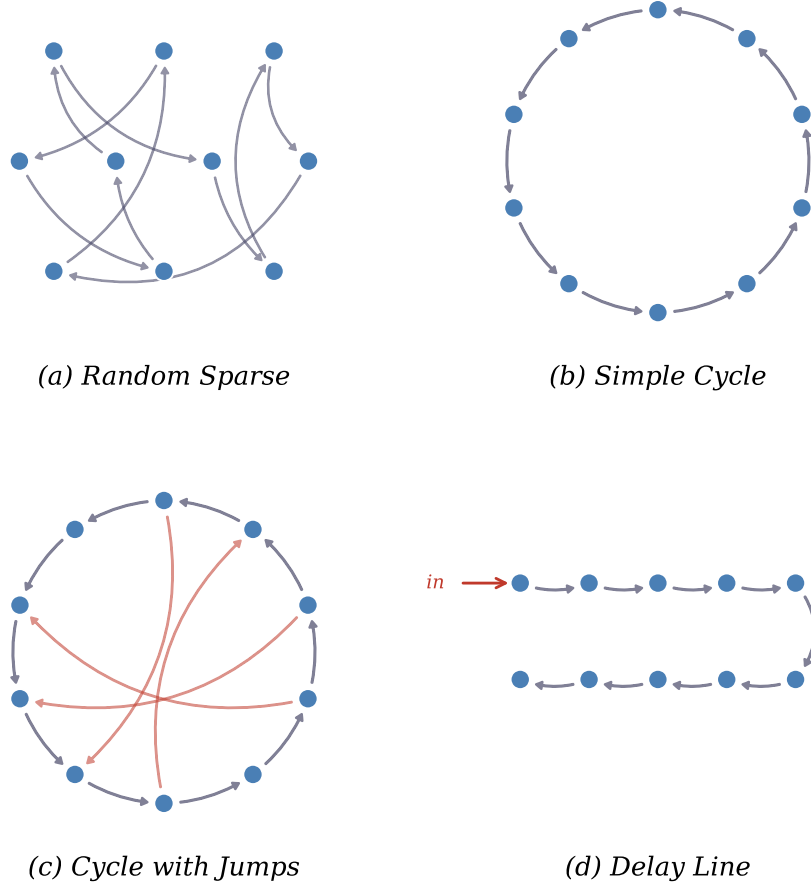


Figure 1.2: Comparison of common reservoir topologies. (a) Random sparse with Erdős-Rényi connectivity. (b) Simple cycle reservoir (SCR): a single directed ring. (c) Cycle with jumps (CRJ): cycle plus long-range skip connections. (d) Delay line reservoir (DLR): a unidirectional chain.

Small-world and other structured topologies. Several studies have explored small-world networks [71], scale-free topologies [72], and other graph-theoretic structures as reservoir connectivity patterns. Small-world reservoirs, which combine local clustering with a small number of long-range connections, have been shown to enhance the echo state property and signal propagation [71]. These topological investigations highlight that the reservoir's graph structure is not merely a structural detail but a design parameter that shapes the network's dynamical repertoire.

The relationship between reservoir topology and computational performance is a central theme of this thesis. In Chapter 2, we investigate how the topological structure—specifically, the presence of cycles and their organization— affects the transient dynamics and convergence rate of the ESP, as well as its impact on forecast performance.

1.5 Training Echo State Networks

The training of an ESN proceeds in two distinct phases that reflect the fundamental separation between the fixed reservoir and the trainable readout (Figure 1.3).

1.5.1 Open-Loop Phase: State Collection

In the *open-loop* (or *teacher-forcing*) phase, the reservoir is driven by the input signal $\mathbf{u}(n)$ for $n = 1, \dots, T$, and the resulting state trajectory $\{\mathbf{x}(n)\}_{n=1}^T$ is recorded. If the task is time series forecasting—predicting the future evolution of a signal—then the “input” at each step is the current value of the signal and the “target output” is its value at the next time step.

An initial segment of the state trajectory, typically the first T_0 steps, is discarded to eliminate the influence of the arbitrary initial state $\mathbf{x}(0)$. This is the *transient washout*: by the time the discarded segment has passed, the ESP ensures (assuming it holds) that the reservoir state has converged to the unique trajectory determined by the input. The choice of T_0 is a practical matter that is often decided heuristically, but it should in principle depend on the convergence rate of the ESP—a point that is rarely addressed in the literature and which we examine systematically in Chapter 2.

The concept of *teacher forcing* can be understood through an analogy. Imagine a student taking a sequential examination where each question depends on the answer to the previous one. Without teacher forcing, an early mistake propagates and corrupts all subsequent answers. With teacher forcing, the correct answer is provided after each question, so the student always works from accurate information. In the ESN context, the reservoir is always driven by the true input signal during the open-loop phase, ensuring that the states used for training the readout are computed from the actual input history, not from the network’s own (potentially erroneous) predictions.

After discarding the initial transient (washout) of length T_0 , the reservoir states $\{\mathbf{x}(n)\}_{n=T_0+1}^T$ and the corresponding target outputs $\{\mathbf{d}(n)\}_{n=T_0+1}^T$ are collected and stacked into matrices, where $\mathbf{d}(n)$ denotes the desired output at time n as defined by the specific training task (e.g., next-step prediction, multi-step forecasting, or classification labels):

$$\mathbf{Z} = \begin{bmatrix} \mathbf{z}(T_0 + 1) & \cdots & \mathbf{z}(T) \end{bmatrix}^\top, \quad \mathbf{D} = \begin{bmatrix} \mathbf{d}(T_0 + 1) & \cdots & \mathbf{d}(T) \end{bmatrix}^\top, \quad (1.15)$$

where $\mathbf{z}(n)$ is the extended state vector (possibly including input, bias, and nonlinear transformations of the state). The readout weights are then computed by solving the regularized least-squares problem:

$$\mathbf{W}_{\text{out}} = \mathbf{D}^\top \mathbf{Z} (\mathbf{Z}^\top \mathbf{Z} + \beta \mathbf{I})^{-1}, \quad (1.16)$$

where $\beta > 0$ is the Tikhonov regularization parameter [11]. The regularization prevents overfitting and improves numerical stability, particularly when the reservoir states are highly correlated (as they tend to be in slow-varying or low-dimensional input regimes).

1.5.2 Closed-Loop Phase: Autonomous Prediction

Once the readout weights $\mathbf{W}_{\text{out}}$ have been computed, the ESN can be operated in *closed-loop* mode for generative tasks such as autonomous forecasting. In this mode, the network’s

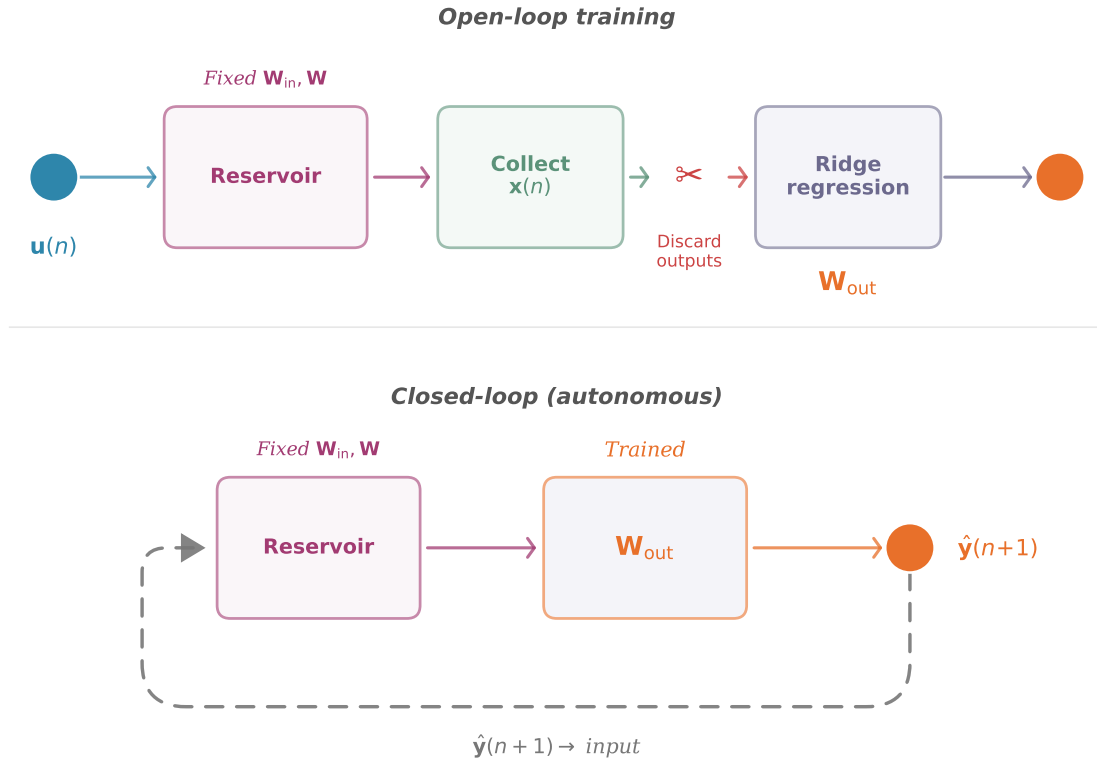


Figure 1.3: The ESN training and inference pipeline. During the open-loop phase, the true input drives the reservoir and states are collected for readout training. During the closed-loop phase, the network's own output is fed back as input for autonomous forecasting.

output at each step is fed back as the input for the next step:

$$\hat{\mathbf{u}}(n+1) = \mathbf{y}(n) = \mathbf{W}_{out} \mathbf{z}(n), \quad \mathbf{x}(n+1) = f(\mathbf{W}_{in} \hat{\mathbf{u}}(n+1) + \mathbf{W} \mathbf{x}(n)). \quad (1.17)$$

The transition from open-loop to closed-loop operation is a critical juncture. During training, the reservoir is driven by the true signal and the readout is optimized to minimize the one-step prediction error under this condition. In closed-loop mode, however, the driving signal is replaced by the network's own predictions, which inevitably contain errors. These errors accumulate over time: a small prediction error at step n shifts the reservoir state, which alters the prediction at step $n+1$, and so on. For chaotic systems, this error amplification follows the positive Lyapunov exponents of the target dynamics, leading to exponential divergence from the true trajectory over a time scale related to the system's predictability horizon.

This distinction between the open-loop and closed-loop regimes is fundamental to the evaluation of ESN performance, as we discuss in Section 1.6.3.

1.6 Critical Issues in the ESN Framework

Despite its conceptual elegance and practical success, the ESN framework presents several methodological challenges that are insufficiently addressed in the existing literature.

We identify three principal issues: hyperparameter sensitivity, time discretization, and benchmarking practices.

1.6.1 Hyperparameter Sensitivity

The performance of an ESN depends on a substantial number of hyperparameters, including but not limited to: the reservoir size N_x , the spectral radius $\rho(\mathbf{W})$, the input scaling (magnitude of $\mathbf{W}_{\text{in}}$), the leak rate α (for leaky integrator ESNs), the topology and sparsity of $\mathbf{W}$, the regularization parameter β , and the transient washout length T_0 . These hyperparameters interact in complex, nonlinear ways, and no general theory exists to prescribe optimal values for a given task.

In practice, hyperparameter selection is performed through grid search, random search, or more sophisticated methods such as Bayesian optimization. The resulting computational cost can be substantial, particularly since ESN performance can be highly sensitive to hyperparameter values—small changes in spectral radius or input scaling can lead to qualitatively different reservoir dynamics and dramatically different forecasting performance. As Lukoševičius [73] noted, practical guidelines exist (e.g., larger spectral radius for tasks requiring longer memory, smaller input scaling for tasks requiring fine discrimination of input amplitude), but these remain heuristic rules of thumb rather than principled design criteria.

In Chapter 5, we develop a principled hyperparameter optimization framework that integrates the insights from the preceding chapters on transient times, time discretization, and benchmarking to systematically navigate the ESN design space.

1.6.2 Time Discretization

When the target system evolves in continuous time and the ESN receives a sampled version of the signal, the choice of sampling interval Δt has profound consequences for the reservoir’s dynamics and computational capacity. This issue is frequently overlooked in the ESN literature, where the time discretization is typically inherited from the dataset or chosen for numerical convenience, with no principled analysis of its implications.

If Δt is too large, aliasing may corrupt high-frequency components of the signal, and the geometric structure of the attractor may be distorted. Rapid dynamical events—such as the sharp transitions in chaotic systems like the Lorenz attractor—may be inadequately resolved, leading to a loss of information that no amount of reservoir processing can recover.

Conversely, if Δt is too small, successive samples become highly correlated, and the input presents little new information at each step. This wastes the reservoir’s finite memory capacity on slowly changing, redundant data. The reservoir’s state trajectory becomes “compressed” along the temporal axis, with the effective dynamics operating on a slow manifold that underutilizes the network’s nonlinear processing capacity.

There exists an optimal regime where Δt is matched to the time scales of the target dynamics and the reservoir’s memory capacity. Identifying this regime requires analyzing the interplay between the input’s spectral content, the reservoir’s fading memory characteristics, and the

geometry of the underlying attractor. We develop a principled approach to this problem in Chapter 3, where we introduce a sagitta-based discretization criterion that quantifies the geometric distortion induced by sampling and provides a systematic method for selecting Δt .

1.6.3 Benchmarking Practices

A widespread but fundamentally flawed practice in the evaluation of Echo State Networks is the restriction to one-step-ahead prediction as the sole benchmarking protocol. While one-step prediction is useful for analyzing local regression capacity, it does not reflect the actual demands of time series forecasting, where models are expected to operate recursively or over extended horizons. Relying exclusively on one-step metrics systematically overstates ESN performance by eliminating the impact of error accumulation, feedback dynamics, and long-range memory effects—precisely the phenomena that recurrent architectures are meant to address.

Several papers adopt this limited evaluation regime, reporting NRMSEs or similar metrics on one-step prediction tasks while omitting any form of autonomous multi-step forecasting. This renders their benchmarking incomplete at best, and misleading at worst, particularly when conclusions are drawn about the ESN's suitability for real-world forecasting problems. Proper evaluation of ESNs requires testing their behavior under recursive prediction, where outputs are fed back as inputs and the model is exposed to its own errors over time. Without this, any claims about temporal modeling or memory capacity are methodologically unsupported.

The issue is compounded by the common use of loss functions that do not account for the time-dependent nature of forecasting errors. For chaotic systems, the divergence between the predicted and true trajectories grows exponentially in time, following the positive Lyapunov exponents of the system. A loss function that penalizes the prediction equally at all time steps conflates the inherent unpredictability of chaos with the model's forecasting failure. What is needed instead are loss functions that *relax* over time, penalizing the model for structural failures (e.g., loss of attractor topology) rather than for the inevitable divergence dictated by sensitive dependence on initial conditions.

We address this issue comprehensively in Chapter 4, where we introduce the Error Forecast Horizon (EFH) framework for evaluating ESN forecasting performance. The EFH quantifies how long the model's predictions remain useful, providing a time-resolved measure that properly accounts for the dynamics of error growth.

1.7 Outlook

The Echo State Network provides a powerful and computationally efficient framework for temporal information processing. Its theoretical foundation—the Echo State Property—connects it to the rich mathematical theory of nonautonomous dynamical systems and generalized synchronization. Its practical variants—leaky integrator, deep, and parallel architectures—offer flexibility to address diverse application requirements. And its deterministic minimal-complexity variants demonstrate that the essential computational

capacity does not depend on complex random structures, but on the fundamental interplay between recurrence and nonlinearity.

Yet, as we have argued in this chapter, the framework suffers from three interconnected methodological gaps. The transient dynamics of the ESP convergence are poorly understood and rarely accounted for. The time discretization is treated as an afterthought. And the benchmarking practices prevalent in the literature fail to evaluate the aspects of ESN performance that matter most for real forecasting applications.

The remainder of this thesis is devoted to addressing these gaps. In Chapter 2, we study the transient times of ESNs and their dependence on reservoir topology, establishing quantitative relationships between the network structure and the rate of ESP convergence. In Chapter 3, we develop a principled time discretization methodology based on sagitta geometry that ensures the input signal is sampled at a resolution appropriate to the underlying dynamics and the reservoir's capacity. In Chapter 4, we introduce the EFH benchmarking framework that evaluates ESN forecasting in a manner consistent with the dynamical nature of the prediction task. Finally, in Chapter 5, we integrate these contributions into a principled hyperparameter optimization pipeline and demonstrate its effectiveness on representative chaotic systems.

All the simulations and experiments performed in this thesis can be reproduced using the open-source library developed by the author in the repository <https://github.com/El3ssar/ResDAG> [43]. This library provides a comprehensive set of tools for the training of ESN models and forecasting of timeseries with the use of GPU acceleration via PyTorch. It is also integrated with the possibility of constructing different types of ESNs, such as leaky integrator ESNs, deep ESNs, parallel ESNs, ensemble ESNs, etc. It also includes several topologies for the recurrent weight matrix as well as a module for bayesian optimization over the model's hyperparameters.

2 Transient Times in Echo State Networks

Reservoir computing depends on the synchronization of the reservoir’s internal state with the driving signal—a process governed by the Echo State Property introduced in Chapter 1. Before training the readout layer, an initial segment of the input sequence must be discarded as washout, during which the reservoir forgets its initial conditions. This chapter investigates the distribution of these synchronization times across ensembles of randomly constructed reservoirs and analyzes their dependence on hyperparameters such as spectral radius, network size, and link density. While individual reservoirs typically exhibit sharply defined convergence times, ensembles display heavy-tailed distributions when the spectral radius ρ approaches or exceeds unity. For certain realizations, the required synchronization time becomes prohibitively long, limiting effective training data. Analytical bounds are derived for $\rho < 1$, demonstrating how non-linear saturation and input forcing enable convergence even for $\rho > 1$, and structural features of reservoir graphs responsible for slow or absent synchronization are identified. These findings establish that transient behavior is a measurable, engineerable property with clear design principles for robust, data-efficient reservoir computing.

2.1 Motivation

The Echo State Property, as discussed in Chapter 1, ensures that the reservoir state $\mathbf{x}(t)$ synchronizes with the input signal $\mathbf{u}(t)$ in the sense of generalized synchronization [74]. Two trajectories initialized differently but driven by the same input asymptotically satisfy $\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\| \rightarrow 0$. This occurs when the driven system is uniformly contracting along the response direction—equivalently, when the largest conditional Lyapunov exponent is negative. Under these conditions, the reservoir state becomes a unique function of the input history, which corresponds to the ESP for the given driver.

The synchronization transient is not merely a prerequisite for the ESP but a practical bottleneck for training and prediction. In open-loop (teacher-forced) training, any initial segment of the input discarded as washout directly reduces the amount of usable data

available for estimating the readout weights. When the transient-time distribution has heavy tails, the training budget becomes dominated by rare but extremely slow reservoirs, reducing statistical efficiency. In closed-loop prediction, the internal state at the moment of switching from teacher forcing to autonomous prediction must lie close to the input-driven orbit; otherwise small mismatches grow under the dynamics at a rate governed by the conditional Lyapunov exponent of the trained system. Reservoirs with long synchronization times therefore begin closed-loop prediction from a poorly aligned internal state, which immediately shortens the prediction horizon.

The spectral radius ρ of the recurrent weight matrix W is a key factor influencing reservoir dynamics and the likelihood of satisfying the ESP [9, 31, 75]. However, having $\rho < 1$ does not ensure the ESP, and in practice, reservoirs with $\rho > 1$ can still exhibit the property—in some cases improving performance [12, 13, 31]. The choice of hyperparameters (N , n_{links} , ρ), even when fixed, yields drastically different performance across random realizations [76–78]. This variability motivates the need for early-stage diagnostics to identify and discard poorly conditioned reservoirs.

This chapter focuses on the synchronization phase preceding readout training—specifically, the time required for the internal state to converge to the input-driven trajectory. The main result is that the distribution of synchronization times has long tails when the spectral radius is close to or exceeds unity. Individual realizations with identical hyperparameters can require extremely long synchronization times. In such cases, it is recommended to stop training and generate a new reservoir, which implies testing whether the internal state has already synchronized with the input.

This work complements Gallicchio’s ESP index [76], which quantifies the fraction of reservoirs that fail to converge within a large but fixed time. The approach here differs in two respects: first, the focus is on the *distribution* of synchronization times rather than a binary convergence criterion; second, a graph-theoretic explanation is provided for why certain reservoir realizations fail to synchronize under zero input, decomposing the recurrent matrix into strongly connected components and classifying nodes by their long-term behavior.

Transient dynamics in high-dimensional networks have been studied extensively in contexts such as contraction to fixed points, synchronization under common driving, and stability conditions expressed via spectral radii or Lyapunov exponents [2, 79, 80]. This analysis builds on these foundations but targets a different object: the distribution of synchronization times across random, sparse architectures under the hyperparameter choices typical of reservoir computing. The contributions are threefold: (i) an explicit estimate of the ϵ -synchronization time as a function of ρ , N , and tolerance, tested in the nonlinear, driven regime; (ii) demonstration that as ρ approaches or exceeds unity, synchronization times become heavy-tailed and depend systematically on link density, system size, and leak rate; (iii) a graph-theoretic explanation of non-convergent zero-input dynamics via decomposition into strongly connected components, identifying core, driven, and quiescent nodes. This rephrases known dynamical mechanisms—spectral contraction, conditional Lyapunov exponents, and SCC structure—in a setting where the finite synchronization time itself is the limiting resource.

The results inform the hyperparameter optimization framework developed in Chapter 5 and provide design principles that extend beyond the standard washout heuristic. By quantifying transient behavior, pre-screening strategies can filter out slow-synchronizing realizations

before training, improving data efficiency and robustness. The graph-based perspective also connects to the geometric discretization methods introduced in Chapter 3, where temporal step sizes are chosen to normalize information content—a procedure that implicitly assumes the reservoir has already synchronized with the input signal on the timescale of interest.

2.2 Quantification

The objective is to quantify the synchronization time of the reservoir’s internal state when driven by a time series input. For brevity, this interval is referred to as the transient time, in analogy to the period a deterministic trajectory, starting from an arbitrary point in phase space, requires to settle onto the attractor in a dissipative dynamical system. An Echo State Network in the sense of Jaeger [9] is indeed a dissipative system, since Jaeger’s requirements of “initial condition forgetting” and “uniformly state contracting” describe exactly what dynamical systems theory calls the existence of a globally attracting set [81]. This goes beyond mere contraction of phase space volume: for instance, the simple system

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} ax \\ by \end{pmatrix} \quad \text{with } a > 0, \quad ab < 0 \quad (2.1)$$

is volume contracting but leads to diverging $|x(t)|$ as $t \rightarrow \infty$. In a reservoir computing system, the globally attracting set to which arbitrary initial states are attracted asymptotically reduces to a fixed point when the input is switched off. Under non-trivial driving, the echo state property guarantees that the dependence on the initial condition $\mathbf{x}(0)$ vanishes asymptotically: after a sufficient transient, the reservoir state is uniquely determined by the input history, regardless of initialization. In the language of random dynamical systems [82, 83], this unique asymptotic state constitutes a point-like pullback attractor.

To determine the transient time for a given reservoir setup and a given input sequence u_t (restricting to scalar inputs), a given reservoir is run in parallel with two different internal states using two different initial conditions. Random initial conditions are chosen for $\mathbf{x}_1(0)$, and $\mathbf{x}_2(0) = \mathbf{0}$ is set, following the procedure in Ref. [76]. These internal states are then evolved in time using the same input sequence u_t , and their Euclidean distance is measured at every time step. The transient time is identified as the first time t for which the distance $\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\| < \epsilon = 10^{-7}$.

To quantify synchronization, we track the distance $D(t) = \|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|$ between two reservoir states evolving from different initial conditions under the same input. We first verify that $D(t)$ decreases monotonically for a single input sequence and a single pair of initial conditions (Fig. 2.1a). To confirm that this decay is not an artifact of a particular input realization, we repeat the computation for multiple input sequences drawn from the same dynamical system and average $D(t)$ over all pairs of random initializations (Fig. 2.1b). In both cases, the curves exhibit a uniformly decreasing trend.

This robustness motivates defining the transient time T_ϵ as the first time t at which $\langle D(t) \rangle < \epsilon = 10^{-7}$, where $\langle \cdot \rangle$ denotes the average over all initial-condition pairs and input

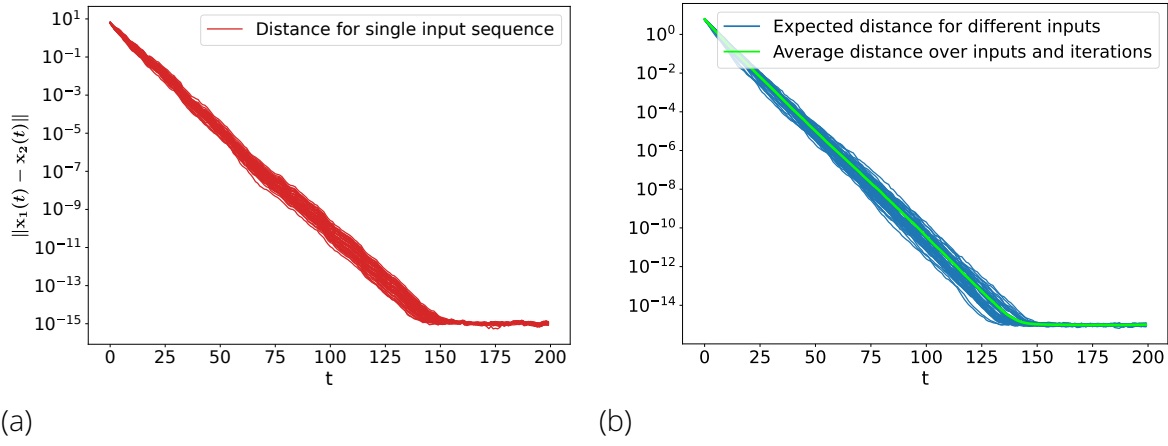


Figure 2.1: Synchronization of reservoir states from different initial conditions. (a) $D(t) = \|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|$ for a single input sequence across several random initializations. (b) $D(t)$ averaged over random initializations for each of several input sequences (blue), and averaged over both initializations and input sequences (green).

sequences. The resulting quantity is a single smooth function of the reservoir parameters (Fig. 2.1b, green), capturing the typical synchronization timescale for a given reservoir.

Unless otherwise stated, averaging is performed over 100 independent reservoir realizations and 50 input sequences per realization; ensemble studies in Sec. 2.5 use 10^4 reservoirs per setting.

Three different input sequences are employed in the following numerical experiments. A deterministic, chaotic trajectory is created by iteration of the Ulam map (logistic equation) $u_{t+1} = 1 - 2u_t^2$. By itself it is normalized to the interval $[-1, 1]$ and has a positive Lyapunov exponent $\lambda = \ln 2$. In a second approach, a random sequence of uniformly distributed white noise is used. While such a sequence is unpredictable and therefore useless for prediction experiments, it serves as a valid input for studying the ESP. As the third input, the constant signal $u_t = 0$ is used. This is equivalent to not driving the system at all. In this case, as can be seen from the update rule introduced in Chapter 1 and the fact that $\tanh(0) = 0$, the internal state $\mathbf{x}_2(0) = \mathbf{0}$ is a fixed point of the dynamics. However, if the reservoir lacks the ESP, this fixed point might not be globally attractive [9].

2.3 Analysis

If the spectral radius satisfies $\rho < 1$, an analytic estimate—and an upper bound (see Appendix A)—for the transient time can be derived from the linearized dynamics. Recall from Chapter 1 that the update rule with leaky integrator is

$$\mathbf{x}(t) = (1 - \alpha)\mathbf{x}(t - \Delta t) + \alpha f(W_{in}\mathbf{u}(t) + W\mathbf{x}(t - \Delta t)), \quad (2.2)$$

where $\alpha \in [0, 1]$ is the leak rate and $f(\cdot) = \tanh(\cdot)$ is the activation function applied element-wise. The parameter α controls the timescale over which the reservoir's internal dynamics evolve: small values of α make the system respond slowly, effectively increasing its memory of past states, while $\alpha \rightarrow 1$ corresponds to a fully updated state at each timestep. Δt is the time step of the discrete-time system.

The $\tanh$ function is approximately linear in the vicinity of zero, so for small arguments $|\mathbf{x}| \ll 1$ the update rule can be linearized to

$$\begin{aligned}\mathbf{x}(t) &= (1 - \alpha) \mathbf{x}(t - \Delta t) + \alpha (W_{in} \mathbf{u}(t) + W \mathbf{x}(t - \Delta t)) \\ &= \underbrace{((1 - \alpha)\mathbb{I} + \alpha W)}_{W_{\text{eff}}} \mathbf{x}(t - \Delta t) + \underbrace{\alpha W_{in}}_{W_{in, \text{eff}}} \mathbf{u}(t) \\ &= W_{\text{eff}} \mathbf{x}(t - \Delta t) + W_{in, \text{eff}} \mathbf{u}(t).\end{aligned}\tag{2.3}$$

Introducing a leaky integrator modifies the effective recurrent matrix acting on the internal state. The leak blends the identity with the recurrent connections, effectively shifting all eigenvalues of W toward unity as α decreases. In practice, when using $\alpha \neq 1$, the matrix W is rescaled so that the spectral radius of W_{eff} matches the desired ρ . From this point forward, unless otherwise stated, ρ refers to the spectral radius of the effective recurrent matrix W_{eff} , which coincides with the spectral radius of W when $\alpha = 1$.

From this linearization, the evolution of the internal states can be analyzed in the special case $\mathbf{u}_t = \mathbf{0}$ as repeated multiplication of the modified recurrent weight matrix W_{eff} with the internal state vector. The norm of the state vector $\mathbf{x}(t)$ is then bounded by its initial length $|\mathbf{x}_0|$ multiplied by the largest eigenvalue of W_{eff} raised to the power t , where this largest eigenvalue is itself bounded by the spectral radius ρ of W_{eff} . Consequently, $|\mathbf{x}(t)|$ evolves asymptotically as ρ^t , and for $\rho < 1$ it decays exponentially over time.

It is important to note that W_{eff} , since it has non-zero spectral radius, cannot be nilpotent—that is, it cannot map the initial state vector exactly to the zero vector after a finite number of multiplications. Due to the linearity of this system, the same exponential decay applies to the distance between any two trajectories of internal states that start from different initial conditions. Moreover, this decay rate is entirely independent of the specific input signal and holds regardless of whether any input is applied at all.

Thus, with this uniform contraction property for $\rho < 1$, an input-driven system will converge within the same characteristic time to a unique time-dependent state determined solely by the input, as the effect of the initial condition is exponentially suppressed. In the absence of input, the system will instead relax to the fixed point $\mathbf{x}(t \rightarrow \infty) = \mathbf{0}$.

For $\rho \rightarrow 1$, the decay time $1/\ln(\rho)$ diverges. However, even for $\rho > 1$, reservoirs can exhibit the Echo State Property, since the activation function prevents the states from diverging to infinity even when the linear system would diverge. Therefore $1/\ln(\rho)$ characterizes the typical decay time for every individual transient of a reservoir with spectral radius $\rho < 1$, where the value of the bound depends on the required precision and on the initial length of the difference $D(0) = \|\mathbf{x}_1(0) - \mathbf{x}_2(0)\|$. If the elements of the initial state are drawn from a uniform distribution on the interval $[-1, 1]$, together with the required precision of 10^{-7} , the following estimate is obtained:

$$T \approx \frac{\ln(10^{-7}/\sqrt{2N/3})}{\ln \rho}.\tag{2.4}$$

This provides an estimate of the typical transient time in the full driven reservoir system. The derivation is given in Appendix A.1.

The exponential decay of the distance $D(t) = \|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|$ between two trajectories driven by the same input is governed by the largest *conditional Lyapunov exponent* (CLE) of the response subsystem. For the linearized update rule $\mathbf{x}(t) = \mathbf{W}_{\text{eff}} \mathbf{x}(t - \Delta t)$, the Jacobian along the driven orbit is the constant matrix $\mathbf{W}_{\text{eff}}$, and thus the conditional Lyapunov exponent is simply

$$\lambda_{\text{cond}} = \ln \rho. \quad (2.5)$$

Therefore, the decay rate entering the denominator of the analytical bound in Eq. (2.4) is precisely $|\lambda_{\text{cond}}| = |\ln \rho|$: for $\rho < 1$, the transient time is governed entirely by the conditional Lyapunov exponent of the linearized dynamics.

For $\rho \gtrsim 1$, the linearized dynamics is unstable, but the nonlinear activation function modifies the effective Jacobian. Saturation can render $\lambda_{\text{cond}} < 0$ even when the linear part would diverge, explaining why many reservoirs with $\rho > 1$ still synchronize under driving. The heavy-tailed distributions of transient times observed in Sec. 2.4 arise from realizations for which λ_{cond} lies close to zero, implying arbitrarily slow convergence rates. This mechanism is consistent with prior observations that near-zero conditional Lyapunov exponents limit attractor reconstruction and reduce closed-loop prediction horizons in reservoir computing [33, 84].

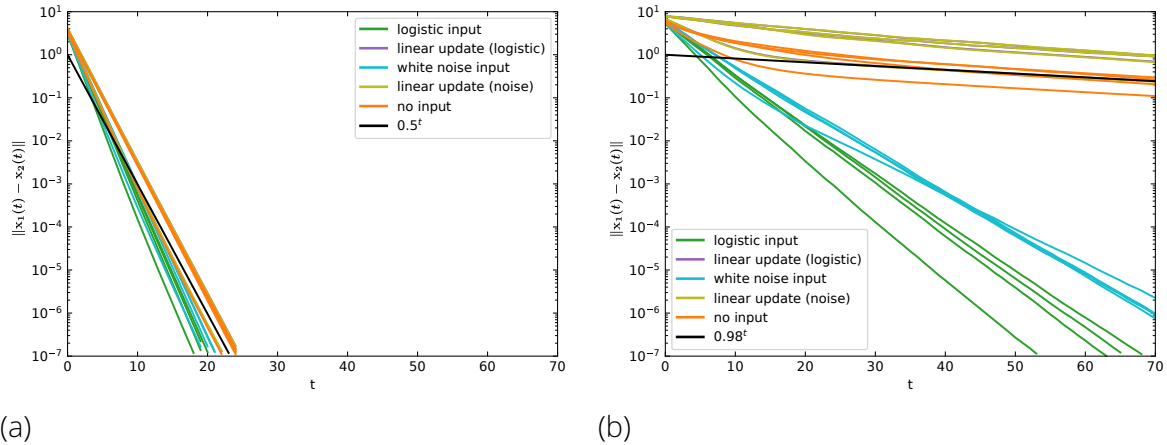


Figure 2.2: The convergence of pairs of trajectories of internal states $\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|$ where one starts from random initial conditions, the other from the zero-state, for four randomly picked reservoirs each. While under the purely linear evolution Eq. (2.3) all distances decay with ρ^t , a faster decay is observed for the non-linear system. For $\rho = 0.5$ (a), the two types of decay are very similar. For $\rho = 0.98$ (b), the damping due to the activation function becomes much more relevant, and the non-linear system's decay depends more strongly on the individual reservoir.

Figure 2.2 shows, for four randomly constructed reservoirs at two different spectral radii ($\rho = 0.5$ and $\rho = 0.98$), the convergence of the full non-linear system under different inputs compared to the convergence of the linear part, Eq. (2.3). For the linear system, an exponential decay is observed with rate $\ln \rho$ for all networks with identical ρ at large times. In contrast, non-linear systems with a sigmoidal activation function exhibit faster decay. For small ρ , different realizations of the reservoirs behave very similarly, while for ρ closer to unity, their convergence can have quite different speeds.

Although for the linear problem Eq. (2.3) the transient time is independent of the type of input signal, the transient time of non-linear reservoirs depends on it. Three different driving

signals are compared: a deterministic chaotic trajectory of the Ulam map, a sequence of independent random numbers (white noise), and the constant signal $u_t = 0$. While for spectral radii much smaller than unity the convergence under all drivings appears with a rate close to $\ln \rho$, for $\rho \rightarrow 1$ the decay under chaotic and white noise driving is much faster than without driving or the decay of the linear dynamics.

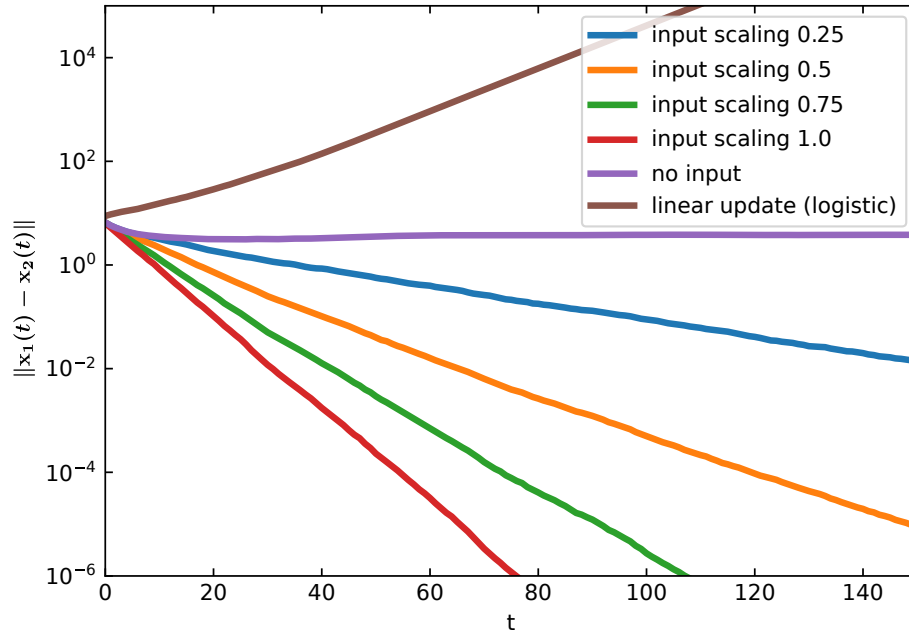


Figure 2.3: The convergence of pairs of trajectories of internal states $\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|$ for a single reservoir with spectral radius $\rho = 1.1$. While under the purely linear dynamics the distances grow asymptotically like ρ^t , the distances in the non-linear system without input remain constant, and with driving (here: logistic equation), this system exhibits the ESP and the internal states converge faster as the input scaling $a \rightarrow 1$.

For $\rho > 1$ the contrast between linear and nonlinear dynamics is even more striking (Fig. 2.3). Under the linearized update, distances grow asymptotically as ρ^t . In the full nonlinear system without driving, they saturate at a finite value instead of diverging, owing to the boundedness of $\tanh$. When the reservoir is driven by a chaotic or stochastic signal, distances decay; the rate is comparable for different input types applied to the same reservoir, but varies considerably across reservoir realizations. Increasing the input scaling a accelerates this convergence, as stronger driving pushes both trajectories deeper into the saturation regime of the activation function, where nonlinear contraction is strongest [13].

The improvement of convergence in the full non-linear update by driving the system with a non-zero input sequence is related to the saturation effect of the activation function. Two trajectories of internal states can converge because, together with the driver, they are both pushed into the same saturation region of the sigmoid function, so that the updated state vectors are closer together than their pre-images. Note, however, that the performance of the reservoir's forecast ability is not being tested here. Despite having the ESP, systems with $\rho > 1$ might not perform well in forecasts. The hypothesis about the effect of non-zero inputs and saturation is verified by a sequence of tests where the amplitude of the inputs is shrunk to intervals $[-a, a]$ with $a < 1$. Indeed, as shown in Fig. 2.3, the smaller the amplitude of the driving signal, the slower the convergence. For Fig. 2.3, the deterministic logistic equation was used as a driver, but qualitatively equivalent results were found when using white noise. These results are coherent with the findings from Ref. [76], where the ESP index value

decreased as the input scaling a was increased, even in the presence of spectral radii larger than unity, evidencing regions of parameters where the ESP was possibly satisfied.

2.4 Transient Time Distributions

While Ref. [76] studied how many randomly constructed reservoirs do *not* lead to convergence in finite time as a function of hyperparameters, the focus here lies in the distributions of times until convergence. Knowledge of this time is relevant for choosing the length of the initial part of the training phase before the training of the output weights can begin.

As mentioned earlier, even for fixed hyperparameters, the transient time distributions may vary significantly from one reservoir to another. This variability, determined by the actual choice of the hyperparameters, is evident in Fig. 2.4, where transient times for reservoirs driven by the same signal can differ by more than an order of magnitude. Since histograms provide limited resolution in the tails, especially for such rare long transients, the complementary cumulative distribution function (CCDF) of the mean transient times is computed next. This representation allows resolving the tail behavior over several decades and quantifying the probability of encountering reservoirs with exceptionally long synchronization times.

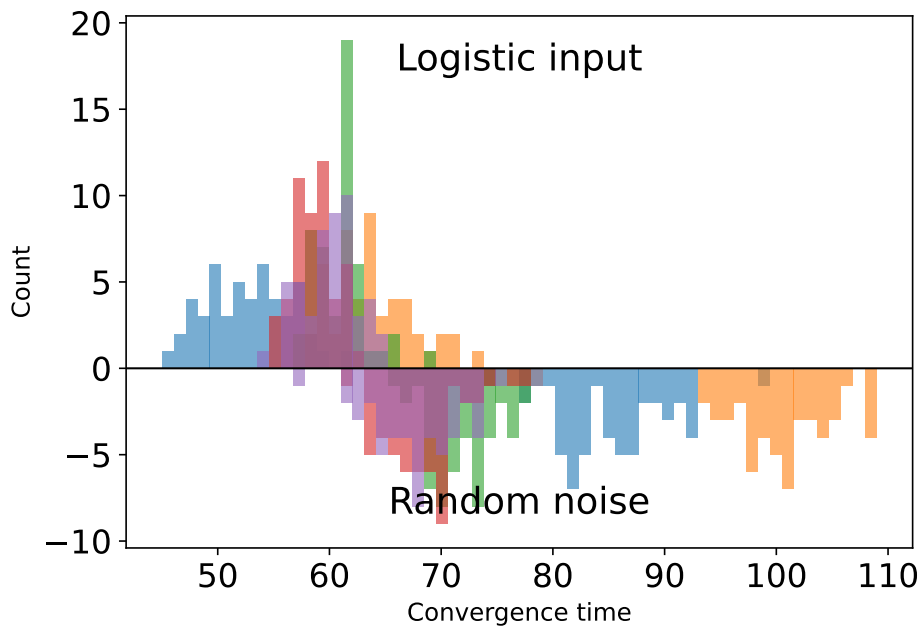


Figure 2.4: A histogram of transient times for four different reservoirs (each color represents one reservoir) with identical hyperparameters but different random weight initializations. For the upper panel, the driving signal is the logistic map; for the lower panel, the driving signal is white noise.

A single mean transient time is calculated for every given reservoir by averaging over 100 random initial conditions for the state $\mathbf{x}_1(0)$. If at least one of these does not converge within 10000 update steps, the reservoir is excluded from the statistics. This procedure is repeated for 10000 randomly constructed reservoirs per hyperparameter setting.

Because the probability density function $p(t)$ of such a distribution suffers from insufficient statistics in the tails, we instead report the complementary cumulative distribution function $F(t) = 1 - \int_0^t p(t') dt'$, which gives the probability of exceeding a given transient time t . In practice, the mean transient times are rank-ordered from largest to smallest, and rank is plotted against magnitude (Figs. 2.5–2.9).

In addition to these distributions, the expected transient time of the linear update Eq. (2.3) is indicated, which was shown to be proportional to $1/\ln \rho$, and which exists only if $\rho < 1$.

Table 2.1: The mean transient time $\bar{T}$ (average over 10000 reservoirs) in dependence on spectral radius ρ . Input data is the logistic map, $N = 100$, $n_{\text{links}} = 200$.

ρ	% convergence	$\bar{T}$	linear estimate
0.5	100.0	21.02	27.58
0.75	100.0	34.96	66.44
0.9	100.0	49.58	181.41
0.99	100.0	64.90	1901.81
1.1	99.85	90.00	N/A
1.5	93.65	233.95	N/A
2.5	68.41	979.80	N/A

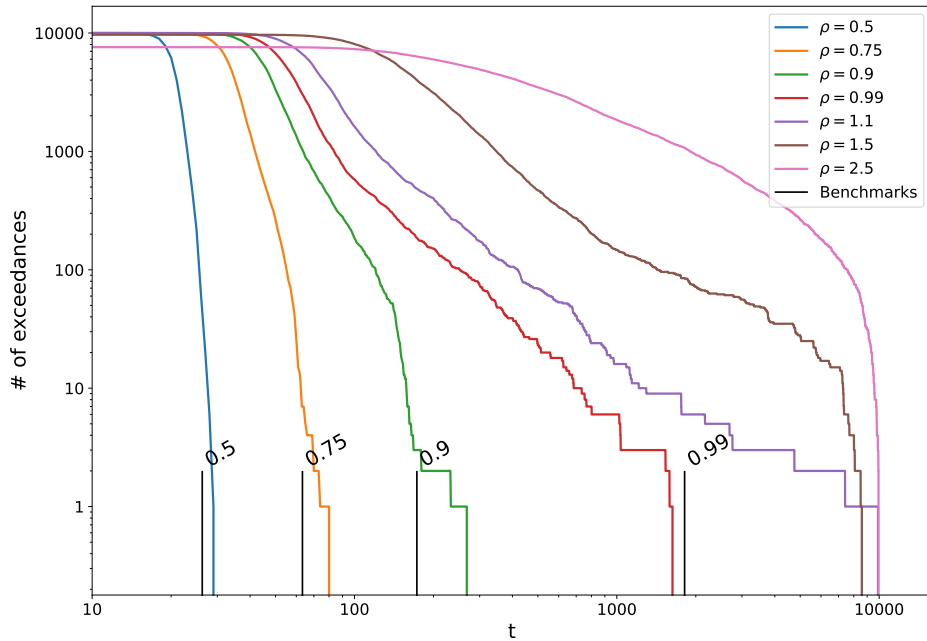


Figure 2.5: The number of exceedances of mean transient times obtained for 10000 reservoirs, for different spectral radii ρ ($N = 100$ nodes, $n_{\text{links}} = 200$, $\alpha = 0.1$). The black vertical markers indicate the benchmarks derived for the linear update Eq. (2.3) for $\rho < 1$. Driving signal: logistic equation. See also Table 2.1.

The results for varying the spectral radius but keeping all other hyperparameters fixed are shown in Fig. 2.5. When ρ comes close to unity, the decay time of the linear system diverges, and consequently, the non-linear updates can require long convergence times. As a consequence, the shown distributions exhibit long tails for $\rho \geq 1$. Focus on $\rho = 0.99$ (red): while 90% of all random realizations of the reservoir converge within 100 time steps, there are “unlucky” choices which take more than a factor of 10 longer. This sensitivity to reservoir realization becomes even more pronounced for $\rho > 1$, where the probability of encountering impractically long transients grows rapidly, see Table 2.1. In practice, this means that for

spectral radii near or above unity, pre-screening reservoirs for fast convergence can be crucial to avoid wasting large portions of the training sequence on synchronization alone.

2.5 Parameter Dependency

The distributions presented in Fig. 2.5 have finite mean values, making them natural objects for systematic investigation under parameter variations. The upper bound derived for the linear system when $\rho < 1$ exhibits a simple dependence: it scales with ρ and the system size N through the initial separation, but remains insensitive to other architectural details such as link density or topology. No comparable theoretical prediction exists for $\rho > 1$, where the interplay between spectral instability and nonlinear saturation becomes nontrivial. To explore this regime systematically, $\rho = 1.5$ is selected as a representative value above unity, with $\rho = 0.9$ serving as a control to verify the expected independence from architectural parameters in the contracting regime.

The analysis concentrates on the two most significant architectural parameters: the number of nodes N and the link density (or equivalently, the total number of connections n_{links}). Throughout these experiments, the deterministic chaotic logistic map serves as the driving signal.

For $\rho < 1$ (exemplified by $\rho = 0.9$), the spectral radius alone governs the maximal growth rate of perturbations between trajectories. In this regime, convergence follows the purely linear estimate in Eq. (2.4), and variations in link density affect only secondary details of the coupling structure. Modifying n_{links} alters the mean transient time by at most a factor of order unity and never violates the bound: the system contracts uniformly regardless of connectivity, so additional links merely redistribute the input more evenly and accelerate synchronization slightly.

Table 2.2: Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the number of links n_{links} , for reservoirs of size $N = 100$ with spectral radius $\rho = 1.5$. The driving signal is the logistic map. The fraction of reservoirs that converged within 10^4 steps is also reported.

n_{links}	% convergence	$\bar{T}$
50	35.7	111.97
100	78.16	141.96
150	90.92	183.34
200	94.6	227.90
250	96.01	267.87
500	98.76	486.80

For $\rho = 1.5$, the picture becomes more intricate (Fig. 2.6). Without external input, the linearized dynamics is unstable, and convergence relies entirely on nonlinear saturation and input forcing to suppress divergence. Table 2.2 reveals that only about one-third of reservoirs with $n_{\text{links}} = 50$ achieve convergence, whereas nearly all reservoirs with $n_{\text{links}} = 500$ succeed. Sparse connectivity produces many weakly coupled or effectively decoupled subgraphs. In such architectures, certain modes can persist in a near-autonomous regime, delaying the propagation of input-driven stabilization and leading to either very long transient times or outright failure to converge. Increasing the link density improves global coupling and

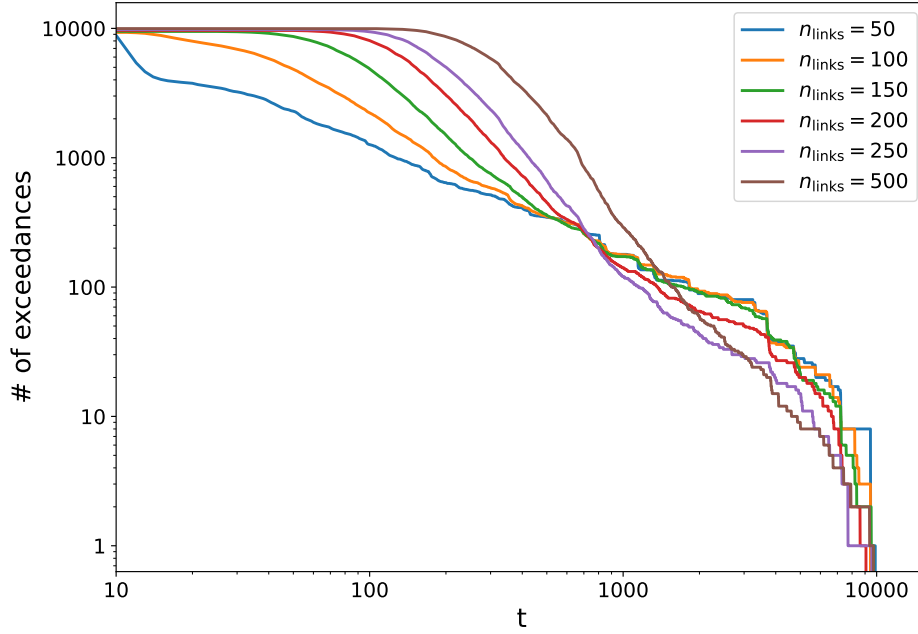


Figure 2.6: The number of exceedances of mean transient times obtained for 10000 reservoirs for different link numbers n_{links} , $N = 100$, $\rho = 1.5$, driving signal is logistic map.

raises the fraction of modes directly influenced by the input, thereby increasing convergence probability. However, beyond approximately 1% density, additional connections create feedback loops with larger gains, pushing modes closer to marginal stability and reducing the effectiveness of nonlinear saturation. This explains why the mean transient time continues to grow for very dense reservoirs even as the fraction of converging reservoirs rises.

Table 2.3: Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the number of nodes N , with link density fixed at 1% ($n_{\text{links}} = 0.01N^2$). Spectral radius is $\rho = 1.5$ and the driving signal is the logistic map. The fraction of reservoirs that converged within 10^4 steps is reported, along with the mean transient time $\bar{T}$ of successful cases.

N	n_{links}	% convergence	$\bar{T}$
50	25	35.42	122.59
100	100	77.89	145.83
150	225	91.35	179.34
200	400	94.61	243.27
250	625	98.57	331.21

When varying N at fixed link density of 1% ($n_{\text{links}} = 0.01N^2$), the average degree $\bar{k} = n_{\text{links}}/N$ scales linearly with system size (Fig. 2.7). This enhances input propagation but simultaneously increases the number of feedback loops with large local gains. For small N , even at constant density, the absolute number of links remains low, increasing the likelihood of disconnected components or weakly coupled subgraphs—this accounts for the lower convergence percentages observed in Table 2.3. As N grows, the probability of drawing at least one slow mode increases, making the right tail of the exceedance distribution more prominent. Nevertheless, because all configurations share the same spectral radius and rely on the same nonlinear damping mechanism, the slowest modes converge on similar timescales, causing the curves to collapse near the same cutoff time.

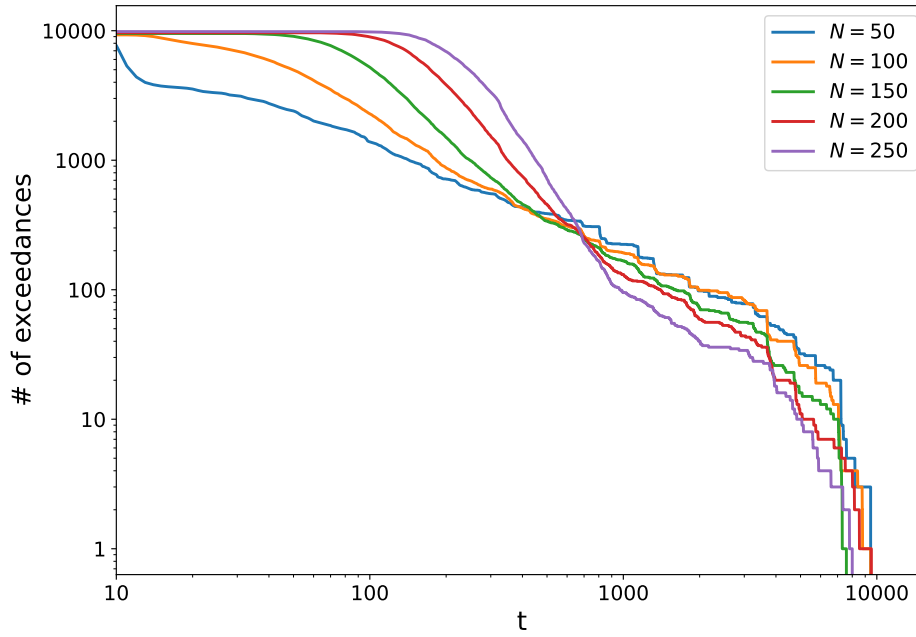


Figure 2.7: The number of exceedances of mean transient times obtained for 10000 reservoirs with different numbers of nodes N but fixed link density of 1%, spectral radius $\rho = 1.5$, driving signal logistic map.

Table 2.4: Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the number of nodes N , with the total number of links fixed at $n_{\text{links}} = 200$. Spectral radius is $\rho = 1.5$ and the driving signal is the logistic map. Reported are the fraction of reservoirs that converged within 10^4 steps and the mean transient time $\bar{T}$ of successful cases.

N	n_{links}	% convergence	$\bar{T}$
50	200	99.7	471.241
100	200	94.8	266.352
150	200	90.1	177.556
200	200	78.8	97.460
250	200	63.7	86.557

Holding $n_{\text{links}} = 200$ constant while increasing N produces the opposite trend: the average degree decreases, reducing feedback loop density and lowering the probability of marginally stable modes (Table 2.4, Fig. 2.8). This shifts the population toward faster convergence and decreases the mean transient time. However, the convergence percentage declines for large N , as sparser graphs become more likely to contain weakly connected regions. The results are qualitatively similar to those in Fig. 2.6 and Table 2.2 when examined in terms of average degree, convergence percentage, and mean transient time.

Maintaining $N = 100$, $n_{\text{links}} = 100$ (1% density), and $\rho(W_{\text{eff}}) = 1.5$ fixed while varying the leak rate α reveals a systematic trend: increasing α shortens transients and raises the fraction of convergent reservoirs (Table 2.5, Fig. 2.9). Mechanistically, two effects operate in tandem. First, the $(1 - \alpha)\mathbf{x}(t - \Delta t)$ term in Eq. (2.2) determines how much of the previous state persists at each step: small α yields per-step updates of order α , thereby slowing the forgetting process, while larger α refreshes the state more aggressively. Second, $W_{\text{in,eff}} = \alpha W_{\text{in}}$ increases linearly with α , so the same input drives units deeper into the $\tanh$ saturation regime where nonlinear contraction is stronger. Even when W is rescaled to maintain constant $\rho(W_{\text{eff}})$ across α , these two factors combine to produce heavier right tails in the exceedance curves

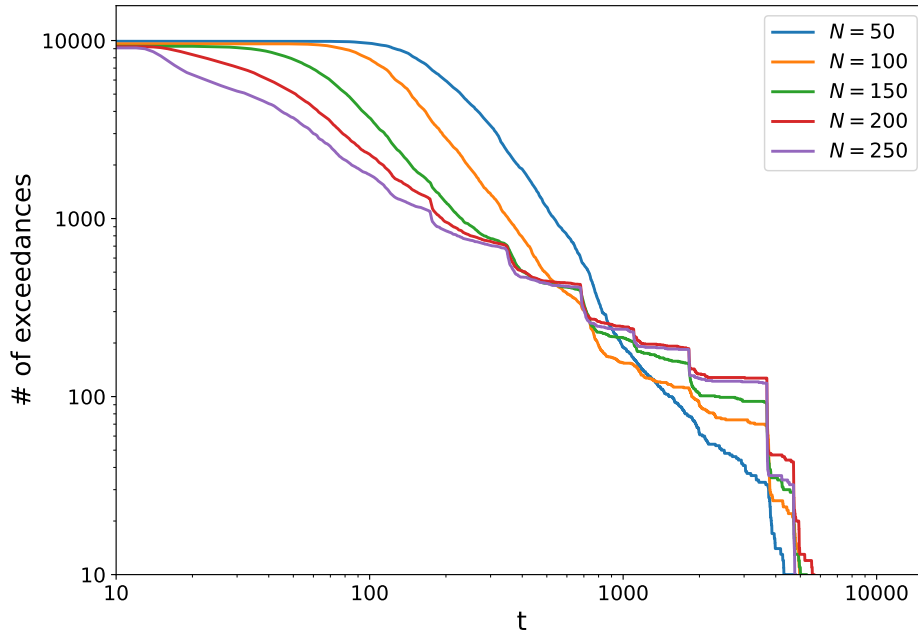


Figure 2.8: The number of exceedances of mean transient times obtained for 10000 reservoirs with different numbers of nodes N but fixed number of links $n_{\text{links}} = 200$, spectral radius $\rho = 1.5$, driving signal logistic map.

Table 2.5: Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the leak rate α , for reservoirs of size $N = 100$ with link density 1% ($n_{\text{links}} = 100$) and spectral radius $\rho = 1.5$. The driving signal is the logistic map. Reported are the fraction of reservoirs that converged within 10^4 steps and the mean transient time $\bar{T}$ of successful cases.

α	% convergence	$\bar{T}$
0.00	0.0	—
0.25	34.12	623.14
0.50	59.21	462.23
0.75	69.38	391.56
0.90	77.62	312.86
0.95	82.18	282.46
0.99	88.19	241.34
1.00	93.63	232.86

for small α and lighter tails for large α . At $\alpha = 0$ the dynamics freeze and no convergence under driving is observed; as $\alpha \rightarrow 1$ the system reacts on the fastest timescale permitted by the discrete update, and synchronization typically becomes most rapid.

The trends in Figs. 2.6–2.9 reflect the joint influence of link density, node count, and spectral radius on the effective contraction rate and stability of the reservoir dynamics. For $\rho < 1$, the spectral radius dominates and connectivity plays a minor role. For $\rho > 1$, topology becomes crucial: too few links lead to weak coupling and poor convergence; too many links increase feedback gain and slow contraction.

An additional structural explanation for non-convergent reservoirs emerges from interpreting the recurrent weight matrix as a directed graph. Decomposition into strongly connected components reveals distinct classes of nodes with different long-term behaviors, as analyzed in detail in Sections 2.6 and 2.7. This graph-theoretic perspective provides a mechanistic understanding of which reservoir realizations fail to synchronize and why.

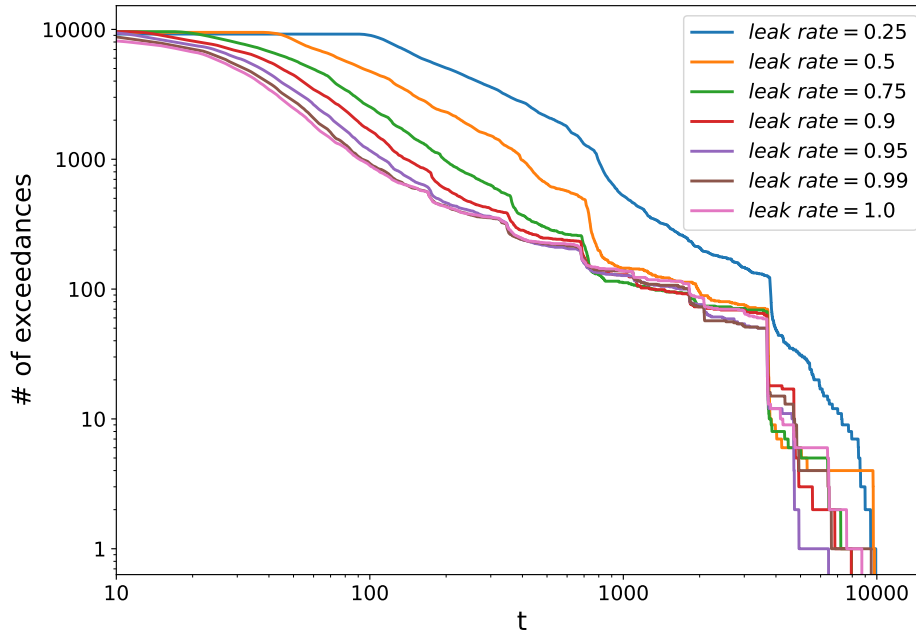


Figure 2.9: Number of exceedances of mean transient times obtained for 10^4 reservoirs for different leak rates α ($N = 100$, $n_{\text{links}} = 100$ (1%), $\rho = 1.5$). Driving signal: logistic map.

2.6 Zero-Input Dynamics and Graph Structure

A legitimate input for studying reservoir synchronization is a constant signal $u_t = u_0$ for all t . The special case $u_0 = 0$ is equivalent to allowing the reservoir's internal state to evolve autonomously without external driving. Given the possibility that certain reservoirs fail to relax to the trivial fixed point, it becomes natural to investigate whether reservoirs can exhibit non-trivial dynamical behavior in the absence of input. According to the Echo State Property, the internal state should relax to the fixed point $\mathbf{x}(t) = \mathbf{0}$ in this scenario, which is indeed observed whenever $\rho < 1$. As reported in Tables 2.1–2.4, for $\rho > 1$ there always exists a certain percentage of reservoirs where the distances $\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|$ fail to converge to zero within the probed 10000 iteration steps. Numerical experiments reveal that reservoirs which do not exhibit convergence when driven by the logistic equation also fail to converge to a fixed point without driving. The internal dynamics of the state vector $\mathbf{x}(t)$ in reservoirs that violate the ESP are therefore examined here, again using $\rho = 1.5$.

One can readily verify from the update rule introduced in Chapter 1 that the state vector $\mathbf{0}$ constitutes a fixed point of the dynamics. However, for $\rho > 0$ this fixed point may be unstable. The state vector emerging from a random initial condition is therefore studied after it has failed to converge to the zero-vector following 10^5 iterations without inputs. A typical outcome under zero input is shown in Fig. 2.10 for $N = 50$ and $n_{\text{links}} = 50$ over $t \in [10000, 10100]$. Some nodes i satisfy $x^i(t) \equiv 0 \forall t$, while others display $x^i(t)$ fluctuating in a quasi-periodic manner over time. This indicates that the reservoir possesses at least one attractive limit cycle. Different initial conditions of the internal state were verified to settle onto the same limit cycle, suggesting that for this particular example, the limit cycle attractor is unique. The coexistence of silent and oscillating units bears resemblance to chimera-like patterns in coupled oscillator networks [85, 86].

The hypothesis is that the coexistence of silent and oscillating nodes can be explained purely from the graph structure of the reservoir. The weight matrix W defines a directed graph $G(W)$, with an edge $i \rightarrow j$ present whenever $w_{ij} \neq 0$. This graph naturally decomposes into strongly connected components (SCCs). Every node belongs to one of three categories: (i) it lies on a cycle, (ii) it is reachable from a cycle, or (iii) it is not influenced by any cycle at all. This observation leads to the following classification:

- **Core nodes** are those that lie on at least one directed cycle of $G(W)$. Their state receives recurrent feedback after a finite number of steps, so their activity never dies out. If the effective loop gain within the cycle is sufficiently large, the nonlinearity of $\tanh$ prevents divergence and stabilizes non-trivial attractors (periodic or quasi-periodic).
- **Driven nodes** are not themselves part of a cycle, but they are downstream from one. Their activity inherits the structure of the cycle, so they oscillate as long as the cycle persists.
- **Quiescent nodes** are neither on a cycle nor reachable from one. They form part of the acyclic periphery of the graph and therefore receive only transient signals. After finitely many steps, their input vanishes, forcing them to collapse to the trivial fixed point $s_i(t) \equiv 0$.

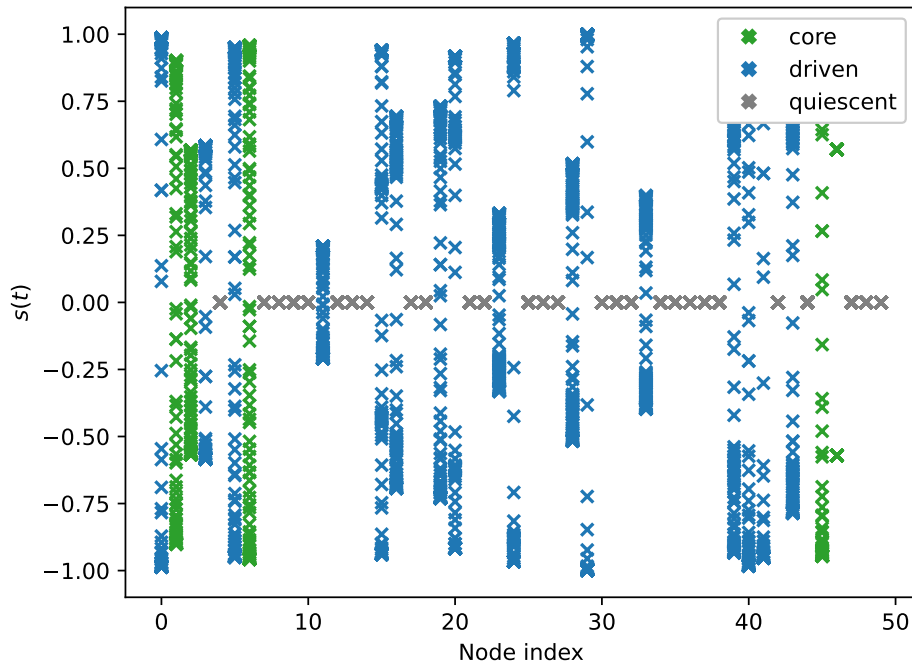


Figure 2.10: Classification of nodes according to their graph-theoretic role. Core nodes (green) belong to cycles and sustain autonomous activity; driven nodes (blue) inherit oscillations from cores; quiescent nodes (gray) collapse to zero. Parameters: $\rho = 1.5$, $N = 50$, $n_{\text{links}} = 50$.

This classification matches the numerical results precisely. A cycle-based classifier was implemented that computes the strongly connected components of $G(W)$, identifies cycles, and colors nodes accordingly. Figure 2.10 demonstrates an exact correspondence: core (green) and driven (blue) nodes oscillate quasi-periodically, while quiescent (gray) nodes converge to zero. The same one-to-one correspondence appears in all other trials exhibiting mixed silent-oscillatory patterns.

The structural interpretation developed above concerns exclusively the autonomous case $u_t = 0$, where the reservoir evolves without external driving. In this setting, the directed graph defined by W determines which nodes can sustain activity indefinitely. Nodes belonging to directed cycles (core nodes) receive recurrent feedback and therefore support persistent oscillations; nodes downstream from such cycles inherit this activity; and nodes that are not reachable from any cycle collapse to the fixed point of the activation function. This explains the coexistence of oscillatory and quiescent units observed in the numerical experiments.

This classification pertains specifically to zero-input dynamics and does not constitute a performance guideline for driven operation. When the reservoir is continuously driven by a time-varying signal, all nodes remain active regardless of whether they belong to a cycle; indeed, acyclic or tree-like topologies can perform very well on certain tasks [36, 77]. The role of cycles is restricted to explaining why some realizations fail to converge when $u(t) = 0$.

2.7 Topology and Forecast Performance

The preceding analysis establishes that the graph-theoretic classification of reservoir nodes—core, driven, and quiescent—determines their long-term dynamical behavior under zero input. The question now arises whether this structural perspective translates into measurable differences in closed-loop forecasting performance on a standard benchmark task, namely short-term prediction of the Lorenz system.

To systematically vary the proportions of core, driven, and quiescent nodes while maintaining a controlled graph structure, the *dendrocycle* topology is introduced (Fig. 2.11), implemented in the open-source library **ResDAG** [43]. This architecture comprises three components:

1. **Core cycle.** A directed ring of $C = \lfloor c \cdot N \rfloor$ nodes (indices $0, \dots, C - 1$), forming the unique strongly connected component and the only directed cycle in the graph.
2. **Dendritic chains.** $D = \lfloor d \cdot N \rfloor$ nodes organized into directed acyclic chains emanating outward from evenly spaced anchor points on the core ring. These nodes receive input from the core but provide no feedback to it.
3. **Quiescent subgraph.** The remaining $A = N - C - D$ nodes form a directed acyclic subgraph with random forward connections. By construction, these nodes are neither part of any cycle nor downstream of one.

The fractions c and d satisfy $c + d \leq 1$, with the quiescent fraction given by $1 - c - d$. This parameterization allows interpolation between reservoirs dominated by recurrent feedback ($c \rightarrow 1$), feedforward memory ($d \rightarrow 1 - c$), or transient-only dynamics ($c, d \rightarrow 0$).

Forecast performance is evaluated on the Lorenz 63 system [38]

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho_L - z) - y, \quad \dot{z} = xy - \beta z, \quad (2.6)$$

with standard parameters $\sigma = 10$, $\rho_L = 28$, $\beta = 8/3$, integrated using a fourth-order Runge-Kutta scheme with time step $\Delta t = 0.02$. The positive Lyapunov exponent $\lambda \approx 0.91$ sets the characteristic timescale for predictability.

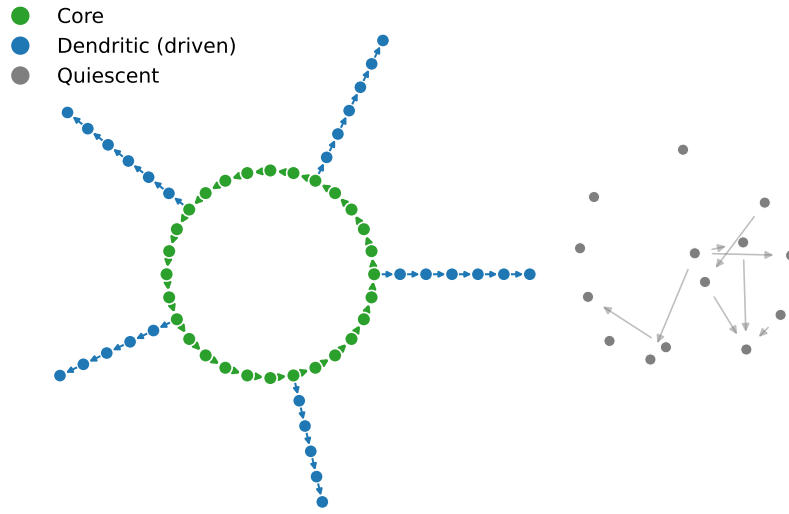


Figure 2.11: Schematic of the dendrocycle topology. Core nodes (green) form a directed cycle providing recurrent feedback. Dendritic nodes (blue) are arranged in chains emanating from the core, receiving but not returning activity. Quiescent nodes (gray) form an acyclic subgraph disconnected from any cycle. Arrows indicate edge directions.

For each parameter pair (c, d) on a 20×20 grid (restricted to $c + d \leq 1$), 20 independent reservoir random realizations are constructed with $N = 400$ nodes, spectral radius $\rho = 0.9$, and leak rate $\alpha = 1$. Each reservoir is trained on 10000 time steps following a 1000-step washout, using ridge regression to fit the readout weights. Closed-loop forecasts are generated for 1000 steps and compared against the ground truth across 20 independent trajectories, each using a different random initial reservoir state.

Performance is quantified by the *valid prediction time* T_{valid} , defined as the time at which the normalized forecast error first exceeds a threshold of 20%. T_{valid} is reported in units of Lyapunov times λt , averaged over all reservoir realizations and trajectories for each (c, d) pair.

Figure 2.12 displays the mean valid prediction time as a function of core fraction c and dendritic fraction d . The results reveal a clear trend: forecast performance improves monotonically as the combined fraction $c + d$ approaches unity, corresponding to reservoirs with fewer quiescent nodes.

The diagonal ridge along $c + d \approx 1$ confirms that both core and dendritic nodes contribute positively to forecast skill, while quiescent nodes—which collapse to zero under autonomous dynamics—provide negligible computational utility. Notably, the improvement is not confined to cycle-rich reservoirs alone: dendritic chains, despite being acyclic, enhance performance by extending the effective memory through feedforward pathways downstream of the core.

Reservoirs in the lower-left region ($c, d \rightarrow 0$) exhibit the poorest performance. In this regime, most nodes belong to the quiescent class and receive only transient input before decaying,

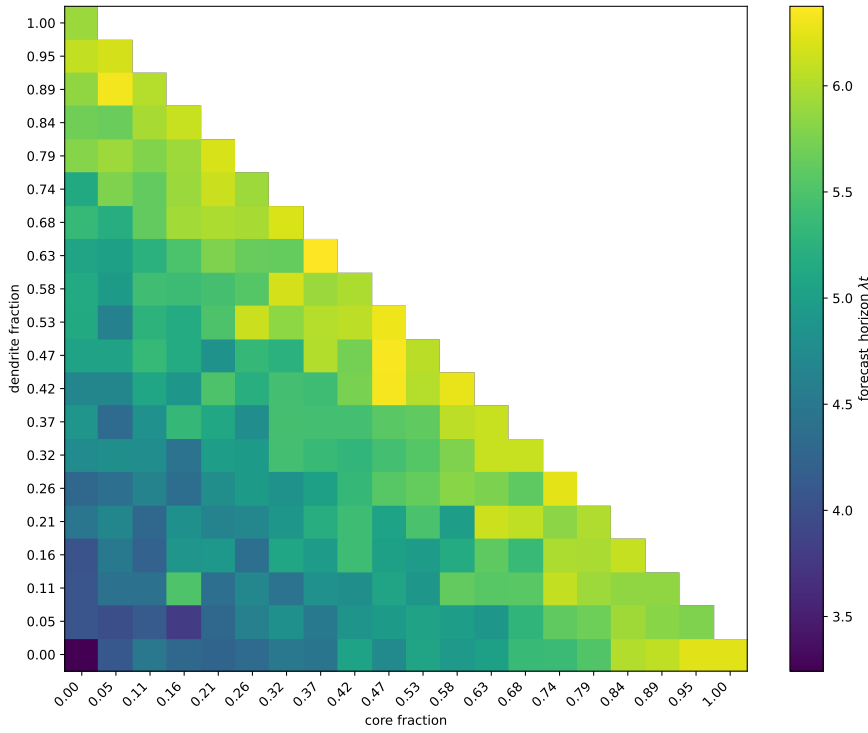


Figure 2.12: Mean valid prediction time (in Lyapunov times λt) for Lorenz forecasting as a function of core fraction c and dendritic fraction d . Performance improves along the diagonal $c + d \rightarrow 1$, where quiescent nodes are minimized. Each cell averages over 20 reservoir realizations and 20 independent trajectories.

consistent with the theoretical prediction that such nodes do not participate in sustained computation.

These results complement the transient-time analysis of Sec. 2.6, but reveal an important distinction. Under zero input with $\rho > 1$, it is the cycle-containing subgraph that fails to synchronize—core and driven nodes sustain autonomous oscillations while quiescent nodes decay to zero (Fig. 2.10). Yet under driven operation, quiescent nodes contribute negligibly to forecast skill precisely because they lack sustained recurrent activity. The dendrocycle experiments thus suggest a design trade-off: cycle-rich or dendritic architectures maximize the computational substrate and improve forecast performance, but when $\rho > 1$ these same cycles can sustain self-oscillations that prolong or prevent synchronization. Practitioners should therefore favor topologies with large active fractions ($c + d \rightarrow 1$) for performance, while monitoring transient times or controlling the spectral radius to mitigate the long transients that cycles can induce.

This empirical validation supports the practical design principles articulated in Refs. [36, 77]: cycle-rich topologies, or those with extensive dendritic structure, make more efficient use of reservoir capacity. Pre-screening strategies based on transient monitoring can therefore serve a dual purpose, filtering out both slow-synchronizing and poorly performing realizations before training. The dendrocycle topology and its performance across diverse tasks will be explored more comprehensively in Chapter 5, where it serves as a controlled substrate for hyperparameter optimization experiments using the **ResDAG** library.

2.8 Summary and Design Principles

Synchronization transients in reservoir computing have been quantified at three complementary levels: (i) an analytical upper bound for $\rho < 1$ derived from the linearized dynamics, (ii) ensemble statistics revealing heavy-tailed transient-time distributions when ρ approaches or exceeds unity, and (iii) a structural explanation of non-convergence at zero input in terms of the reservoir graph. Taken together, these results transform the informal notion of “washout” into a measurable system property with clear design levers.

From a dynamical perspective, three intertwined factors control transient behavior. First, the spectral contraction rate yields the bound in Eq. (2.4) for $\rho < 1$, predicting the dependence on tolerance and initial separation. Second, nonlinear saturation under driving shortens transients even when the linear part would diverge, accounting for the strong influence of input amplitude. Third, topology matters: by interpreting W as a directed graph and decomposing it into strongly connected components, three structural classes with distinct long-term behaviors at zero input were identified. Nodes on cycles (core) sustain autonomous activity; nodes downstream of cycles (driven) inherit oscillations; and nodes with no cycle upstream (quiescent) collapse to zero in finitely many steps. This cycle-based classification reproduces the time-series evidence node by node, as demonstrated in Fig. 2.10.

From a statistical perspective, ensembles with fixed hyperparameters show narrow variability within each reservoir but broad variability across reservoirs once $\rho \gtrsim 1$. The exceedance curves demonstrate that extremely slow realizations are not anomalous outliers but intrinsic to the parameter regime. Link density and system size both modulate the tails: when too sparse, the network fragments into weakly coupled components; when too dense, high-gain feedback loops proliferate, slowing contraction despite higher convergence rates.

From a practical perspective, these insights suggest concrete design principles. Reservoirs should be pre-screened by monitoring $D(t)$ or a related ESP index, discarding slow realizations rather than sacrificing data to washout. Topologies rich in cycles—or with many nodes downstream of cycles—make better use of available neurons, maximizing the active computational substrate. Input scaling and leakage should be tuned to exploit nonlinear saturation without overwhelming the reservoir. At the same time, operating points that combine large ρ with very low or very high degrees should be avoided, as both extremes promote long transients.

The cycle-based perspective provides a structural lens for understanding and engineering reservoirs. It explains why some realizations fail the ESP at zero input yet synchronize once driven, and it points toward topology-aware construction as a route to more reliable and data-efficient reservoir computing. As demonstrated in Sec. 2.7, controlling the abundance of cycles and downstream nodes directly improves forecast performance; together with pre-screening strategies, this reduces training waste and improves robustness.

The results of this chapter inform the hyperparameter optimization framework developed in Chapter 5, where pre-screening based on transient times will be incorporated as a standard step before training. They also connect to the geometric discretization methods introduced in Chapter 3: the sagitta-based approach for selecting temporal step sizes implicitly assumes the reservoir has synchronized with the input signal on the timescale of interest, making the washout quantification presented here a necessary prerequisite for fair performance

comparisons across dynamical systems. Future work should aim at sharper theoretical conditions for $\rho > 1$ based on local loop gains, and at connecting cycle architecture directly to task-specific performance metrics beyond chaotic forecasting.

3 Geometric Time Discretization

The previous chapters established that Echo State Networks, like all data-driven models for dynamical systems, operate on discretely sampled trajectories. The transient analysis of Chapter 2 revealed how reservoir dynamics depend sensitively on the interplay between network parameters and the temporal structure of the input. Yet one fundamental preprocessing choice has remained implicit throughout: the sampling step Δt at which continuous-time dynamics are rendered discrete. This chapter addresses that gap by developing a principled, geometry-based criterion for selecting Δt directly from trajectory data.

The core observation is that the error introduced by piecewise-linear interpolation—the simplest model one could fit between consecutive samples—is controlled by the local curvature of the trajectory. Rather than estimating curvature through numerical differentiation, which amplifies noise and requires careful tuning, we measure it indirectly via the *sagitta*: the orthogonal deviation of a sample point from the chord connecting its temporal neighbors. This quantity admits a clean asymptotic expansion linking it to the normal component of acceleration, scales quadratically with the time step in the small-step regime, and can be computed robustly from data without differentiation.

From the sagitta we derive a selection rule that identifies the coarsest admissible Δt for which geometric distortion remains below a prescribed tolerance. The rule is idempotent: once the trajectory is resampled at the recommended step, reapplying the same procedure returns the same answer. This fixed-point property stabilizes downstream analyses and ensures that the chosen resolution genuinely reflects the intrinsic timescale of the dynamics rather than artifacts of the selection process. The tolerance parameter ε becomes a single, interpretable knob controlling the geometric fidelity of the discretization, enabling fair comparisons of predictive performance across dynamical systems with disparate timescales.

3.1 The discretization problem in forecasting

Data-driven modeling of dynamical systems begins with a sampled trajectory. Whether the goal is short-term prediction, attractor reconstruction, or learning a surrogate model, the

raw material is a sequence of state vectors $y_0, y_1, \dots, y_{N-1}$ recorded at uniform intervals Δt . For systems governed by ordinary differential equations, this sequence arises from evaluating (or numerically integrating) the flow at discrete times $t_i = i \Delta t$. The choice of Δt is rarely discussed in detail, yet it exerts a profound influence on every subsequent step of the modeling pipeline.

Consider first the regime of very fine sampling. When Δt is much smaller than any characteristic timescale of the dynamics, consecutive samples differ only marginally. The trajectory appears nearly linear at the scale of a single step, and the information content per sample collapses. In this regime, even a trivial predictor—one that simply extrapolates the local tangent—can achieve impressive one-step accuracy, not because it has learned the dynamics but because the dynamics have been rendered trivial by oversampling. Metrics such as mean-squared prediction error become misleading: they reward models for interpolating what is essentially a straight line. Reservoir computers and recurrent networks trained on such data may appear to perform well while capturing little of the underlying nonlinear structure.

The opposite extreme is equally problematic. When Δt exceeds the timescales over which the trajectory curves appreciably, the sampled sequence no longer faithfully represents the continuous flow. Sharp turns are missed, folding structures are aliased, and the geometric skeleton of the attractor is distorted beyond recognition. Models trained on such data fail not because of limited capacity but because the training signal itself has been corrupted by undersampling. Forecasts diverge rapidly, and any comparison with models trained at different resolutions becomes meaningless.

Between these extremes lies a regime where the sampling step resolves the geometry of the trajectory without redundancy. Identifying this regime is nontrivial because different dynamical systems possess different intrinsic timescales. The Lorenz system at standard parameters evolves on timescales of order unity [38], while a turbulent fluid or a physiological rhythm may span milliseconds or hours. Comparing the forecasting performance of a reservoir computer on two such systems requires that both be discretized at comparable geometric fidelity; otherwise, differences in accuracy reflect differences in preprocessing rather than in modeling capacity.

The literature offers few systematic guidelines [2, 4]. Common practice is to choose Δt by intuition, by matching conventions in prior work, or by trial and error guided by validation error. These approaches are difficult to reproduce and impossible to transfer across systems. What is needed is a criterion that adapts automatically to the dynamics encoded in the data, that is interpretable in geometric terms, and that yields the same answer regardless of who applies it or on what computing platform. The remainder of this chapter develops such a criterion, grounded in the observation that the natural scale for Δt is set by the curvature of the trajectory itself.

3.2 Linearization error and curvature

Let $y : [0, T] \rightarrow \mathbb{R}^d$ denote a sufficiently smooth trajectory generated by a continuous-time dynamical system $\dot{y} = f(y)$. We seek to understand how the choice of sampling

step Δt controls the error incurred when the trajectory is approximated by piecewise-linear interpolation between samples. This is the simplest possible model one could fit to consecutive observations, and bounding its error provides a natural criterion for admissible discretization.

Fix a time t and consider the segment of trajectory between $y(t)$ and $y(t + \Delta t)$. Linear interpolation assumes that the system evolves along the secant joining these two points. The interpolant at any intermediate time $t + \tau$, with $0 \leq \tau \leq \Delta t$, is

$$y_{\text{lin}}(t + \tau) = y(t) + \frac{\tau}{\Delta t} [y(t + \Delta t) - y(t)]. \quad (3.1)$$

The true trajectory, expanded in Taylor series around t , reads

$$y(t + \tau) = y(t) + \dot{y}(t)\tau + \frac{1}{2}\ddot{y}(t)\tau^2 + \mathcal{O}(\tau^3). \quad (3.2)$$

Writing $v = \dot{y}(t)$ for the velocity and $a = \ddot{y}(t)$ for the acceleration, the difference between the true position and the linear interpolant at the midpoint $\tau = \Delta t/2$ becomes

$$e(t; \frac{\Delta t}{2}) = y(t + \frac{\Delta t}{2}) - y_{\text{lin}}(t + \frac{\Delta t}{2}) = \frac{1}{8}a(\Delta t)^2 + \mathcal{O}((\Delta t)^3). \quad (3.3)$$

The leading-order error is thus quadratic in Δt and proportional to the acceleration. However, not all components of a contribute equally to the geometric deviation from the interpolating line.

To see this, decompose the acceleration into components parallel and perpendicular to the velocity. Define the orthogonal projector onto the span of v as

$$P_v = \frac{vv^\top}{|v|^2}, \quad (3.4)$$

where $|\cdot|$ denotes the Euclidean norm. The tangential and normal components of the acceleration are then

$$a_{\parallel} = P_v a, \quad a_{\perp} = (I - P_v) a. \quad (3.5)$$

The tangential component $a_{\parallel}$ represents changes in speed along the direction of motion; it stretches or compresses the trajectory but does not bend it. The normal component $a_{\perp}$ represents the rate at which the velocity vector rotates; it is this component alone that causes the trajectory to deviate from a straight line.

The distinction matters because linear interpolation, by construction, lies along the chord connecting $y(t)$ and $y(t + \Delta t)$. Any deviation of the true trajectory in the direction of this chord is partially captured by the interpolant's own displacement. What linear interpolation cannot capture is deviation orthogonal to the chord—precisely the contribution from $a_{\perp}$. A careful analysis (detailed in Appendix B) shows that the magnitude of the interpolation error, measured orthogonally to the chord, satisfies

$$|e_{\perp}(t; \Delta t)| = \frac{1}{2} |a_{\perp}(t)| (\Delta t)^2 + \mathcal{O}((\Delta t)^3). \quad (3.6)$$

The factor of $1/2$ differs from the $1/8$ in (3.3) because here we measure the maximal orthogonal deviation over the arc, which occurs at the midpoint of a symmetric triplet $(t - h, t, t + h)$ with $h = \Delta t/2$. The essential point is that the geometric error of linearization is controlled by $|a_{\perp}|$, the magnitude of the normal acceleration.

The normal acceleration admits a direct interpretation in terms of the curvature $\kappa(t)$ of the trajectory. For a curve parametrized by time, the curvature is defined as the rate of rotation of the unit tangent vector per unit arc length. Standard differential geometry yields the relation [87]

$$|a_{\perp}(t)| = \kappa(t) |v(t)|^2. \quad (3.7)$$

Substituting into (3.6) gives

$$|e_{\perp}(t; \Delta t)| = \frac{1}{2} \kappa(t) |v(t)|^2 (\Delta t)^2 + \mathcal{O}((\Delta t)^3). \quad (3.8)$$

This expression makes explicit that the linearization error grows with both the curvature of the trajectory and the square of the time step. High curvature demands finer sampling; low curvature permits coarser discretization.

To ensure that the interpolation error remains below a prescribed tolerance $\varepsilon > 0$, one imposes

$$\frac{1}{2} \kappa(t) |v(t)|^2 (\Delta t)^2 \leq \varepsilon, \quad (3.9)$$

which, solving for Δt , yields

$$\Delta t \leq \sqrt{\frac{2\varepsilon}{\kappa(t) |v(t)|^2}} = \sqrt{\frac{2\varepsilon}{|a_{\perp}(t)|}}. \quad (3.10)$$

The bound (3.10) varies along the trajectory: regions of high curvature or high speed impose tighter constraints. A global discretization step must respect the most stringent local requirement, suggesting that one aggregate the pointwise bounds over the entire trajectory. The structure of the bound (3.10) is analogous to the local error estimators used in adaptive step-size control for numerical integration of ordinary differential equations [88], though here the goal is data preprocessing rather than numerical solution. The question then becomes how to estimate $|a_{\perp}(t)|$ or $\kappa(t)$ from sampled data without resorting to numerical differentiation, which is notoriously sensitive to noise and discretization artifacts.

The resolution lies in observing that the orthogonal deviation from a chord—the very quantity bounded in (3.6)—can be measured directly from three consecutive samples without computing any derivatives. This geometric quantity is the *sagitta*, and its relation to curvature

provides the derivative-free estimator we seek. The next section develops this connection in detail.

3.3 The sagitta as a derivative-free curvature proxy

The bound (3.10) expresses the admissible time step in terms of the normal acceleration $|a_{\perp}(t)|$, a quantity that would ordinarily require estimating the second derivative of the trajectory. Numerical differentiation of sampled data is problematic: it amplifies high-frequency noise and depends sensitively on the finite-difference scheme employed. We now show that the same geometric information can be extracted directly from triplets of consecutive samples, without differentiation, through a classical construction known as the sagitta [89, 90].

Consider three points along the trajectory at times $t - h$, t , and $t + h$, for some small $h > 0$. Denote these points by $A = y(t - h)$, $B = y(t)$, and $C = y(t + h)$. The chord connecting the outer points has direction

$$u = \frac{C - A}{|C - A|}, \quad (3.11)$$

assuming $C \neq A$. The *sagitta* at time t with half-width h is defined as the orthogonal distance from the midpoint B to the line through A and C :

$$s(t; h) = |(I - uu^{\top})(B - A)|. \quad (3.12)$$

Geometrically, $(I - uu^{\top})$ projects onto the subspace orthogonal to the chord, and the sagitta measures how far the trajectory at the central time departs from the straight path joining its neighbors. For a perfectly linear segment, the sagitta vanishes; any nonzero value reflects curvature of the arc.

Figure 3.1 illustrates the construction. Panel (a) depicts a smooth planar arc with a triplet (A, B, C) : the dashed chord connects the outer points, and the sagitta $s(t; h)$ is the orthogonal distance from B to this chord. The shaded region between the true trajectory and the linear interpolant represents the geometric distortion that the sagitta quantifies. Panel (b) applies the pointwise computation to a segment of the Lorenz attractor, coloring each sample by its local sagitta value. The curvature concentrates at the outer rims of the lobes and at the lobe-switching transitions, precisely the regions where piecewise-linear interpolation incurs the largest error.

To connect the sagitta to the normal acceleration, we expand A , B , and C in Taylor series around t . Writing $v = \dot{y}(t)$, $a = \ddot{y}(t)$, and $j = y^{(3)}(t)$ for the velocity, acceleration, and jerk, respectively, we have

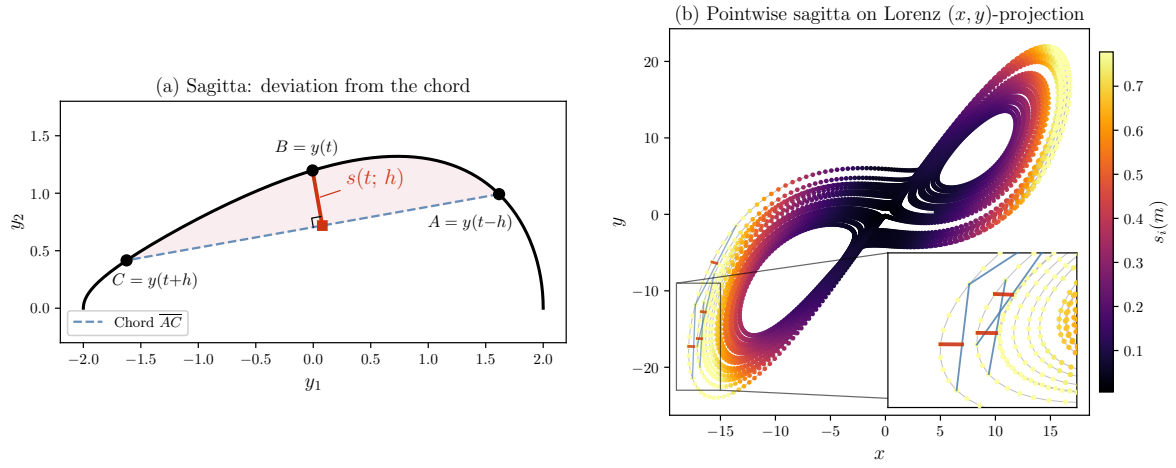


Figure 3.1: Geometric definition of the sagitta. (a) A smooth arc with triplet (A, B, C) spaced by half-width h . The sagitta $s(t; h)$ is the orthogonal distance from B to the chord $\overline{AC}$; the shaded area shows the region where the trajectory deviates from linear interpolation. (b) Pointwise sagitta $s_i(m)$ computed along a Lorenz trajectory (projected onto the x - y plane), with $m = 4$ and $\Delta t_0 = 0.005$. Red segments mark the sagitta at selected high-curvature triplets.

$$A = y(t) - vh + \frac{1}{2}ah^2 - \frac{1}{6}jh^3 + \mathcal{O}(h^4), \quad (3.13)$$

$$B = y(t), \quad (3.14)$$

$$C = y(t) + vh + \frac{1}{2}ah^2 + \frac{1}{6}jh^3 + \mathcal{O}(h^4). \quad (3.15)$$

The chord vector and the displacement from A to the midpoint B are then

$$C - A = 2vh + \frac{1}{3}jh^3 + \mathcal{O}(h^5), \quad (3.16)$$

$$B - A = vh - \frac{1}{2}ah^2 + \frac{1}{6}jh^3 + \mathcal{O}(h^4). \quad (3.17)$$

The even powers of h cancel in the chord (3.16) by symmetry, leaving only odd terms. From (3.16), the chord direction satisfies

$$u = \frac{v}{|v|} + \mathcal{O}(h^2) = \hat{v} + \mathcal{O}(h^2), \quad (3.18)$$

where $\hat{v} = v/|v|$ is the unit tangent. To leading order, the chord aligns with the velocity.

Applying the orthogonal projector $(I - uu^\top)$ to $(B - A)$ and using (3.18), the terms proportional to v are annihilated:

$$(I - uu^\top)(B - A) = -\frac{1}{2}(I - \hat{v}\hat{v}^\top)a h^2 + \mathcal{O}(h^3) = -\frac{1}{2}a_\perp h^2 + \mathcal{O}(h^3). \quad (3.19)$$

Taking norms yields the fundamental relation

$$s(t; h) = \frac{1}{2} |a_\perp(t)| h^2 + \mathcal{O}(h^3). \quad (3.20)$$

The sagitta scales quadratically with the half-width h and is proportional to the magnitude of the normal acceleration. This is precisely the quantity appearing in the error bound (3.6). Measuring the sagitta from the data is therefore equivalent to measuring the linearization error, but without any need for differentiation.

Substituting the curvature relation $|a_\perp| = \kappa|v|^2$ from (3.7) and noting that the chord length satisfies $L = |C - A| = 2|v|h + \mathcal{O}(h^3)$, we obtain the classical geometric formula

$$s(t; h) = \frac{\kappa(t) L^2}{8} + \mathcal{O}(h^3). \quad (3.21)$$

This expression connects the sagitta directly to the curvature and the chord length, independent of the parametrization. Equations (3.20) and (3.21) provide the theoretical foundation for using the sagitta as a curvature proxy: in the small- h regime, the sagitta encodes the same geometric information as the normal acceleration or the curvature, but it is computable from point coordinates alone.

We now pass to the discrete setting. Suppose the trajectory is sampled at times $t_i = i\Delta t_0$ for $i = 0, 1, \dots, N - 1$, where Δt_0 is a base sampling step assumed fine enough to oversample the dynamics. For an integer span $m \geq 1$, define the discrete sagitta at index i by forming the triplet (y_{i-m}, y_i, y_{i+m}) and computing

$$u_i = \frac{y_{i+m} - y_{i-m}}{|y_{i+m} - y_{i-m}|}, \quad s_i(m) = |(I - u_i u_i^\top)(y_i - y_{i-m})|. \quad (3.22)$$

This mirrors the continuous definition (3.12) with half-width $h = m\Delta t_0$. The collection $\{s_i(m)\}$ for $i = m, \dots, N - 1 - m$ samples the local curvature of the trajectory at resolution $m\Delta t_0$.

To summarize these pointwise values into a single representative statistic, we apply a robust aggregate operator $\mathcal{A}$ —for instance, a high percentile or a trimmed mean—that is monotone in its arguments and insensitive to outliers. The aggregated sagitta at span m is

$$S(m) = \mathcal{A}(\{s_i(m) : i = m, \dots, N - 1 - m\}). \quad (3.23)$$

For the relative variant needed in the scalar case (discussed below), we also define the median chord length

$$L_{\text{med}}(m) = \text{median}\{|y_{i+m} - y_{i-m}| : i = m, \dots, N - 1 - m\}, \quad (3.24)$$

and the relative sagitta

$$R(m) = \frac{S(m)}{L_{\text{med}}(m)}. \quad (3.25)$$

The discrete aggregates inherit the scaling behavior of their continuous counterparts. The following proposition, proved in Appendix B, makes this precise.

Proposition 3.1 (Small-span asymptotics). *Let $y \in C^3([0, T], \mathbb{R}^d)$ with $\dot{y} \neq 0$ almost everywhere, and suppose Δt_0 is small enough that the expansion (3.20) holds for $h = m\Delta t_0$ over the range of spans considered. Then*

$$S(m) = c(m\Delta t_0)^2 + \mathcal{O}((m\Delta t_0)^3), \quad c = \frac{1}{2} \mathcal{A}(|a_\perp(t_i)|), \quad (3.26)$$

$$R(m) = \tilde{c}(m\Delta t_0) + \mathcal{O}((m\Delta t_0)^2), \quad \tilde{c} \propto \mathcal{A}\left(\frac{|a_\perp(t_i)|}{|v(t_i)|}\right). \quad (3.27)$$

The proposition confirms that $S(m)$ grows quadratically with the effective time step $m\Delta t_0$, mirroring the continuous sagitta (3.20), while $R(m)$ grows linearly. Both functions are monotonically increasing in the small-span regime, a property essential for the selection rules that follow.

The quadratic scaling predicted by Proposition 3.1 is verified empirically in Figure 3.2. For both the Lorenz and Rössler systems, the aggregated sagitta $S(m)$ is plotted against the effective half-width $h = m\Delta t_0$ on logarithmic axes, with $\Delta t_0 = 0.005$ and 95th-percentile aggregation. In the small-span regime (roughly $h < 0.05$ for Lorenz, $h < 0.2$ for Rössler), the data follow the h^2 reference line closely: the fitted log-log slopes are 1.88 and 2.00, respectively, consistent with the leading-order term in (3.26). At larger spans the Lorenz curve saturates as $S(m)$ approaches the scale of the attractor itself, marking the breakdown of the small-span asymptotics. For the Rössler system, whose trajectory has more uniform curvature, the quadratic regime extends over nearly two decades. The horizontal line marks $\varepsilon = 0.1$; the dash-dotted vertical line indicates the corresponding Δt^* , which differs by nearly an order of magnitude between the two systems ($\Delta t_{\text{Lorenz}}^* \approx 0.020$, $\Delta t_{\text{Rössler}}^* \approx 0.145$), reflecting the much higher curvature of the Lorenz attractor.

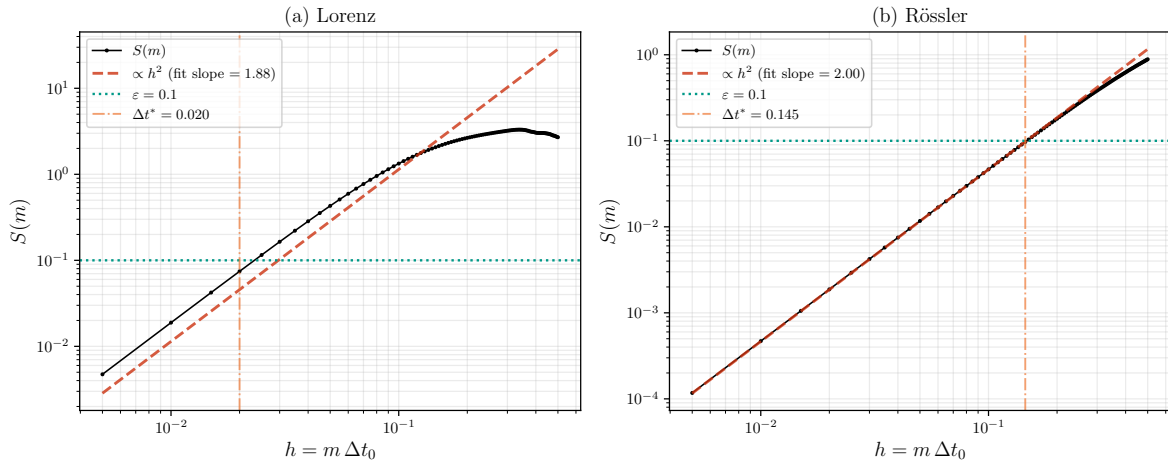


Figure 3.2: Verification of the quadratic scaling $S(m) \propto h^2$ (Proposition 3.1) for (a) the Lorenz system and (b) the Rössler system. The base sampling step is $\Delta t_0 = 0.005$ with 95th-percentile aggregation. Dashed red lines show the h^2 power law fitted to the first 20 spans; the fitted slopes are indicated in the legend. The horizontal dotted line marks $\varepsilon = 0.1$, and the dash-dotted vertical line marks the selected Δt^* .

With these aggregates in hand, the selection of Δt reduces to finding the largest span m for which the aggregate sagitta remains below a tolerance. For multivariate observations ($d \geq 2$), we work directly with the absolute sagitta and impose

$$m^* = \max\{m \in \mathbb{N}_{\geq 1} : S(m) \leq \varepsilon\}, \quad \Delta t^* = m^* \Delta t_0. \quad (3.28)$$

The monotonicity of S ensures that this maximum exists and is unique. The tolerance ε is specified in standardized units: before computing sagittae, each coordinate of the trajectory is divided by its sample standard deviation (with Bessel's correction) so that distances are comparable across dimensions and the threshold has a consistent interpretation.

For scalar observations ($d = 1$), geometry must first be reconstructed via delay embedding. Given a scalar series $\{x_i\}$, one forms vectors [3, 91]

$$X_i = (x_i, x_{i+\ell}, \dots, x_{i+(d_e-1)\ell}) \in \mathbb{R}^{d_e}, \quad (3.29)$$

where ℓ is the embedding lag and d_e the embedding dimension. The lag is chosen as the first local minimum of the average mutual information between x_i and $x_{i+\ell}$ [92], and the dimension is determined by the false nearest neighbors criterion [93]; details are given in Appendix B. On the embedded trajectory $\{X_i\}$, we compute $S(m)$ and $L_{\text{med}}(m)$ as above and impose a relative bound:

$$m^* = \max\{m \in \mathbb{N}_{\geq 1} : R(m) \leq \varepsilon\}, \quad \Delta t^* = m^* \Delta t_0. \quad (3.30)$$

The relative criterion normalizes by chord length, rendering the threshold dimensionless and preventing pathological behavior when absolute sagittae are uniformly small on nearly linear arcs.

Figure 3.3 validates both scaling laws for the scalar case on the x -component of the Lorenz system, delay-embedded with lag $\ell = 17$ samples and dimension $d_e = 3$. The absolute sagitta $S(m)$ exhibits the same quadratic growth (fitted slope 1.91) as in the multivariate case, confirming that the embedding preserves the asymptotic structure. The relative sagitta $R(m)$ grows linearly (fitted slope 0.96), as predicted by (3.27). The nonmonotonic bump visible in $R(m)$ near $h \approx 0.4$ corresponds to spans comparable to the orbital period of the Lorenz system, where the chord length L_{med} decreases as the triplet wraps around a lobe; this departure from monotonicity occurs well beyond the small-span regime and does not affect the selection rule, which operates at much smaller spans.

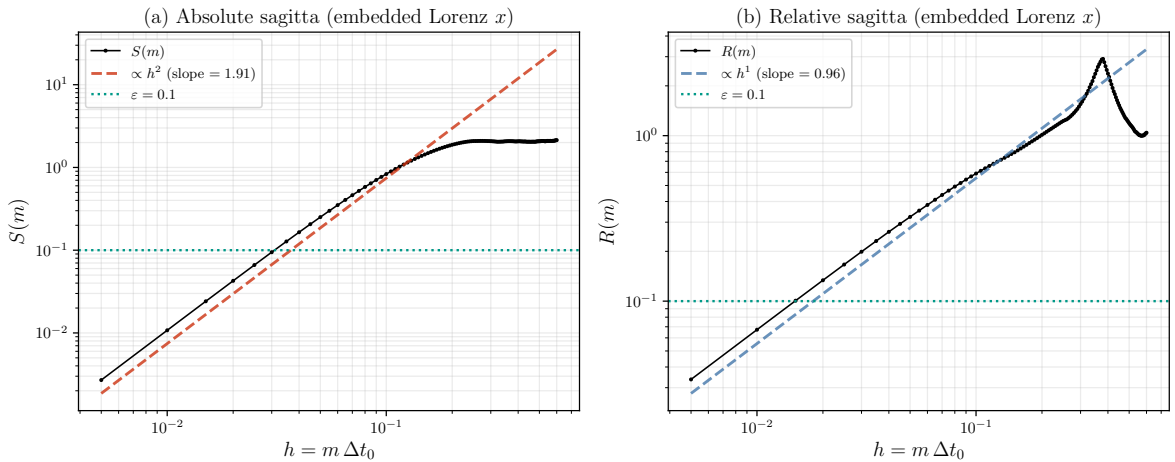


Figure 3.3: Scaling of the absolute and relative sagitta for scalar observations. The x -component of the Lorenz system is delay-embedded ($\ell = 17$, $d_e = 3$) and standardized. (a) $S(m) \propto h^2$ with fitted slope 1.91. (b) $R(m) \propto h$ with fitted slope 0.96, confirming the linear scaling of Proposition 3.1. The horizontal line marks $\varepsilon = 0.1$.

Both rules share the same structure: a monotone function of the span is compared against a tolerance, and the largest admissible span is selected. The next section establishes that these rules are idempotent—resampling the trajectory at Δt^* and reapplying the procedure yields the same step—and discusses practical aspects of their implementation.

3.4 Selection rules and idempotence

The selection rules (3.28) and (3.30) identify a time step $\Delta t^* = m^* \Delta t_0$ by finding the largest span for which the aggregated sagitta remains within tolerance. A natural question arises: what happens if we resample the trajectory at this recommended step and apply the same procedure again? If the recomputed step differed substantially from Δt^* , the rule would be unstable—its output would depend on the resolution at which we happened to start, undermining reproducibility. A desirable property is *idempotence*: resampling at Δt^* and reapplying the rule should return the same step, up to higher-order corrections.

We now establish that both the absolute and relative rules possess this fixed-point property. The key insight is that decimation by factor m^* transforms the optimal span from m^* (on the original series) to 1 (on the decimated series), while preserving the tolerance inequalities that defined optimality in the first place.

Let $\{y_i\}_{i=0}^{N-1}$ denote the original trajectory sampled at step Δt_0 , and suppose the absolute rule (3.28) selects m^* as the maximal admissible span. Define the decimated series

$$\hat{y}_k = y_{km^*}, \quad k = 0, 1, \dots, \hat{N} - 1, \quad (3.31)$$

where $\hat{N} = \lfloor (N - 1)/m^* \rfloor + 1$. The decimated series has base step $\hat{\Delta}t_0 = m^* \Delta t_0 = \Delta t^*$. For any span $k \geq 1$ on the decimated series, let $\hat{S}(k)$ denote the aggregated sagitta computed from $\{\hat{y}_k\}$ in the same manner as $S(m)$ was computed from $\{y_i\}$.

The connection between aggregates on the original and decimated series is captured by the following result.

Lemma 3.2 (Span mapping under decimation). *Let $y \in C^3([0, T], \mathbb{R}^d)$ and suppose the small-span asymptotics (3.26) hold. For $k \in \{1, 2\}$, the aggregated sagitta on the decimated series satisfies*

$$\hat{S}(k) = S(km^*) + \mathcal{O}((k\Delta t^*)^3). \quad (3.32)$$

The analogous relation holds for the relative sagitta:

$$\hat{R}(k) = R(km^*) + \mathcal{O}((k\Delta t^*)^2). \quad (3.33)$$

The lemma states that computing the sagitta at span k on the decimated series is equivalent, up to higher-order terms, to computing it at span km^* on the original series. This is intuitive: the triplet $(\hat{y}_{j-k}, \hat{y}_j, \hat{y}_{j+k})$ on the decimated series corresponds to the triplet $(y_{(j-k)m^*}, y_{jm^*}, y_{(j+k)m^*})$ on the original, which is precisely the triplet at span km^* . The proof, given in Appendix B, formalizes this correspondence and tracks the remainder terms through the aggregate operator.

With this machinery, idempotence follows from the gap between $S(m^*)$ and $S(m^* + 1)$ that defines optimality.

Theorem 3.3 (Idempotence of the absolute rule). *Let $y \in C^3([0, T], \mathbb{R}^d)$ with $\dot{y} \neq 0$ almost everywhere, and assume the map $m \mapsto S(m)$ is strictly increasing on $\{1, 2, \dots, M\}$ for some $M \geq m^* + 1$. Let m^* be the maximal span satisfying $S(m^*) \leq \varepsilon < S(m^* + 1)$, and set $\Delta t^* = m^* \Delta t_0$. If Δt_0 is sufficiently small, then reapplying the selection rule to the decimated series $\{\hat{y}_k\}$ yields $\hat{m}^* = 1$. Consequently,*

$$\hat{\Delta t}^* = \hat{m}^* \cdot \hat{\Delta t}_0 = 1 \cdot \Delta t^* = \Delta t^*, \quad (3.34)$$

and the rule is idempotent up to $\mathcal{O}((\Delta t^*)^3)$ corrections.

Sketch of argument. The maximality of m^* implies a gap $\delta = S(m^* + 1) - \varepsilon > 0$. By Lemma 3.2, choosing Δt_0 small enough ensures that

$$|\hat{S}(1) - S(m^*)| \leq \frac{\delta}{4} \quad |\hat{S}(2) - S(2m^*)| \leq \frac{\delta}{4}$$

Then

$$\hat{S}(1) \leq S(m^*) + \delta/4 \leq \varepsilon + \delta/4 < \varepsilon + \delta/2$$

so span 1 is admissible on the decimated series. For span 2, monotonicity gives $S(2m^*) \geq S(m^* + 1) = \varepsilon + \delta$, hence $\hat{S}(2) \geq \varepsilon + 3\delta/4 > \varepsilon$, so span 2 fails. Strict monotonicity then excludes all $k \geq 2$, forcing $\hat{m}^* = 1$. The full argument appears in Appendix B. $\square$

Figure 3.4 demonstrates the idempotence property on the Lorenz system with $\Delta t_0 = 0.002$. Panel (a) shows $S(m)$ on the original series: the threshold $\varepsilon = 0.1$ is crossed between $m^* = 11$ (admissible) and $m^* + 1 = 12$ (inadmissible), yielding $\Delta t^* = 0.022$. Panel (b) shows $\hat{S}(k)$ computed on the decimated series at the new base step $\hat{\Delta t}_0 = \Delta t^*$. As predicted by Theorem 3.3, span $k = 1$ yields $\hat{S}(1) = 0.0896 \leq \varepsilon$ while span $k = 2$ yields $\hat{S}(2) = 0.3387 > \varepsilon$, so the reapplied rule returns $\hat{m}^* = 1$ and $\hat{\Delta t}^* = 1 \cdot \Delta t^* = \Delta t^*$. The fixed point is exact at the level of integer spans, confirming that the selected resolution is stable under reprocessing.

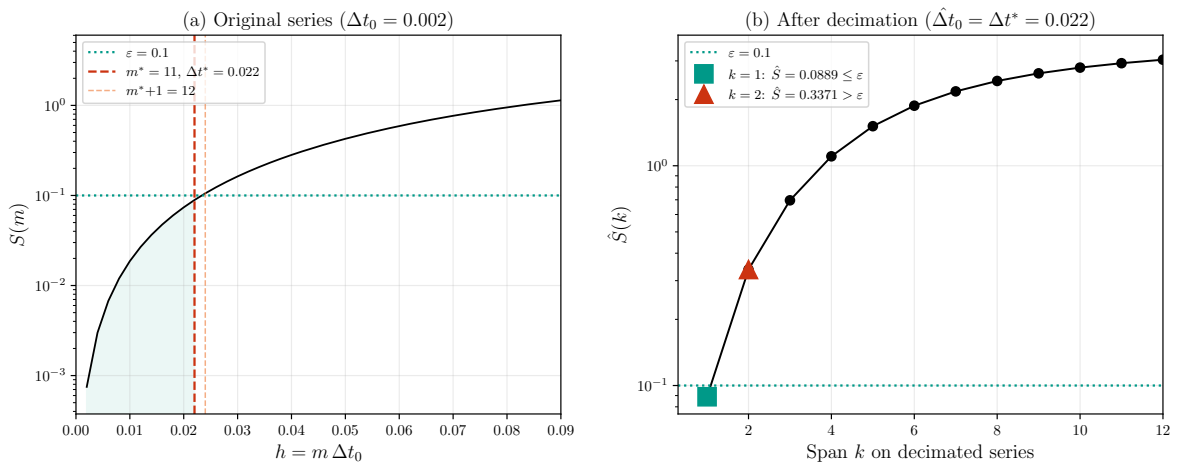


Figure 3.4: Idempotence of the absolute selection rule on the Lorenz system. (a) $S(m)$ on the original series ($\Delta t_0 = 0.002$); the shaded region indicates admissible spans, with $m^* = 11$ and $\Delta t^* = 0.022$. (b) $\hat{S}(k)$ on the decimated series: $k = 1$ (square) satisfies $\hat{S}(1) = 0.0889 \leq \varepsilon$, while $k = 2$ (triangle) exceeds the tolerance at $\hat{S}(2) = 0.3387$, so $\hat{m}^* = 1$ and the rule returns Δt^* unchanged.

The relative rule enjoys an analogous property.

Theorem 3.4 (Idempotence of the relative rule). *Under the same smoothness and monotonicity assumptions, let m^* be the maximal span satisfying $R(m^*) \leq \varepsilon < R(m^* + 1)$, and set $\Delta t^* = m^* \Delta t_0$. If Δt_0 is sufficiently small, then reapplying the relative selection rule to the decimated series yields $\hat{m}^* = 1$, so that*

$$\hat{\Delta t}^* = \Delta t^*, \quad (3.35)$$

up to $\mathcal{O}((\Delta t^)^2)$ corrections.*

The proof is identical in structure, substituting R for S and adjusting the remainder order; see Appendix B.

The idempotence theorems have a simple geometric interpretation. Decimation by m^* coarsens the sampling so that what was previously the optimal half-width $h^* = m^* \Delta t_0$ becomes the unit span on the new grid. The tolerance ε was chosen precisely so that this half-width lies at the boundary of admissibility: the sagitta at span m^* just satisfies the bound, while at span $m^* + 1$ it exceeds it. After decimation, span 1 inherits the former role and span 2 inherits the latter. The gap between them, preserved up to higher-order terms by Lemma 3.2, ensures that span 1 passes and span 2 fails, leaving no room for the optimum to shift.

The practical consequence is that the selected Δt^* is stable under reprocessing. If an analyst receives a trajectory already resampled at some step and applies the sagitta criterion, the output will be consistent with what would have been obtained from the original high-resolution data. This reproducibility is essential when comparing results across studies or when the preprocessing history of a dataset is uncertain.

A few remarks on implementation are in order. The monotonicity assumption—that $S(m)$ or $R(m)$ is strictly increasing over the relevant range of spans—holds in the small-span regime by the asymptotics of Proposition 3.1. In practice, one verifies monotonicity empirically during the search. The search itself is efficient: rather than evaluating all spans up to some maximum, one employs geometric bracketing—testing spans $m = 1, \lfloor g \rfloor, \lfloor g^2 \rfloor, \dots$ for a growth factor $g > 1$ (typically $g = 1.5$)—followed by bisection to pinpoint the threshold crossing. This yields a logarithmic number of span evaluations. Each evaluation is vectorized over all admissible triplets and runs in time linear in the trajectory length times the dimension, making the entire procedure lightweight even for long time series.

The tolerance ε remains the sole free parameter. Its choice depends on the application: smaller values enforce finer resolution and higher geometric fidelity, at the cost of retaining more samples; larger values permit coarser discretization, compressing the data but risking greater interpolation error. Because the trajectory is standardized before computing sagittae, ε has a consistent interpretation across systems: it bounds the typical orthogonal deviation, in units of per-coordinate standard deviations, that linear interpolation incurs. For forecasting benchmarks, a common ε ensures that all trajectories are discretized at comparable geometric difficulty, enabling meaningful cross-system comparisons.

3.5 Practical considerations

The developments of the preceding sections provide a principled answer to a question that pervades data-driven modeling of dynamical systems: at what temporal resolution should a trajectory be discretized? The sagitta-based criterion replaces arbitrary choices with a geometrically grounded rule that adapts to the intrinsic curvature of the data, is stable under reapplication, and admits a single interpretable parameter. We now discuss the practical implications of this framework, its limitations, and its role in the forecasting experiments that follow.

The most immediate benefit is interpretability. The tolerance ε directly bounds the orthogonal deviation—measured in standardized units—that piecewise-linear interpolation incurs between consecutive samples. When $\varepsilon = 0.1$, for instance, the typical geometric error of treating the trajectory as locally linear is at most ten percent of the per-coordinate variability. This is a statement about the data themselves, not about any particular model, and it holds uniformly along the trajectory up to the robustness of the aggregate. An analyst can therefore reason about the fidelity of the discretization without reference to downstream tasks: the bound on interpolation error is intrinsic to the geometry of the sampled curve.

This intrinsic character enables meaningful comparisons across dynamical systems with disparate timescales. The Lorenz system, the Rössler oscillator [39], and a turbulent flow may evolve on timescales differing by orders of magnitude, yet once each is resampled at its own Δt^* under a common ε , all three share a comparable level of local curvature per step. Differences in forecasting performance can then be attributed to the complexity of the dynamics or the capacity of the model, rather than to incidental choices of temporal resolution. Without such normalization, a model that excels on an oversampled trajectory—where predictions reduce to near-linear extrapolation—cannot be fairly compared against one evaluated on a coarser, geometrically richer discretization.

Figure 3.5 quantifies how the selected Δt^* varies with tolerance ε for three systems: Lorenz, Rössler, and Chen [42]. All three curves are monotonically increasing— a looser tolerance permits coarser sampling— but they occupy different vertical bands, reflecting the distinct curvature profiles of each attractor. At the reference tolerance $\varepsilon = 0.1$, the Rössler system admits $\Delta t^* \approx 0.146$, the Lorenz system requires $\Delta t^* \approx 0.022$, and the Chen system, whose dynamics involve sharper folding, demands the finest resolution at $\Delta t^* \approx 0.012$. The staircase structure arises from the integer constraint $m^* \in \mathbb{N}$; each plateau corresponds to a range of ε values for which the same integer span is optimal. These curves serve as a diagnostic: they reveal the intrinsic geometric difficulty of each system and make explicit the consequences of any particular tolerance choice. The order-of-magnitude separation between the Rössler and Chen curves at fixed ε underscores why comparing forecasting performance without controlling for discretization is fundamentally misleading.

The relevance to reservoir computing is direct. Echo State Networks, as discussed in Chapter 1, process input sequences one step at a time, updating an internal state that encodes a fading memory of past inputs. The utility of this memory depends critically on the information content of each step. If the sampling is too fine, consecutive inputs are nearly identical and the reservoir’s recurrent dynamics add little beyond what a simple linear extrapolator could achieve. The effective memory span, measured in physical time, shrinks

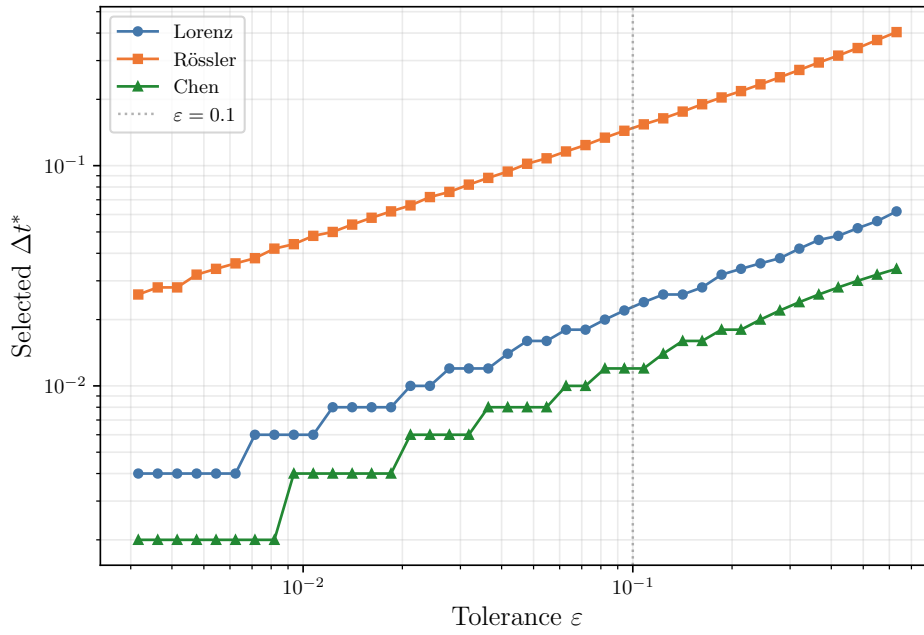


Figure 3.5: Selected time step Δt^* as a function of tolerance ε for the Lorenz, Rössler, and Chen systems ($\Delta t_0 = 0.002$, 95th-percentile aggregation). The vertical dotted line marks $\varepsilon = 0.1$. Each system occupies a distinct band reflecting its characteristic curvature; the staircase structure arises from the integer constraint on the span m^* .

because many steps are needed to traverse a meaningful arc of the attractor. At the other extreme, if the sampling is too coarse, the trajectory between steps curves so sharply that the linear readout cannot capture the true evolution. The reservoir’s internal representation, however rich, is undermined by geometric distortion in the target signal itself.

The sagitta criterion identifies the regime between these failure modes. By bounding the linearization error, it ensures that each step carries nontrivial geometric information—the trajectory bends enough to challenge a linear model—while preserving fidelity to the underlying continuous dynamics. Forecasting performance measured under this discretization reflects the genuine capacity of the reservoir to learn and extrapolate nonlinear structure, rather than artifacts of preprocessing. When comparing architectures, hyperparameter settings, or training procedures, this guarantee is essential: it isolates the contribution of the model from the contribution of the data representation.

The spatial distribution of curvature on the Lorenz attractor is visualized in Figure 3.6, which colors each point of the Lorenz trajectory by its pointwise sagitta $s_i(m)$ with $m = 3$. The curvature is far from uniform: it peaks at the outer edges of both lobes, where the trajectory reverses direction before spiraling inward, and at the saddle region near the origin where lobe-switching transitions produce sharp bends. The interior of each lobe, by contrast, consists of nearly concentric arcs with low curvature, where the dynamics are essentially linear at the scale of a few time steps. This heterogeneity is precisely what the 95th-percentile aggregation captures: it sets the resolution requirement by the most geometrically demanding regions—the high-curvature tails of the distribution—while remaining insensitive to the abundant low-curvature samples in the lobe interiors. The figure also illustrates why a single, spatially uniform Δt is appropriate despite the nonuniform curvature: the high-curvature events, though localized, are dynamically critical because they correspond to the

folding and stretching mechanisms that generate the attractor’s fractal structure and that any faithful predictive model must resolve.

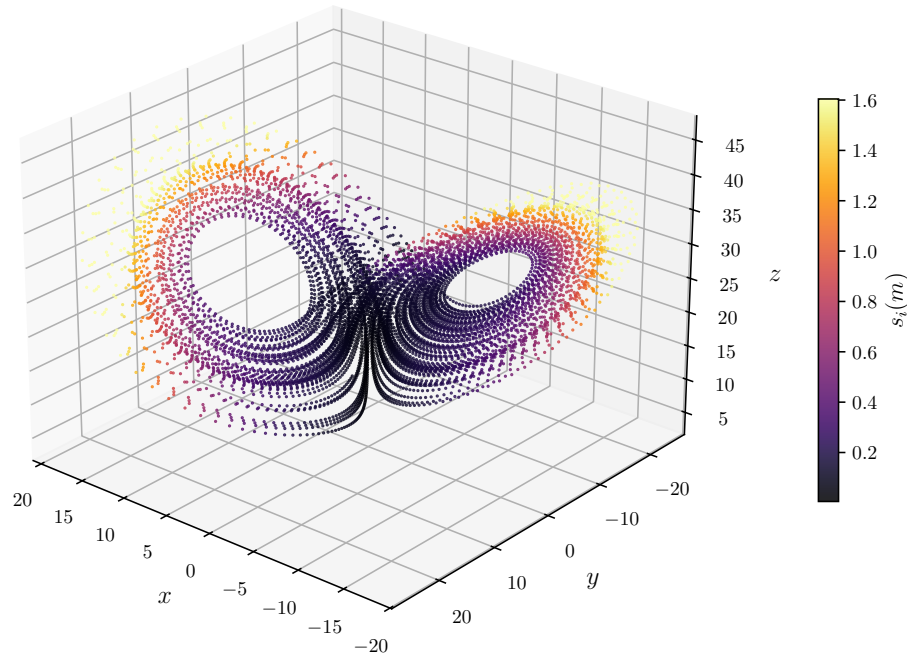


Figure 3.6: Lorenz attractor colored by pointwise sagitta $s_i(m)$ with $m = 3$ and $\Delta t_0 = 0.01$. High curvature (bright) concentrates at the outer rims of the lobes and at lobe-switching transitions near the origin; the lobe interiors (dark) are nearly linear at this time scale. Color scale clipped at the 97th percentile.

A subtlety worth noting is that the high sagitta values at the outer rims of the lobes are not purely a consequence of geometric curvature: the trajectory moves fastest in these regions, so a fixed Δt produces longer chords and larger orthogonal deviations even if the local radius of curvature were comparable to the interior. The sagitta conflates both effects—speed and curvature enter through $|a_\perp| = \kappa|v|^2$ —which is precisely appropriate for discretization purposes: what matters for linearization error is the actual geometric distortion per step, regardless of whether it arises from sharp bending or rapid traversal of a mildly curved arc.

3.5.1 Possible caveats

Several caveats merit attention. First, the method presupposes access to an initial sampling step Δt_0 that is fine enough to oversample the dynamics. The theoretical results—the quadratic scaling of the sagitta, the monotonicity of the aggregate, and the idempotence of the selection rule—all rest on the small-span asymptotics derived in Section 3.3. If Δt_0 is already too coarse, these asymptotics may not hold, and the procedure can return misleading results. In practice, one should verify that the chosen Δt_0 lies well within the regime where $S(m)$ grows smoothly and monotonically with m ; a diagnostic plot of $S(m)$ against m readily reveals departures from the expected quadratic trend.

Second, the framework is most naturally suited to synthetic data, where the sampling step can be controlled at will. Numerical integration of ordinary differential equations

produces trajectories at arbitrarily fine resolution, and the sagitta criterion then identifies the coarsest admissible decimation. For experimental data, the situation is more constrained: the measurement apparatus imposes a fixed sampling rate, and one cannot retroactively refine it. Nevertheless, the method remains valuable in two respects. If the experimental resolution is finer than necessary—as is often the case with modern high-frequency sensors—the sagitta criterion guides subsampling to a more parsimonious representation without loss of geometric fidelity. And in settings where measurement is expensive or invasive, preliminary trials at high resolution can establish the curvature profile of the system, informing the choice of a coarser, more economical sampling rate for subsequent data collection. The criterion thus serves both as a post-hoc diagnostic and as a design tool for instrumentation.

Third, the asymptotic expansions underlying the method assume sufficient smoothness of the trajectory—specifically, three continuous derivatives—and nonvanishing velocity. Systems with impacts, switching surfaces, or other discontinuities violate these assumptions locally, and the sagitta may spike at isolated events without reflecting the typical geometric difficulty. The use of robust aggregates (percentiles or trimmed means) mitigates this issue by down-weighting outliers, but in systems where such events are frequent or physically significant, supplementary criteria may be warranted. Similarly, regions where the velocity approaches zero—near fixed points or slow manifolds—can produce degenerate chords and ill-defined sagittae. The implementation handles these cases by excluding degenerate triplets, but users should be aware that the effective sample size for the aggregate may be reduced in nearly stationary regimes.

3.6 Conclusions

Despite these caveats, the sagitta framework offers a reproducible and transparent standard for temporal discretization. Its idempotence guarantees that the selected step is not an artifact of the initial resolution, and its geometric foundation ensures that the tolerance has a consistent meaning across systems. For the forecasting experiments presented in the remainder of this thesis, we adopt the following convention: all trajectories are preprocessed using the sagitta-based selection rule with tolerance $\varepsilon = 0.1$ and a 95th-percentile aggregation. This corresponds to bounding the interpolation error at roughly ten percent of the standardized variability, with the bound holding for all but the most extreme five percent of local arcs. The choice balances geometric fidelity against data compression and has proven empirically robust across the systems studied.

By fixing this discretization protocol, we establish a level playing field for comparing reservoir computers and other predictive models. Performance differences reported in subsequent chapters reflect differences in modeling capacity and hyperparameter tuning, not differences in how the data were sampled. The sagitta criterion thus serves as a methodological anchor: it standardizes the raw material from which all models learn, enabling the kind of controlled experimentation that reproducible science demands.

4 Chaotic forecasting evaluation

Evaluating forecasting models for chaotic time series is fundamentally a benchmarking problem. The goal is not to measure absolute prediction accuracy—which is intrinsically limited by sensitive dependence on initial conditions [38, 94]—but to compare models in a manner that is meaningful, reproducible, and robust across different systems and parameter regimes.

In much of the existing literature on Echo State Networks (ESNs) [9, 15, 19], performance is reported using pointwise error measures such as the mean squared error (MSE) or root mean squared error (RMSE) evaluated over fixed prediction horizons [20, 21]. While convenient, this practice is poorly aligned with chaotic dynamics. Due to exponential divergence of nearby trajectories [95], pointwise errors inevitably grow and saturate after a characteristic transient, regardless of the quality of the model. Once saturation occurs, the metric ceases to distinguish between predictors and instead reflects properties of the dynamical system being forecast.

From a benchmarking perspective, this behaviour is problematic. Metrics that saturate early compress performance differences and obscure meaningful variation between models. Two ESNs with markedly different short-term predictive capabilities may exhibit nearly identical long-horizon RMSE values, yielding ambiguous or misleading conclusions. Moreover, optimization driven by such metrics often converges toward solutions that perform well only in the initial steps, without improving the model’s ability to track the dynamics over longer horizons.

A further complication arises when comparing performance across different chaotic systems. Fixed-horizon pointwise losses implicitly assume a common notion of difficulty, yet chaotic systems differ widely in predictability timescales. The maximal Lyapunov exponent $\lambda_{\max}$ sets a characteristic timescale $T_{\lambda} = 1/\lambda_{\max}$ beyond which individual trajectory predictions become unreliable [94, 95]. Identical error values may therefore correspond to qualitatively different predictive behaviours when measured against each system’s intrinsic timescale.

To address these issues, benchmarking in chaotic forecasting must shift focus from error magnitude to temporal persistence of meaningful predictions. Rather than asking how large the error becomes after a prescribed horizon, a more informative criterion is *how long* a model can produce predictions that remain dynamically consistent with the reference

trajectory. This reframing aligns the benchmark with the intrinsic limits of predictability and restores discriminative power to performance evaluation.

The concept of a “valid prediction time” (VPT)—defined as the number of consecutive time steps during which a normalized error remains below a fixed threshold—has become a *de facto* standard in evaluating the short-term forecasting performance of models. However, VPT is defined as a hard hitting time: it is discontinuous in the model parameters, non-differentiable, and brittle under finite-sample fluctuations. These properties make it unsuitable as a training objective and fragile as a reported statistic when small perturbations can shift the threshold crossing by several time steps.

In this chapter, we introduce the Expected Forecast Horizon (EFH) as a continuous, differentiable surrogate of the valid prediction time. The EFH quantifies the time interval over which a model’s predictions remain informative before error growth becomes dominated by chaotic divergence. Constructed via soft-thresholding and a cumulative survival process, EFH provides an interpretable measure of predictive skill that is compatible with gradient-based optimization and enables fair comparison across models and dynamical systems. The algebraic form of EFH admits a natural interpretation as a *restricted mean survival time* (RMST) [96] in the language of discrete-time survival analysis [97, 98], establishing a formal connection between forecast horizon estimation and well-studied statistical methodology.

The benchmarking framework developed here is used throughout the remainder of this thesis to evaluate ESNs in chaotic forecasting tasks. By grounding performance assessment in predictability horizons rather than asymptotic error values, we obtain benchmarks that are both dynamically meaningful and methodologically sound.

4.1 Expected forecast horizon: definition and interpretation

This section formalises the EFH as a horizon-like performance measure for long-horizon (closed-loop) forecasting. The guiding idea is that in chaotic regimes the practically relevant quantity is not the terminal error at an arbitrarily chosen time, but the *time span over which the prediction remains within an acceptable tolerance*. EFH is a smooth surrogate of that time-to-threshold notion, designed to be both interpretable as a metric and usable as a differentiable training objective.

4.1.1 Forecasting setting, closed-loop prediction, and normalization

We consider a dynamical system observed through a D -dimensional state (or measurement) vector. For each evaluation instance $i \in \{1, \dots, B\}$ we have a reference trajectory

$$\mathbf{y}_{1:T}^{(i)} = \{\mathbf{y}_t^{(i)}\}_{t=1}^T, \quad \mathbf{y}_t^{(i)} \in \mathbb{R}^D,$$

and a model forecast

$$\hat{\mathbf{y}}_{1:T}^{(i)} = \{\hat{\mathbf{y}}_t^{(i)}\}_{t=1}^T, \quad \hat{\mathbf{y}}_t^{(i)} \in \mathbb{R}^D.$$

The forecast is produced in *closed-loop* mode: after a short initialization phase, the model feeds its own previous outputs as inputs to generate future predictions (See Section 1.5.2).

This is the regime where divergence due to sensitivity to initial conditions is unavoidable, and it is the regime that EFH is intended to evaluate.

A horizon-based metric requires a notion of what constitutes a “small” or “large” error, and that notion must remain stable across datasets, dynamical systems, and parameter regimes. For this reason, we assume that the state coordinates are normalised using a *fixed, task-level convention*. Concretely, normalization parameters (mean and scale) are determined once—typically from the training set or from a long reference simulation—and the same transformation is applied to all trajectories used for training and evaluation. A standard choice is per-coordinate standardization:

$$y'_{t,d} = \frac{y_{t,d} - \mu_d}{\sigma_d}, \quad \hat{y}'_{t,d} = \frac{\hat{y}_{t,d} - \mu_d}{\sigma_d}, \quad (4.1)$$

with μ_d and σ_d fixed constants. After this transformation, each coordinate has comparable scale, and an error tolerance such as $\tau = 0.2$ has a clear meaning: it corresponds to a discrepancy of roughly 0.2 standard deviations in the normalised space. Other normalizations (e.g., scaling to $[-1, 1]$ per coordinate) are also possible; the essential requirement is that the same transformation is applied consistently so that τ is comparable across experiments.

Without such a convention, the tolerance becomes arbitrary: the same numerical threshold could be “tight” for one system and “loose” for another, making EFH values non-portable and cross-system comparisons ill-defined.

4.1.2 Pointwise error signal

Given normalised trajectories, we define a nonnegative pointwise error per sample and time step,

$$\varepsilon_{i,t} := \mathcal{E}(\mathbf{y}_t^{(i)}, \hat{\mathbf{y}}_t^{(i)}), \quad i = 1, \dots, B, \quad t = 1, \dots, T, \quad (4.2)$$

where $\mathcal{E}$ is a differentiable discrepancy measure. The default choice used throughout this thesis is the RMSE across the D channels,

$$\varepsilon_{i,t} = \sqrt{\frac{1}{D} \sum_{d=1}^D (y_{t,d}^{(i)} - \hat{y}_{t,d}^{(i)})^2}. \quad (4.3)$$

This yields an error in the same normalised units as the state coordinates, which is precisely what makes a tolerance τ interpretable. An example of a typical error signal produced in closed-loop forecasting of a chaotic system is shown in Figure 4.1: the error remains small for an initial tracking phase, then undergoes a sigmoidal transition to saturation, consistent with the well-known pattern of exponential growth followed by attractor-bounded saturation [94].

The properties of EFH—its dependence on τ , δ , and the shape of the error curve—are most transparently examined on a controlled test signal whose behaviour is known analytically. For this purpose, we define a synthetic error signal that mimics the qualitative pattern observed in closed-loop chaotic forecasting: an initial phase of near-zero error, followed by a smooth sigmoidal transition to saturation. Concretely,

$$e_t = \sigma(a(t - t_0)), \quad a = 0.2, \quad t_0 = 50, \quad t \in \{0, \dots, 99\}, \quad (4.4)$$

where σ denotes the standard logistic function ($\sigma(z) = 1/(1 + e^{-z})$). The steepness $a = 0.2$ and midpoint $t_0 = 50$ are chosen so that the transition occupies a visible fraction of the horizon while leaving clear tracking and saturated regimes on either side (Figure 4.1). The specific values are otherwise arbitrary; the sensitivity analysis in Sections 4.1.4–4.2.7 will confirm that the conclusions are independent of this particular choice.

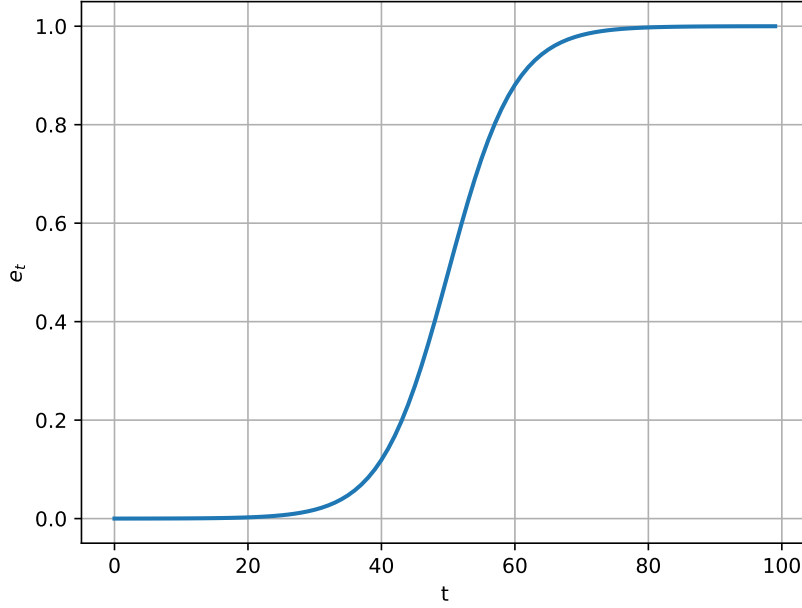


Figure 4.1: Synthetic error signal e_t defined in Eq. (4.4), used as the reference test case for the sensitivity analysis of EFH in Sections 4.1.4–4.2.7.

4.1.3 Single benchmark curve

EFH is defined from a single time series of errors, so we must aggregate the B per-sample errors $\{\varepsilon_{i,t}\}_{i=1}^B$ at each time t into a scalar e_t that reflects the “typical” behaviour of the predictor. We use the geometric mean:

$$e_t := \left(\prod_{i=1}^B \varepsilon_{i,t} \right)^{1/B} = \exp \left(\frac{1}{B} \sum_{i=1}^B \log \varepsilon_{i,t} \right). \quad (4.5)$$

There are two reasons to prefer this over the arithmetic mean in the present context.

Log-scale stability. In chaotic forecasting, the error across different initial conditions often becomes heavy-tailed: a small subset of trajectories may diverge early, while the majority remains accurate longer. The arithmetic mean is dominated by these early blow-ups, which can obscure the typical case. The geometric mean corresponds to averaging *log-errors*

and therefore emphasises multiplicative differences while reducing sensitivity to occasional extreme values.

Compatibility with multiplicative growth. A common empirical observation in closed-loop chaotic forecasting is that errors grow approximately exponentially over an initial window before saturating [94]. Exponential growth is additive in log-space:

$$\log \varepsilon_t \sim \lambda t + \text{const.}$$

Averaging log-errors therefore produces a time series e_t whose growth behaviour is smoother and easier to interpret, which is beneficial both for benchmarking and for optimization.

Numerical remark. The expression (4.5) requires evaluating $\log \varepsilon_{i,t}$. In practice, $\varepsilon_{i,t}$ can be extremely small (near machine precision) for short times. To avoid $\log(0)$ and to keep gradients finite, we clamp the argument:

$$\log \varepsilon_{i,t} \leftarrow \log(\max(\varepsilon_{i,t}, \epsilon)),$$

with $\epsilon > 0$ at machine precision. An additive offset $\log(\varepsilon_{i,t} + \epsilon)$ would also suffice; the clamping form is used here because it leaves all values above ϵ exactly unchanged, avoiding any shift in the physically meaningful range.

4.1.4 Soft acceptability gate

Suppose we want to declare that predictions are “reliable” at time t if e_t remains below a tolerance $\tau > 0$. The hard indicator $\mathbf{1}\{e_t < \tau\}$ is not suitable for gradient-based training—it has zero gradient almost everywhere—and is also brittle as a reported statistic, since small perturbations can flip the indicator. We therefore introduce a smooth acceptability gate

$$g_t(\tau, \delta) := \sigma\left(\frac{\tau - e_t}{\delta}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (4.6)$$

where $\delta > 0$ controls the softness of the transition around the threshold.

This gate has a useful interpretation: if $e_t \ll \tau$, then $g_t \approx 1$ (clearly acceptable); if $e_t \gg \tau$, then $g_t \approx 0$ (clearly unacceptable); if $e_t \approx \tau$, then g_t transitions smoothly between 0 and 1 over a band of width proportional to δ . The parameter δ plays the role of a temperature: smaller δ yields a sharper approximation of the hard threshold; larger δ yields a smoother but more conservative criterion that starts penalising earlier (see Figure 4.2).

Remark 1 (The dimensionless ratio $R = \delta/\tau$). Since δ and τ have the same units (normalised error), their ratio $R := \delta/\tau$ is the natural dimensionless parameter that controls the sharpness of the gate relative to the tolerance. The gate argument can be written as

$$\frac{\tau - e_t}{\delta} = \frac{1}{R} \left(1 - \frac{e_t}{\tau}\right),$$

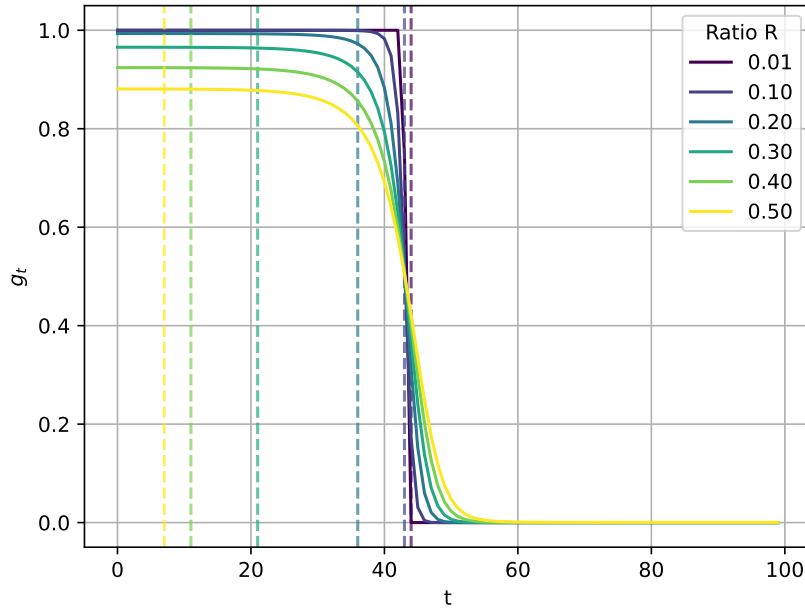


Figure 4.2: Gate function $g_t(\tau, \delta)$ for different softness ratios $R = \delta/\tau$. Dashed vertical lines indicate the corresponding EFH values. As R increases, the transition region broadens and the effective horizon decreases.

showing that R governs how sharply the sigmoid discriminates between errors above and below τ . Small R (e.g., $R \leq 0.1$) approximates a hard threshold; large R (e.g., $R \geq 0.5$) blurs the transition substantially. We use R as the control parameter in the sensitivity analysis of EFH.

4.1.5 Soft survival process and definition of EFH

A horizon is intrinsically a *prefix* concept: a forecast is useful up to time t only if it has remained acceptable at all previous steps. EFH encodes this by defining a soft survival probability

$$s_t(\tau, \delta) := \prod_{k=1}^t g_k(\tau, \delta), \quad t = 1, \dots, T, \quad (4.7)$$

and then taking the discrete area under the survival curve:

$$\text{EFH}(\tau, \delta) := \sum_{t=1}^T s_t(\tau, \delta). \quad (4.8)$$

Since $0 < g_t < 1$, we have $0 < s_t < 1$ and therefore $0 \leq \text{EFH} \leq T$.

Interpretation as an expected valid horizon. If we were to replace g_t by the hard indicator $\mathbf{1}\{e_t < \tau\}$, then s_t would equal 1 exactly until the first time the threshold is violated, and 0 thereafter. The sum (4.8) would then equal the classical hitting time—the number of consecutive acceptable steps—which is precisely the VPT. EFH retains this meaning but replaces the discontinuous indicator by the smooth gate (4.6), producing a

continuous surrogate that can be optimised and that varies smoothly under finite-sample fluctuations. The construction is directly analogous to the *restricted mean survival time* (RMST) in biostatistics [96], which is likewise defined as the area under a survival curve truncated at a finite time horizon.

What EFH measures in practice. With the normalization convention fixed, τ defines what “acceptable tracking” means for the task. EFH then measures, in units of time steps, how long the model typically stays within that tolerance before it consistently departs. In particular, EFH does *not* try to summarise the saturated regime after divergence; by construction, once g_t becomes small for a sustained period, s_t collapses and contributes negligibly to the sum (see Figures 4.3 and 4.4). This is precisely the behaviour desired for chaotic forecasting benchmarks, where late-horizon errors are dominated by divergence rather than model structure.

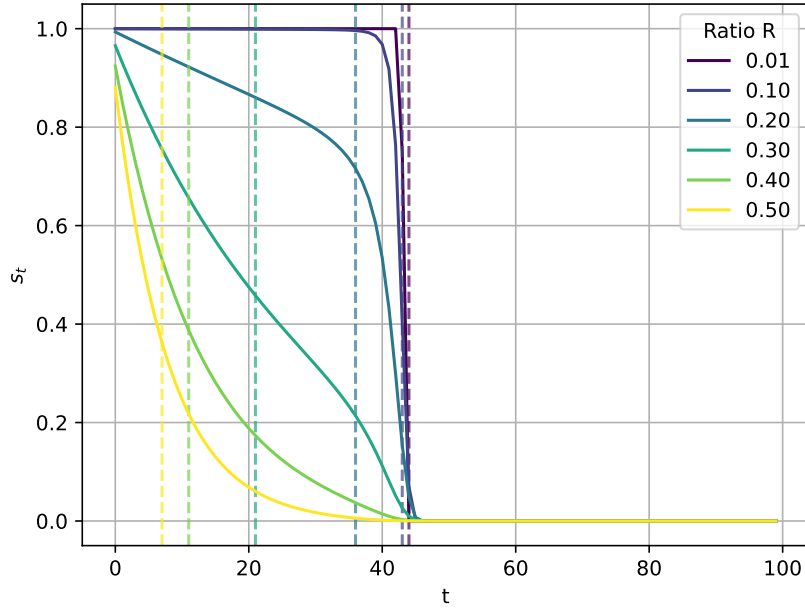


Figure 4.3: Survival process $s_t(\tau, \delta)$ for different softness ratios. The survival curve collapses once the cumulative effect of below-unity gate values accumulates sufficiently. Dashed lines mark the EFH values.

Figure 4.5 summarises the dependence of EFH on R in log scale. The EFH decreases monotonically with R for this error signal (since most of the pre-crossing errors satisfy $e_t < \tau$, so increasing δ reduces most gate values). We observe that EFH is relatively stable for $R < 0.2$ and degrades rapidly for $R \geq 0.2$. This motivates the recommendation of $R \in [0.01, 0.2]$ as the practical operating range.

4.1.6 EFH as a metric and as a differentiable training objective

EFH can be used both for reporting and for learning.

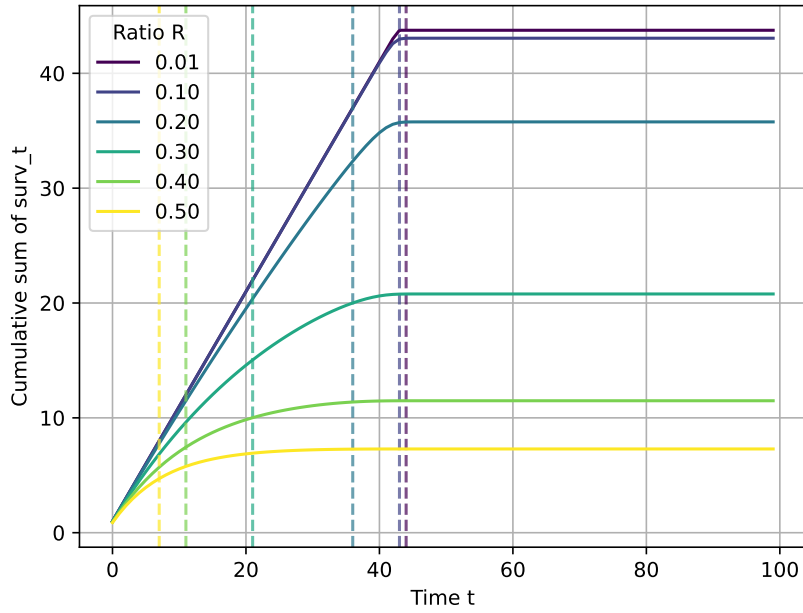


Figure 4.4: Cumulative sum $\sum_{k=1}^t s_k$ as a function of time. The plateau value at $t = T$ equals $\text{EFH}(\tau, \delta)$. Curves are shown for different softness ratios R .

Metric. For evaluation, $\text{EFH}(\tau, \delta)$ is reported either in steps or in physical time units as $\Delta t \text{EFH}(\tau, \delta)$. Because τ is defined in normalised units, the reported EFH values are comparable across datasets and systems under the same normalization protocol.

Loss. For training we maximise EFH, equivalently minimise

$$\mathcal{L}_{\text{EFH}} := -\text{EFH}(\tau, \delta). \quad (4.9)$$

This objective concentrates learning pressure around the *onset of failure*. Time steps for which $e_t \ll \tau$ yield $g_t \approx 1$ and therefore contribute negligibly to the gradient, since the prediction is already well within tolerance. Conversely, time steps far beyond the loss of tracking contribute little because the survival factor s_t has already decayed and suppresses their influence. As a result, the dominant gradient signal arises from the narrow temporal region where the aggregated error approaches and crosses the tolerance τ —a property we formalise in Proposition 4.5 and verify empirically in Figure 4.9.

The practical effect is that optimization focuses on delaying the first sustained threshold crossing, rather than attempting to reduce error in regimes where divergence is either negligible or unavoidable. This behaviour aligns the training objective with the operational meaning of long-horizon performance in chaotic forecasting.

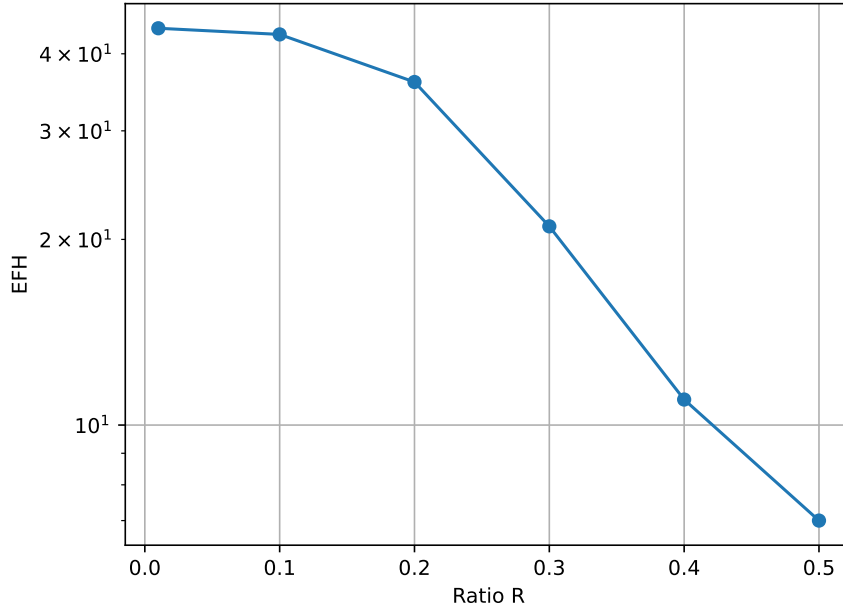


Figure 4.5: EFH as a function of the softness ratio $R = \delta/\tau$ (log scale). The horizon is stable for $R \leq 0.1$ and degrades rapidly for $R \geq 0.3$, motivating the practical recommendation $R \in [0.01, 0.1]$.

4.2 Theoretical properties

This section collects the main structural properties of EFH that are relevant for its use as a benchmarking statistic. The statements below clarify (i) how EFH behaves when the tolerance parameters are changed, (ii) how it relates to the classical notion of a forecast horizon as a hitting time, (iii) why the construction is stable under the normalization convention, and (iv) where the gradient signal concentrates when EFH is used as a training objective.

Recall that EFH is defined from the aggregate error curve e_t (Equation (4.5)) through the gate

$$g_t(\tau, \delta) = \sigma\left(\frac{\tau - e_t}{\delta}\right) \in (0, 1),$$

the survival process

$$s_t(\tau, \delta) = \prod_{k=1}^t g_k(\tau, \delta),$$

and the horizon functional

$$\text{EFH}(\tau, \delta) = \sum_{t=1}^T s_t(\tau, \delta),$$

as introduced in Section 4.1.

4.2.1 Bounds and monotonicity

The first properties are essentially immediate but important in practice: EFH is bounded, and it changes monotonically with the tolerance.

Lemma 4.1 (Bounds). *For any error curve $(e_t)_{t=1}^T$ and any $\tau > 0$, $\delta > 0$,*

$$0 \leq \text{EFH}(\tau, \delta) \leq T. \quad (4.10)$$

Moreover, $\text{EFH}(\tau, \delta)$ is close to T when $e_t \ll \tau$ for all t , and close to 0 when $e_1 \gg \tau$.

Proof. For any $\tau > 0$ and $\delta > 0$, the logistic satisfies $0 < \sigma(z) < 1$ for all $z \in \mathbb{R}$. Hence each $g_t = \sigma((\tau - e_t)/\delta) \in (0, 1)$, so each $s_t = \prod_{k=1}^t g_k \in (0, 1)$, and summing gives $0 < \text{EFH} < T$. In limiting cases ($\delta \rightarrow 0^+$ or $\tau \rightarrow \infty$) the closed bounds $0 \leq \text{EFH} \leq T$ are attained. $\square$

Lemma 4.2 (Monotonicity in τ and δ). *Fix an error curve $(e_t)_{t=1}^T$ and parameters $\tau > 0$, $\delta > 0$.*

1. *For fixed δ , the map $\tau \mapsto \text{EFH}(\tau, \delta)$ is strictly increasing.*
2. *For fixed τ and each fixed t , the map $\delta \mapsto g_t(\tau, \delta)$ is strictly decreasing if $e_t < \tau$, strictly increasing if $e_t > \tau$, and constant $g_t \equiv \frac{1}{2}$ if $e_t = \tau$. Consequently, EFH is not monotone in δ in general.*

Proof. (1) Differentiating the gate with respect to τ gives

$$\frac{\partial g_t}{\partial \tau} = \frac{1}{\delta} g_t(\tau, \delta)(1 - g_t(\tau, \delta)) > 0$$

since $g_t \in (0, 1)$. Products of strictly increasing positive functions preserve strict monotonicity, so s_t is strictly increasing in τ for each t , and summing yields the claim.

(2) Differentiating with respect to δ :

$$\frac{\partial g_t}{\partial \delta} = \frac{e_t - \tau}{\delta^2} g_t(\tau, \delta)(1 - g_t(\tau, \delta)).$$

The sign equals $\text{sign}(e_t - \tau)$. Since a typical error curve has $e_t < \tau$ for early times and $e_t > \tau$ for late times, the product s_t experiences competing effects, and EFH is not monotone in δ . $\square$

These monotonicity statements formalise the basic calibration intuition: increasing τ relaxes the acceptability criterion and increases the reported horizon. Empirical verification of the monotonicity in τ is provided in Figure 4.6.

4.2.2 Limiting cases and the connection to a hitting time

A central design requirement is that EFH should reduce to the conventional valid prediction time (VPT) in an appropriate limit. This is captured by the hard-threshold regime $\delta \rightarrow 0^+$.

Lemma 4.3 (Hard-threshold limit). *Fix $\tau > 0$ and an error curve $(e_t)_{t=1}^T$. Define the first violation index*

$$t^* := \min\{t \in \{1, \dots, T\} : e_t > \tau\}, \quad (4.11)$$

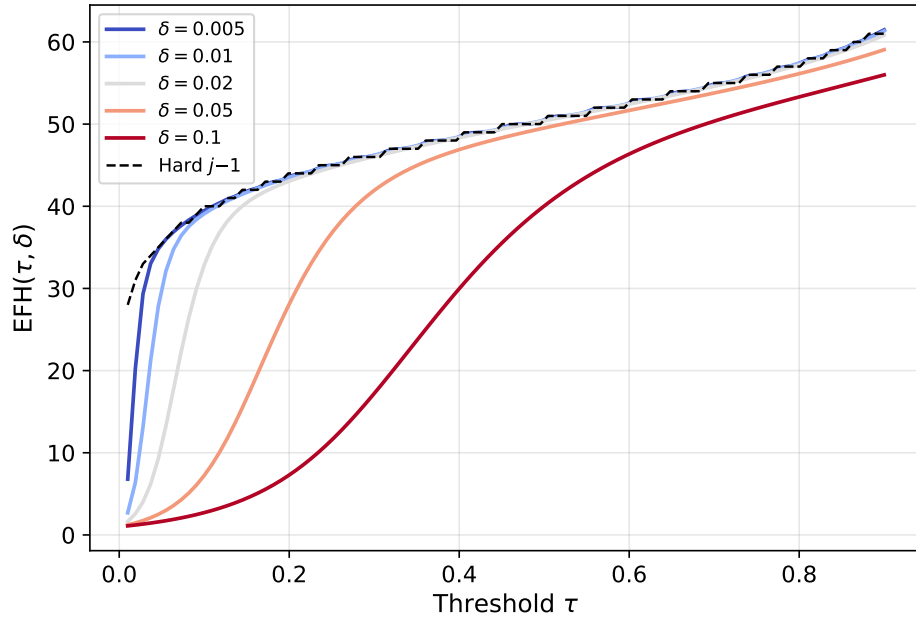


Figure 4.6: Monotonicity of $\text{EFH}(\tau, \delta)$ with respect to τ , for several values of δ . The black dashed curve shows the hard hitting time $t^* - 1$. All EFH curves are monotonically nondecreasing in τ , as guaranteed by Lemma 4.2 (1), and converge to the hard hitting time as δ decreases.

with the convention $t^* = T + 1$ if no such t exists. Assume $e_t \neq \tau$ for all t (the generic case). Then, as $\delta \rightarrow 0^+$,

$$g_t(\tau, \delta) \rightarrow \mathbf{1}\{e_t < \tau\}, \quad s_t(\tau, \delta) \rightarrow \begin{cases} 1, & t < t^*, \\ 0, & t \geq t^*, \end{cases} \quad \text{EFH}(\tau, \delta) \rightarrow t^* - 1.$$

Proof. If $e_t < \tau$, then $(e_t - \tau)/\delta \rightarrow -\infty$ as $\delta \rightarrow 0^+$, so $g_t \rightarrow 1$. If $e_t > \tau$, then $(e_t - \tau)/\delta \rightarrow +\infty$, so $g_t \rightarrow 0$. For $t < t^*$, all factors in s_t tend to 1, so $s_t \rightarrow 1$. For $t \geq t^*$, the product contains $g_{t^*} \rightarrow 0$, so $s_t \rightarrow 0$. Summing gives $\text{EFH} \rightarrow t^* - 1$. $\square$

Thus $\lim_{\delta \rightarrow 0^+} \text{EFH}(\tau, \delta) = \text{VPT}$, confirming that EFH is a smooth generalization of the hard-threshold forecast horizon. The convergence is verified numerically in Figure 4.7.

At the opposite extreme, excessively large softness destroys horizon information.

Lemma 4.4 (Large-softness limit). Fix $\tau > 0$ and an error curve $(e_t)_{t=1}^T$. Then, as $\delta \rightarrow \infty$,

$$g_t(\tau, \delta) \rightarrow \frac{1}{2}, \quad s_t(\tau, \delta) \rightarrow 2^{-t}, \quad \text{EFH}(\tau, \delta) \rightarrow \sum_{t=1}^T 2^{-t} = 1 - 2^{-T} < 1.$$

Proof. As $\delta \rightarrow \infty$, $(e_t - \tau)/\delta \rightarrow 0$ for every t , so $g_t \rightarrow \sigma(0) = 1/2$. Then $s_t \rightarrow 2^{-t}$ and

$$\text{EFH} \rightarrow \sum_{t=1}^T 2^{-t} = 1 - 2^{-T} \xrightarrow{T \rightarrow \infty} 1. \quad \square$$

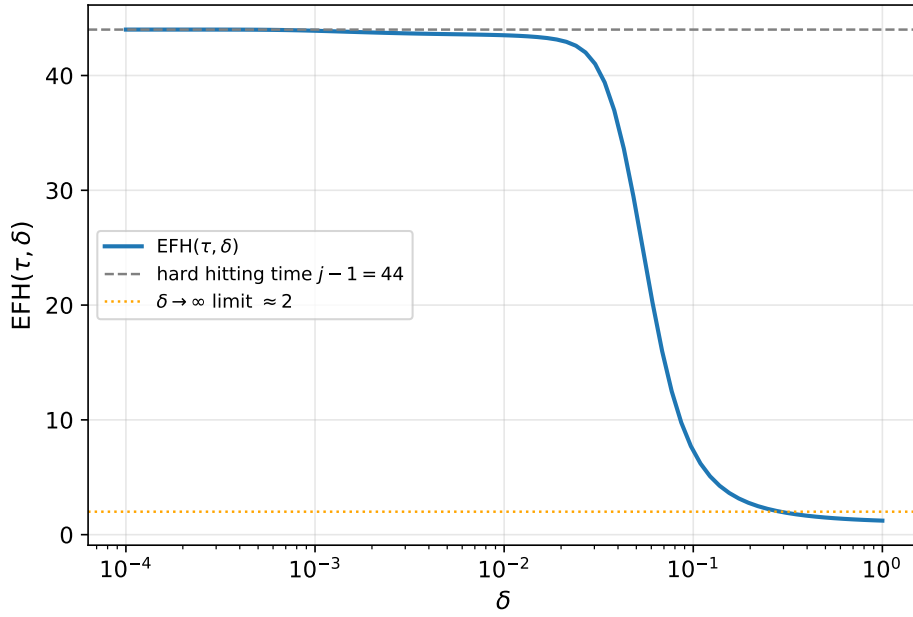


Figure 4.7: Convergence of $\text{EFH}(\tau, \delta)$ to the hard hitting time $t^* - 1$ as $\delta \rightarrow 0^+$. The EFH (blue curve) approaches $t^* - 1 = 39$ for small δ and tends to $1 - 2^{-T} \approx 1$ for large δ , in agreement with Lemmas 4.3 and 4.4.

Remark 2. The two limits provide a useful sanity range: meaningful horizons require δ small enough that the gate transitions near τ , but not so small that the statistic becomes brittle. The ratio $R = \delta/\tau$ (Remark 1) controls this trade-off, with $R \in [0.01, 0.2]$ providing a good operating regime in practice.

4.2.3 Survival and hazard representation

Although EFH was motivated operationally as a horizon, its algebraic form admits a standard interpretation in discrete-time survival analysis [97, 98]. Define the stepwise *hazard* and cumulative hazard by

$$h_t := -\log g_t(\tau, \delta) \geq 0, \quad H_t := \sum_{k=1}^t h_k. \quad (4.12)$$

Then

$$s_t(\tau, \delta) = \prod_{k=1}^t g_k(\tau, \delta) = \exp(-H_t), \quad (4.13)$$

and therefore

$$\text{EFH}(\tau, \delta) = \sum_{t=1}^T \exp(-H_t). \quad (4.14)$$

This representation makes two structural points explicit.

(i) EFH emphasises prefix reliability. Because H_t is cumulative, once the gate becomes small for a sustained period, H_t grows and the survival $\exp(-H_t)$ collapses rapidly. Late-time behaviour after persistent failure therefore has negligible impact on EFH, which is

the intended behaviour. Figure 4.8 visualises h_t , H_t , and s_t for the reference error signal, confirming the rapid collapse of the survival process once the hazard accumulates.

(ii) **EFH is a soft first-passage functional.** In the hard limit $\delta \rightarrow 0^+$, each h_t becomes either 0 (acceptable) or $+\infty$ (unacceptable), so H_t jumps to $+\infty$ at the first threshold violation and Eq. (4.14) reduces to $t^* - 1$ as in Lemma 4.3.

(iii) **Connection to restricted mean survival time.** In survival analysis, the *restricted mean survival time* (RMST) is defined as the area under the survival curve $S(t)$ up to a truncation time T [96]: $\text{RMST}(T) = \int_0^T S(t) dt$. In our discrete-time setting, $\text{EFH} = \sum_{t=1}^T s_t$ is exactly the discrete analogue. This connection is not merely formal: it means that the extensive statistical literature on RMST estimation, comparison, and inference [96, 98] can inform the design and analysis of EFH-based benchmarks.

Figure 4.8 displays the stepwise hazard $h_t = -\log g_t$, the cumulative hazard $H_t = \sum_{k=1}^t h_k$, and the survival $s_t = \exp(-H_t)$ for the reference error signal. The hazard is negligible during the tracking phase (where $g_t \approx 1$), rises sharply near the threshold crossing, and plateaus at a high level in the divergence phase. The cumulative hazard consequently undergoes a transition from slow linear growth to rapid accumulation, which drives the exponential collapse of s_t . This visualization confirms the survival-analysis interpretation of Section 4.2.3 and provides intuition for why EFH is insensitive to late-horizon behaviour.

4.2.4 Gradient concentration at the failure onset

A key practical property of EFH as a training loss is that its gradients concentrate near the threshold-crossing time, rather than being spread uniformly across the prediction horizon. We now formalise this observation.

Proposition 4.5 (Gradient concentration). *Let $\text{EFH}(\tau, \delta) = \sum_{t=1}^T s_t$ be defined as above, with $s_t = \prod_{k=1}^t g_k$. For any time step $m \in \{1, \dots, T\}$, the derivative of EFH with respect to the error e_m satisfies*

$$\frac{\partial \text{EFH}}{\partial e_m} = -\frac{g_m(1 - g_m)}{\delta} \sum_{t=m}^T \frac{s_t}{g_m}, \quad (4.15)$$

which can be rewritten as

$$\frac{\partial \text{EFH}}{\partial e_m} = -\frac{g_m(1 - g_m)}{\delta} \cdot s_{m-1} \cdot \sum_{t=m}^T \prod_{k=m+1}^t g_k, \quad (4.16)$$

where $s_0 := 1$. In particular, $\partial \text{EFH} / \partial e_m$ is negligible whenever:

1. $e_m \ll \tau$ (then $g_m \approx 1$, so $g_m(1 - g_m) \approx 0$), or
2. $e_m \gg \tau$ (then $g_m \approx 0$, so both $g_m(1 - g_m)$ and $s_{m-1} \cdot (\dots)$ vanish), or
3. the prediction has already failed before step m (then $s_{m-1} \approx 0$).

The gradient magnitude is maximal in the neighbourhood of the first sustained threshold crossing.

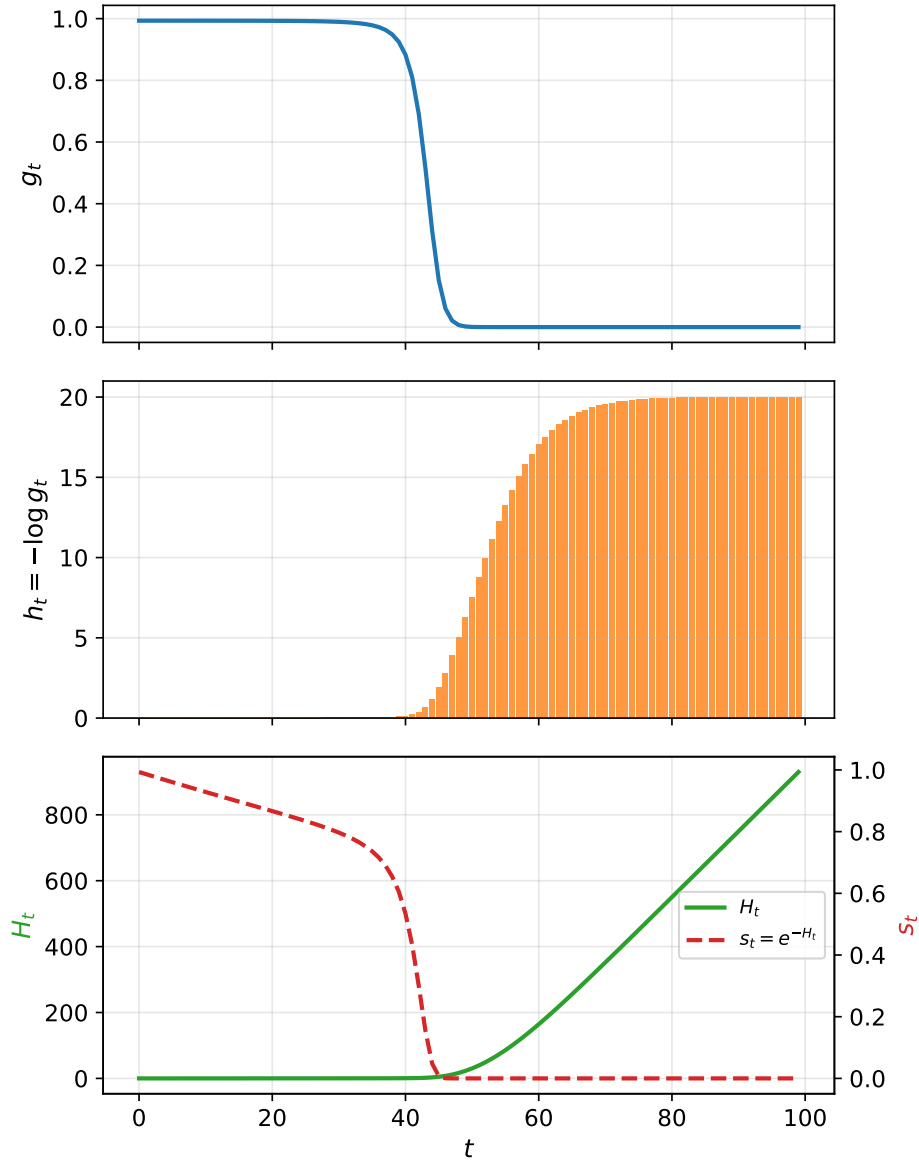


Figure 4.8: Survival-analysis decomposition of EFH for the reference error signal ($\tau = 0.2$, $R = 0.1$). **Top:** Gate g_t . **Middle:** Stepwise hazard $h_t = -\log g_t$. **Bottom:** Cumulative hazard H_t (green, left axis) and survival $s_t = \exp(-H_t)$ (red dashed, right axis). The rapid accumulation of hazard near the threshold crossing drives the exponential collapse of s_t .

Proof. Since $s_t = \prod_{k=1}^t g_k$ and each g_k depends on e_k alone, $\partial s_t / \partial e_m = 0$ for $t < m$. For $t \geq m$,

$$\frac{\partial s_t}{\partial e_m} = s_t \cdot \frac{1}{g_m} \frac{\partial g_m}{\partial e_m} = -\frac{s_t}{\delta} \cdot \frac{g_m(1 - g_m)}{g_m} = -\frac{s_t(1 - g_m)}{\delta}.$$

Summing over t gives (4.15). For the factored form, note $s_t/g_m = s_{m-1} \prod_{k=m+1}^t g_k$ when $t \geq m$. The three vanishing regimes follow directly from the factor structure. $\square$

This gradient concentration property means that optimization naturally focuses computational effort on the critical temporal window where the forecast transitions from “tracking” to “lost”. Steps deep within the tracking phase or deep within the divergence phase produce negligible gradients. Figure 4.9 verifies this concentration numerically for the reference sigmoidal error signal.

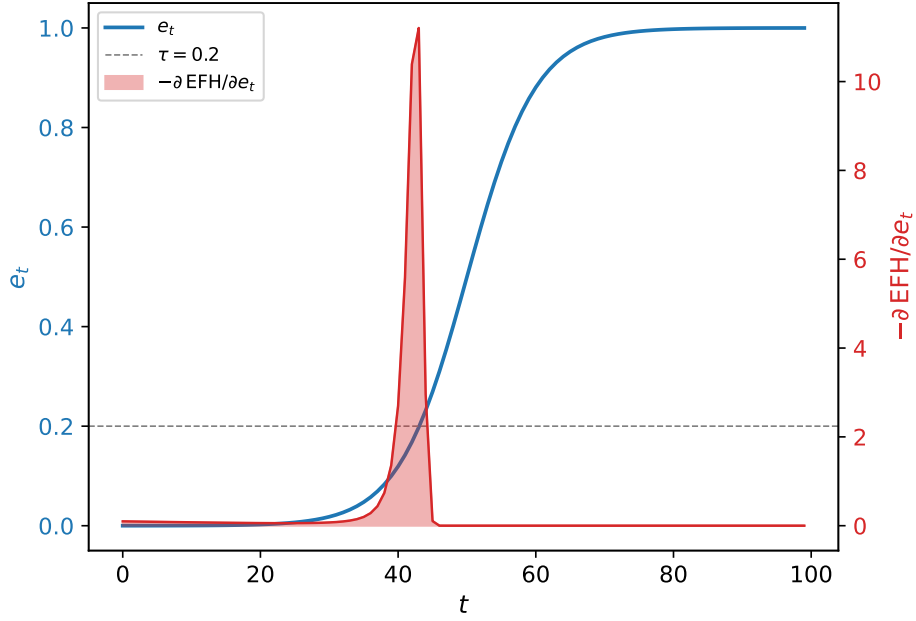


Figure 4.9: Gradient concentration of the EFH loss. The blue curve shows the error signal e_t ; the red shaded region shows $-\partial \text{EFH} / \partial e_t$ (right axis). The gradient is sharply localised near the threshold crossing ($t \approx 39$), confirming Proposition 4.5: optimization pressure concentrates at the onset of failure.

4.2.5 Stability under the normalization convention

EFH depends on the numerical scale of the error through the tolerance parameters (τ, δ) . This dependence is not a weakness; it is unavoidable for any horizon definition that uses an absolute tolerance. What matters is that the tolerance retains consistent meaning across experiments, which is ensured by the task-level normalization introduced in Section 4.1.1.

To make this explicit, consider a uniform rescaling of the state coordinates by a factor $a > 0$. Under RMSE, $\varepsilon_{i,t} \mapsto a \varepsilon_{i,t}$ and therefore $e_t \mapsto a e_t$. If one simultaneously rescales

$$\tau \mapsto a\tau, \quad \delta \mapsto a\delta, \quad (4.17)$$

then the gate arguments $(\tau - e_t)/\delta$ are unchanged, and EFH is invariant.

Lemma 4.6 (Scale equivariance). *Under the simultaneous rescaling $e_t \mapsto a e_t$, $\tau \mapsto a\tau$, $\delta \mapsto a\delta$ for $a > 0$, EFH is invariant: $\text{EFH}(a\tau, a\delta) = \text{EFH}(\tau, \delta)$.*

Proof. The geometric mean rescales as $e'_t = a e_t$. The gate becomes

$$g'_t = \sigma\left(\frac{a\tau - a e_t}{a\delta}\right) = \sigma\left(\frac{\tau - e_t}{\delta}\right) = g_t.$$

Hence $s'_t = s_t$ for all t , and $\text{EFH}' = \text{EFH}$. $\square$

In other words, EFH depends only on the relative position of e_t with respect to the tolerance, provided the tolerance is expressed in the same units as the error. The dimensionless ratio $R = \delta/\tau$ (Remark 1) is itself invariant under this rescaling, confirming that R is the correct parameter for sensitivity analysis. When cross-system comparisons require an intrinsic

timescale, EFH is optionally reported in Lyapunov-time units $\text{EFH} \cdot \Delta t \cdot \lambda_{\max}$; this does not change the definition, only the unit in which the horizon is expressed.

4.2.6 Robustness to noise

A practical concern with hard hitting times is their sensitivity to noise: a single noisy spike in the error signal can shift the reported VPT by many time steps. EFH, by contrast, is inherently smoothed through the sigmoid gate and the cumulative survival product.

To quantify this, we add i.i.d. Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the reference error signal and compute both EFH and the hard hitting time $t^* - 1$ over 500 independent realizations for each noise level $\sigma \in \{0, 0.007, \dots, 0.08\}$ (Figure 4.10). The coefficient of variation (CV = std/mean) of EFH remains smaller than that of the hard hitting time across all noise levels, confirming that EFH is a more stable statistic under perturbation. This stability is particularly valuable in practice, where the error signal is inherently noisy due to finite evaluation batches and the stochastic nature of reservoir initialization.

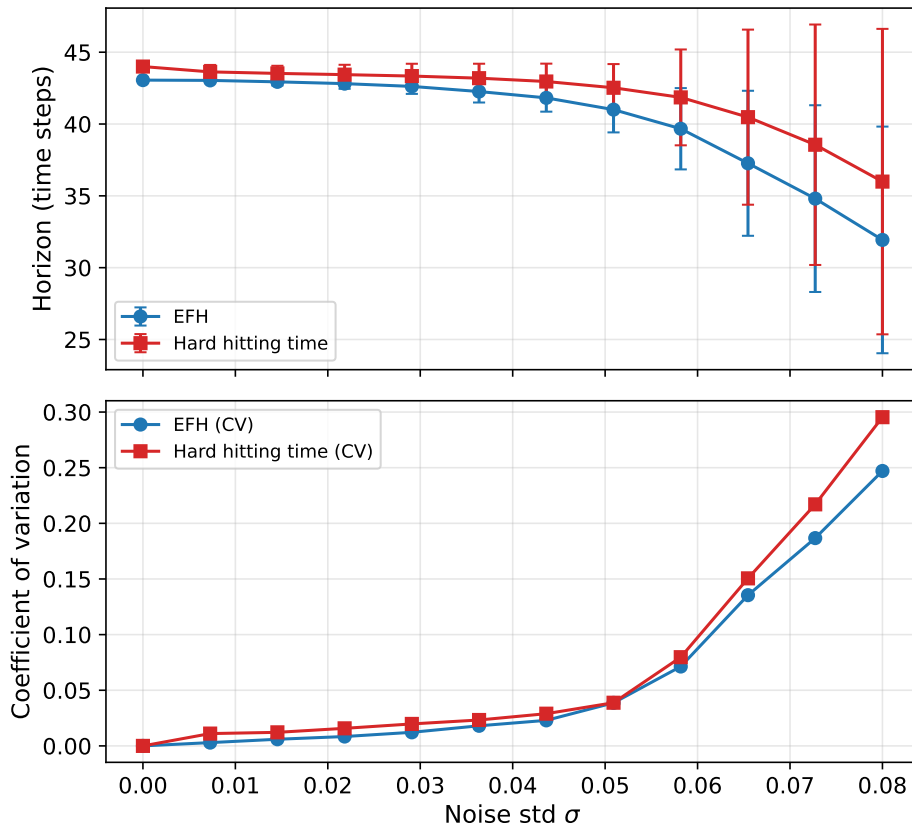


Figure 4.10: Top: Mean $\pm$ standard deviation of EFH and hard hitting time over 500 noise realizations at each noise level σ . Bottom: Coefficient of variation (CV). EFH maintains a substantially lower CV than the hard hitting time across all noise levels, demonstrating its robustness as a benchmarking statistic.

4.2.7 Behaviour across different error profiles

The synthetic sigmoidal error signal, while representative, is only one possible divergence pattern. To verify that EFH produces sensible horizons across qualitatively different error profiles, we evaluate it on four prototypical shapes (Figure 4.11):

1. **Exponential:** $e_t = \min(0.005 e^{0.08t}, 1)$, mimicking pure Lyapunov divergence.
2. **Linear:** $e_t = \min(t/80, 1)$, representing a slow drift.
3. **Sigmoidal:** the reference signal (4.4).
4. **Step-like:** $e_t = \frac{1}{2}(1 + \tanh(0.4(t - 40)))$, approximating a sudden failure.

In each case, EFH assigns a horizon that is close to, but slightly larger than, the hard hitting time $t^* - 1$, reflecting the soft gate's tolerance for marginal exceedances. The survival curve s_t (shaded region) confirms that EFH captures the “effective tracking duration” regardless of the shape of the transition from tracking to divergence.

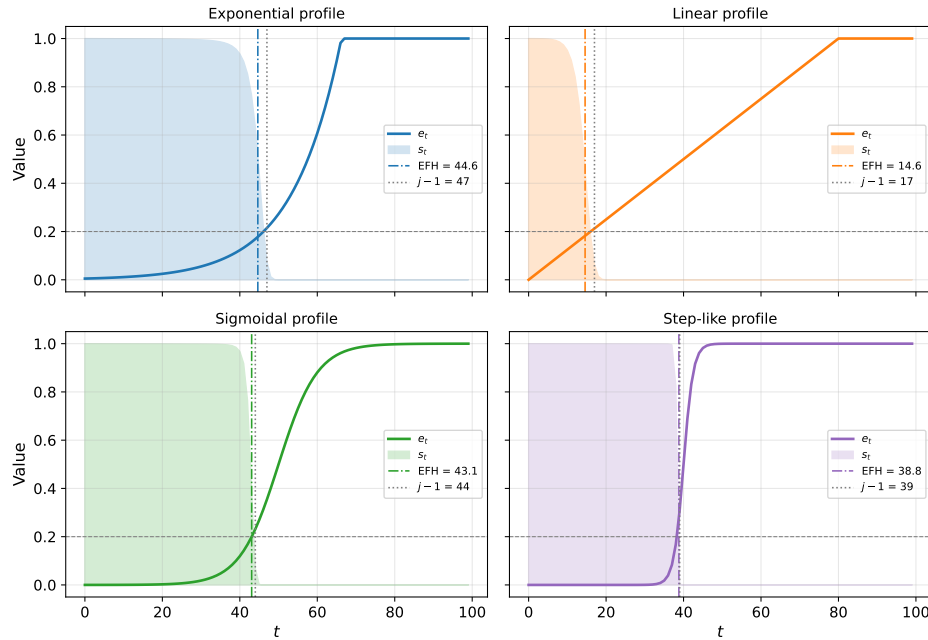


Figure 4.11: EFH evaluated on four qualitatively different error profiles: exponential growth, linear ramp, sigmoidal transition, and step-like failure. In each panel, the coloured curve is e_t , the shaded region shows s_t , the dot-dashed line marks the EFH, and the dotted line marks the hard hitting time $t^* - 1$. EFH produces sensible horizons across all divergence patterns.

4.3 Conclusions

This chapter introduced the Expected Forecast Horizon as a continuous, differentiable measure of predictive skill for chaotic time series forecasting. The construction replaces the hard threshold crossing of the classical valid prediction time with a soft acceptability gate and a cumulative survival process, yielding a functional that is bounded, monotone in the tolerance τ , scale-equivariant, and formally equivalent to a restricted mean survival time in discrete-time survival analysis.

The theoretical analysis established three properties that make EFH suitable both as a benchmarking statistic and as a training objective. First, EFH recovers the hard hitting time in the limit $\delta \rightarrow 0^+$ (Lemma 4.3), so it generalises rather than replaces the standard notion of forecast horizon. Second, its gradients concentrate at the onset of failure (Proposition 4.5), directing optimization pressure precisely where it matters: the temporal region where tracking is lost. Third, the soft survival product confers robustness to noise that the hard hitting time lacks, as demonstrated by the substantially lower coefficient of variation under perturbation (Figure 4.10).

The sensitivity analysis identified $R = \delta/\tau \in [0.01, 0.1]$ as the practical operating range in which EFH is stable and closely approximates the hard horizon while retaining differentiability. Outside this range, EFH either degenerates to the brittle hard threshold ($R \rightarrow 0$) or collapses to uninformative values near unity ($R \rightarrow \infty$). The evaluation across multiple error profiles confirmed that EFH produces sensible horizons regardless of the qualitative shape of the tracking-to-divergence transition.

The EFH loss $\mathcal{L}_{\text{EFH}} = -\text{EFH}(\tau, \delta)$ serves as the objective function for the Bayesian hyperparameter optimization of Echo State Networks developed in Chapter 5. Because EFH directly measures the quantity of interest— how long a model tracks the reference trajectory— optimising it aligns the search with the operational goal of chaotic forecasting, avoiding the pitfalls of fixed-horizon pointwise losses discussed at the opening of this chapter.

5 Hyperparameter Optimization

The preceding chapters developed three independent methodological contributions: a quantitative understanding of transient times and their dependence on reservoir topology (Chapter 2), a geometry-based discretization protocol that normalises the temporal resolution across dynamical systems (Chapter 3), and a smooth, differentiable forecast evaluation metric that replaces the brittle valid prediction time (Chapter 4). Each was motivated by a specific gap in the Echo State Network literature, and each was presented as a self-contained tool. This chapter brings the three together in the setting for which they were ultimately designed: systematic hyperparameter optimization (HPO) of Echo State Networks for chaotic forecasting.

The central claim of this chapter is not that HPO discovers dramatically superior ESN configurations—indeed, the optimization landscapes exhibit rapid saturation, with performance plateauing well before the trial budget is exhausted. Rather, the contribution is threefold. First, the integration of sagitta preprocessing, dendrocycle topology, and EFH-based evaluation into a single automated pipeline demonstrates that the tools of Chapters 2–4 compose naturally into a principled workflow. Second, the marginal relationships between hyperparameters and forecast performance, revealed by analysing hundreds of trials, expose the structure of the ESN design space in a way that isolated experiments cannot. Third, and most importantly, the Tree Parzen Estimator (TPE) [26] sampler—searching freely over the dendrocycle fractions c and d —converges organically to topologies satisfying $c + d \approx 1$, thereby rediscovering the design principle of Chapter 2 without any structural constraint being imposed.

5.1 Experimental setup

The hyperparameter optimization pipeline consists of four stages, each grounded in the developments of a preceding chapter. The benchmarking process is summarized in the following steps:

Stage 1: Preprocessing. Each dynamical system is integrated from a random initial condition using a fourth-order Runge-Kutta scheme at a fine base step Δt_0 , typically by the order of $1e-4$, chosen small enough to lie well within the quadratic scaling regime of the sagitta (cf. Section 3.3). The trajectory is then decimated to the step Δt^* selected by the absolute sagitta rule with tolerance $\varepsilon = 0.1$ and 95th-percentile aggregation, following the protocol established in Chapter 3. Then the system is integrated once more using an adaptive step size integrator with relative tolerance and absolute tolerance set to $1e-6$ and $1e-9$ respectively, and afterwards it is sampled at Δt^* using interpolation. Per-coordinate standardization is applied using the training-set statistics, so that the EFH threshold τ has a consistent interpretation across systems. For the Hénon map, which is already discrete, the sagitta step reduces to the native iteration step $\Delta t^* = 1$.

Stage 2: Reservoir construction and training. For each candidate hyperparameter vector, a dendrocycle reservoir is constructed with the specified core fraction c , dendritic fraction d , and their associated weights w_c and w_d controlling the strength of core and dendritic connections respectively. The reservoir is driven by the training segment in open-loop mode, with the first T_0 steps discarded as transient washout. Following the recommendations of Chapter 2, T_0 is set conservatively based on the analytical estimate (2.4) for $\rho < 1$, and extended by a safety factor of 2 for $\rho \geq 1$. The readout weights are computed via ridge regression with Tikhonov parameter β .

Stage 3: Evaluation. The trained ESN is switched to closed-loop mode and generates an autonomous forecast over a fixed horizon of T steps. The normalized pointwise NRMSE is computed against the ground truth, and the EFH is computed with threshold $\tau = 0.2$ and softness ratio $R = \delta/\tau = 0.2$, placing the evaluation firmly within the stable operating range identified in Section 4.2. The resulting EFH is expressed in Lyapunov times, $\text{EFH}_\lambda = \text{EFH} \cdot \Delta t^* \cdot \lambda_{\max}$, providing a system-independent measure of forecast quality. This score is averaged over 20 independently drawn test trajectories from the same system, sharing identical parameters but differing initial conditions, so that the reported EFH reflects typical forecasting performance rather than a single realization.

Stage 4: Optimization. The Optuna framework [29] with the tree-structured Parzen estimator (TPE) [26] serves as the black-box optimiser. TPE fits kernel density estimators to the top and bottom quantiles of observed objectives and proposes configurations that maximise the expected improvement. The objective is $\mathcal{L} = -\text{EFH}_\lambda$, so that minimization corresponds to maximising the forecast horizon. Each study runs for 500 trials with a single random seed per trial. The single-seed protocol is justified by the reduced variance of EFH relative to the hard VPT (cf. Figure 4.10), and by the TPE’s inherent ability to smooth over per-trial noise through density estimation across hundreds of evaluations.

The random seed is fixed identically across all trials; this does not imply that the reservoir matrix is constant—topological parameters such as c , d , w_c , and w_d still alter the connectivity structure—but it ensures that stochasticity is not an independent source of variance, so that performance differences between trials reflect hyperparameter effects rather than initialization noise.

The benchmark suite comprises one discrete map and three continuous flows, chosen to span a range of dynamical complexities and Lyapunov timescales. The Hénon map [41] serves a dual purpose: as a discrete system it requires no temporal discretization, providing a clean test of whether the TPE sampler converges reliably in the absence of the sagitta preprocessing step. The three continuous systems—Lorenz [38], Rössler [39], and Chen [42]—are integrated with a fourth-order Runge-Kutta scheme at a fine base step and subsequently decimated to the sampling interval Δt^* selected by the sagitta rule of Chapter 3 with tolerance $\varepsilon = 0.1$ and 95th-percentile aggregation. All systems are used at their canonical parameter values.

The Hénon map is defined by

$$x_{n+1} = 1 - a x_n^2 + y_n, \quad y_{n+1} = b x_n, \quad (5.1)$$

with $a = 1.4$, $b = 0.3$, yielding a maximal Lyapunov exponent $\lambda_{\max} \approx 0.42$.

The Lorenz system,

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho_L - z) - y, \quad \dot{z} = xy - \beta z, \quad (5.2)$$

with $\sigma = 10$, $\rho_L = 28$, $\beta = 8/3$, has $\lambda_{\max} \approx 0.91$.

The Rössler system,

$$\dot{x} = -y - z, \quad \dot{y} = x + ay, \quad \dot{z} = b + z(x - c), \quad (5.3)$$

with $a = 0.2$, $b = 0.2$, $c = 5.7$, has $\lambda_{\max} \approx 0.07$.

The Chen system,

$$\dot{x} = a(y - x), \quad \dot{y} = (c - a)x - xz + cy, \quad \dot{z} = xy - bz, \quad (5.4)$$

with $a = 35$, $b = 3$, $c = 28$, has $\lambda_{\max} \approx 2.03$.

These four exponents span nearly two orders of magnitude, ensuring that any systematic relationship between optimization behaviour and predictability timescale becomes visible.

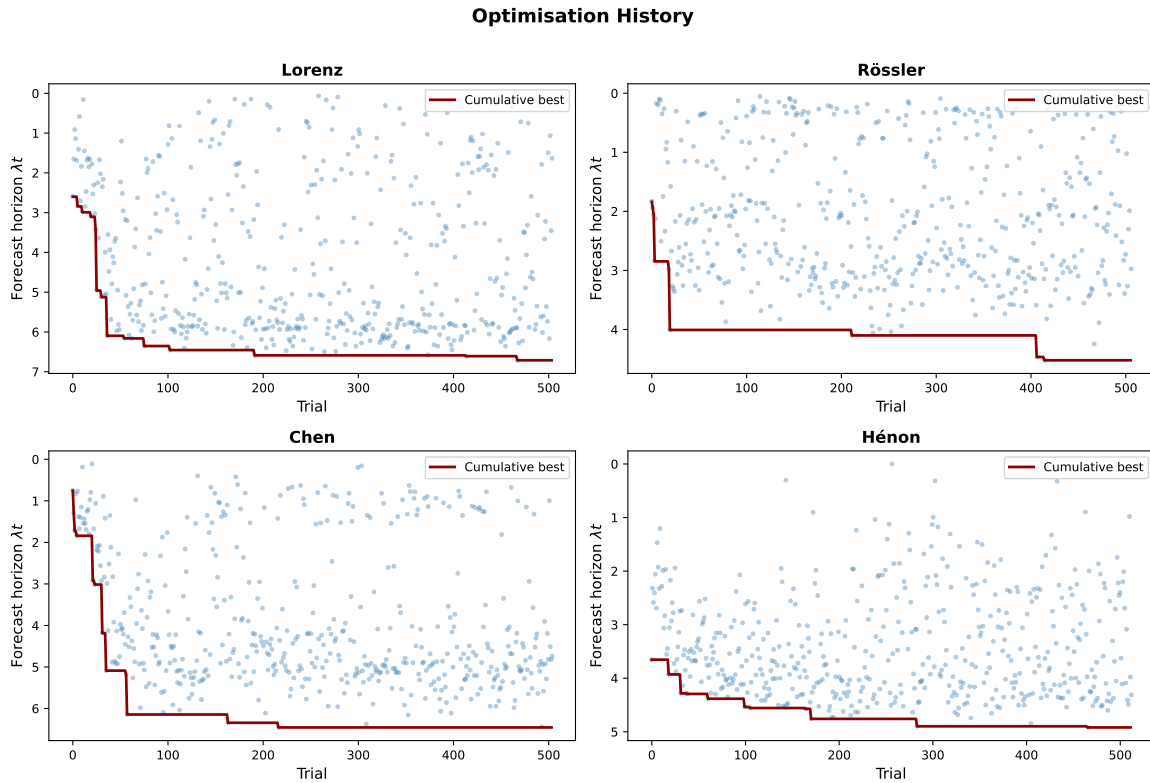
Search space. Table 5.1 specifies the hyperparameter ranges and sampling scales. The ranges were chosen to be deliberately broad, encompassing regimes where the ESP may not hold ($\rho > 1$) and where the reservoir is either very small or very large. This ensures that the optimiser explores the full landscape rather than being confined to a narrow neighbourhood of known good configurations. The spectral radius is sampled on a log scale as it influences the ESN speed dynamics in an exponential way. The input scaling is sampled on a log scale because of its multiplicative nature. The ridge parameter is sampled on a log scale because of its span over several orders of magnitude. The number of neurons of the reservoirs are fixed to $N = 300$.

Table 5.1: Hyperparameter search space for the TPE optimization. Log-scale sampling is used for parameters spanning multiple orders of magnitude.

Hyperparameter	Symbol	Range	Scale
Spectral radius	ρ	[0.5, 1.5]	log
Input scaling	σ_{in}	[0.1, 10.0]	log
Ridge parameter	β	$[10^{-8}, 10^{-2}]$	log
Leak rate	α	[0.1, 1.0]	uniform
Core fraction	c	[0.0, 1.0]	uniform
Dendritic fraction	d	[0.0, $1 - c$]	uniform
Core weight	w_c	[0.0, 2.0]	uniform
Dendritic weight	w_d	[0.0, 2.0]	uniform

5.2 Optimization landscape and convergence

Figure 5.1 displays the optimization history for all four systems. Each panel shows the EFH in Lyapunov times for every trial (scatter) together with the cumulative best (step curve).

**Figure 5.1:** Optimization history for the four benchmark systems. Each point represents a single TPE trial; the step curve tracks the cumulative best EFH in Lyapunov times. All four systems exhibit rapid initial improvement followed by saturation within the first 100–150 trials.

The most prominent feature shared by all four panels is the rapid initial descent of the cumulative best curve: in every system, the majority of the performance gain is captured within the first 50–100 trials, after which the step curve flattens into a plateau punctuated by increasingly rare and marginal improvements. This behaviour is consistent with a search space whose effective dimensionality is low— only a few hyperparameters sharply

discriminate good from poor configurations, a structure that the fANOVA analysis of the next section will confirm.

The absolute forecast horizons attained at convergence differ across systems. The Lorenz system reaches approximately $6.5\text{--}7\lambda t$, the longest normalised horizon among the four benchmarks. The Chen system, despite its substantially larger Lyapunov exponent, achieves a comparable $\sim 6\lambda t$, indicating that the ESN exploits the sagitta-adapted discretization effectively even for fast dynamics. The Hénon map saturates near $5\lambda t$, while the Rössler system plateaus around $4\lambda t$. The relatively shorter Rössler horizon in Lyapunov units is consistent with the system’s low Lyapunov exponent: a small $\lambda_{\max}$ means that each Lyapunov time encompasses many discrete steps, so even a long forecast in absolute terms appears modest once normalised.

The trial-to-trial scatter remains broad throughout the full 500-trial budget in all four systems. This persistent spread does not indicate a failure of the TPE to learn the landscape; rather, it reflects Optuna’s deliberate exploration-exploitation trade-off, whereby a fraction of trials continue to sample from the prior to avoid premature convergence to a local basin. The scatter is densest in the poorly performing region (short horizons), confirming that large portions of the search space yield inadequate reservoirs— precisely the regime dominated by quiescent-heavy topologies and mismatched spectral radii.

Two systems exhibit notable late improvements. The Rössler cumulative best shows a distinct step near trial 400, suggesting that the sampler discovered a qualitatively different configuration after prolonged exploration— possibly a basin with an atypical balance of spectral radius and leak rate suited to the slow Rössler dynamics. The Hénon map displays a similar late step past trial 450, although significantly less pronounced. These isolated jumps do not alter the overall conclusion that the landscape saturates early, but they illustrate that occasional exploration beyond the dominant basin can yield incremental gains, particularly for systems where the optimal operating point occupies a narrow region of the search space.

5.3 Hyperparameter importance

To quantify which hyperparameters most strongly influence the forecast horizon, we apply the functional ANOVA (fANOVA) decomposition [99] as implemented in Optuna. The fANOVA partitions the total variance of the objective into additive contributions from individual hyperparameters and their interactions, providing a model-agnostic importance ranking.

Figure 5.2 reveals that the importance hierarchy is far from universal: the dominant hyperparameter changes from system to system, reflecting how the dynamical regime of the target shapes the sensitivity of the reservoir. Despite this variability, several structural patterns emerge.

The ridge regularization parameter is negligible in every system, accounting for less than 4% of the variance in all four cases. This confirms that once the reservoir’s operating point is determined by the spectral radius, input scaling, and topology, the precise value of the Tikhonov penalty plays a secondary role— consistent with the observation that ridge

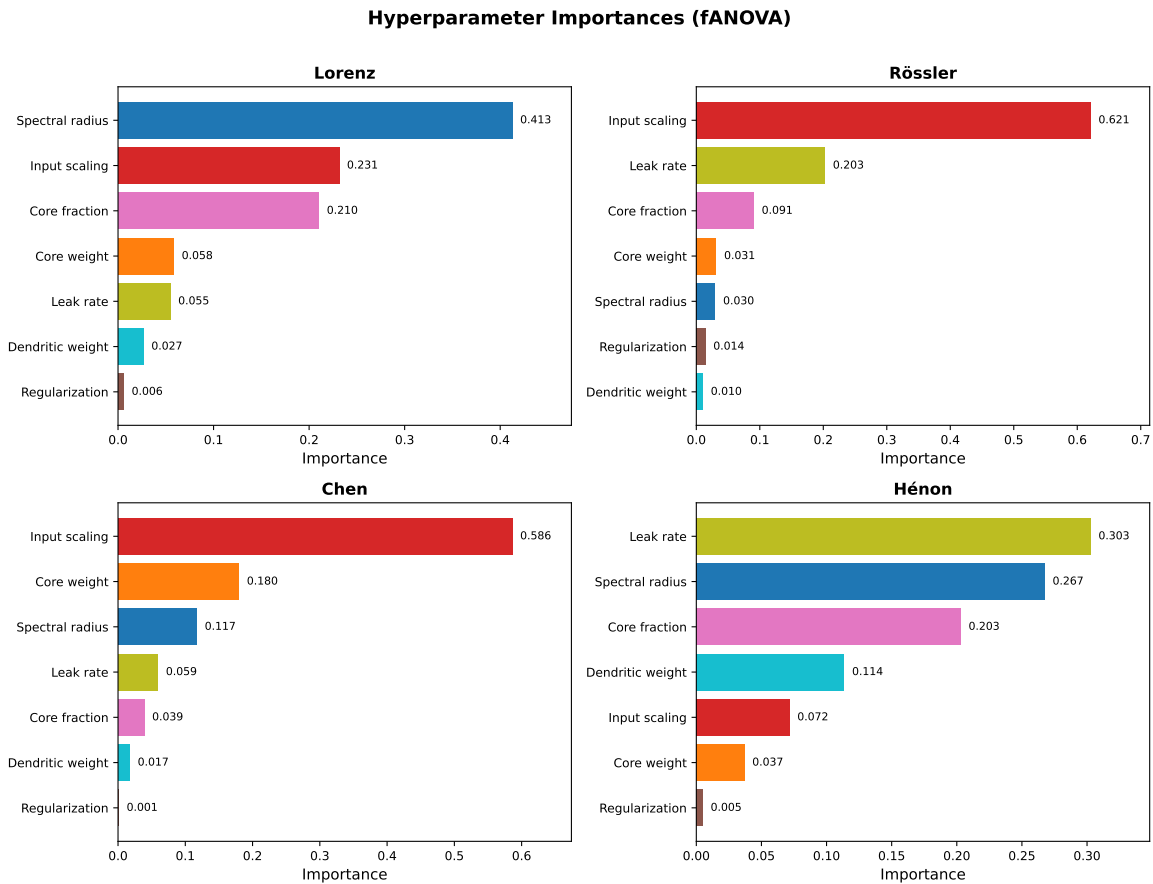


Figure 5.2: Relative hyperparameter importance computed via the fANOVA decomposition for each benchmark system. The ranking varies across systems, reflecting differences in the dynamical regime and the sensitivity of the ESN to each design choice.

regression is primarily a numerical stabiliser rather than a performance-critical design choice.

Input scaling ranks among the two most important hyperparameters in three of the four systems (Rössler, Chen, and Lorenz), and its overall prominence across the benchmark suite is consistent with its well-established role in the ESN literature [24]: it controls the degree to which the driving signal pushes the reservoir’s activation functions into their saturating regime, directly modulating the effective nonlinearity of the state mapping and influencing the satisfaction of the echo state property, as mentioned in Section 2.5.

The input scaling dominates overwhelmingly in both the Rössler (0.65) and Chen (0.51) systems, but we hypothesise that it does so for distinct reasons rooted in the character of each attractor. The Rössler dynamics consist of slow spiralling motion in the x - y plane, punctuated by intermittent excursions in the z variable that produce the attractor’s characteristic folding. With $\lambda_{\max} \approx 0.07$, the dynamics change slowly, and the informationally rich events—the z -excursions that distinguish one orbit from the next—involve relatively small amplitude variations. The input scaling acts as an amplifier for these subtle perturbations: too small a value renders the folding events invisible to the reservoir, while too large a value saturates the activation functions on the dominant spiralling component, drowning out the very features that carry predictive information. The Chen system, by contrast, has $\lambda_{\max} \approx 2.03$ and exhibits fast, tightly wound dynamics with rapid stretching and folding throughout the

attractor. Here the input scaling becomes critical for a different reason: the high Lyapunov exponent demands that the reservoir track a signal whose informational content per time step is large, requiring precise calibration of the nonlinear operating point to balance signal fidelity against activation saturation. In both cases the input scaling gates the information flow into the reservoir, but the Rössler sensitivity is driven by signal subtlety whereas the Chen sensitivity is driven by signal intensity.

For the Lorenz system, the spectral radius emerges as the single most important parameter (0.44), followed by the core fraction and input scaling. The Lorenz attractor is characterised by two distinct lobes connected by rapid switching transitions— the regions of highest curvature identified by the sagitta analysis of Chapter 3. Tracking these transitions requires the reservoir to maintain coherent internal dynamics across the switching timescale, and the spectral radius governs precisely this: it sets the effective memory length and determines how close the reservoir operates to the edge of stability where its dynamical range is maximal. A spectral radius that is too small contracts the internal state too aggressively, erasing the memory of the pre-switch trajectory; one that is too large destabilises the autonomous dynamics and prolongs the transient, as established in Chapter 2. The Lorenz system, with its moderate Lyapunov exponent and topologically complex attractor, appears to occupy a regime where this balance is particularly delicate.

The Hénon map presents a qualitatively different importance profile: the leak rate (0.29), spectral radius (0.26), and core fraction (0.25) share the top rank nearly equally. The prominence of the leak rate is a direct consequence of the system being a discrete map. The leaky integrator update (Eq. 1.11) blends the current activation with the previous state at rate $(1 - \alpha)$; for a map that updates fully at each iteration, retaining a large fraction of the previous internal state artificially slows the reservoir’s response and introduces a timescale mismatch that has no continuous-time analogue to compensate for. Small values of α are therefore actively detrimental, and the optimizer must learn to avoid them— a constraint that elevates the leak rate to a performance-critical parameter. We will see in Section 5.4 that the TPE sampler discovers this organically, concentrating its best trials near $\alpha \approx 1$.

A notable structural pattern is the role of the core fraction c . It appears among the three most important hyperparameters in three of the four systems: second in Lorenz (0.21), third in both Rössler (0.08) and Hénon (0.25). The exception is the Chen system, where the core fraction accounts for only 3.5% of the variance and ranks sixth. We defer a detailed interpretation of this anomaly to Section 5.5, where the joint scatter of c and d will suggest that for the Chen system, the *sum* $c + d$ matters as in the other systems, but the individual partition between core and dendritic nodes is more fungible, distributing the importance across both fractions and the associated weight parameters rather than concentrating it on c alone.

5.4 Marginal effects of individual hyperparameters

The fANOVA importances identify *which* hyperparameters matter; the marginal scatter plots reveal *how* they influence performance. Figures 5.3–5.7 display the EFH in Lyapunov times against six of the search dimensions, with all four systems shown side by side for direct comparison.

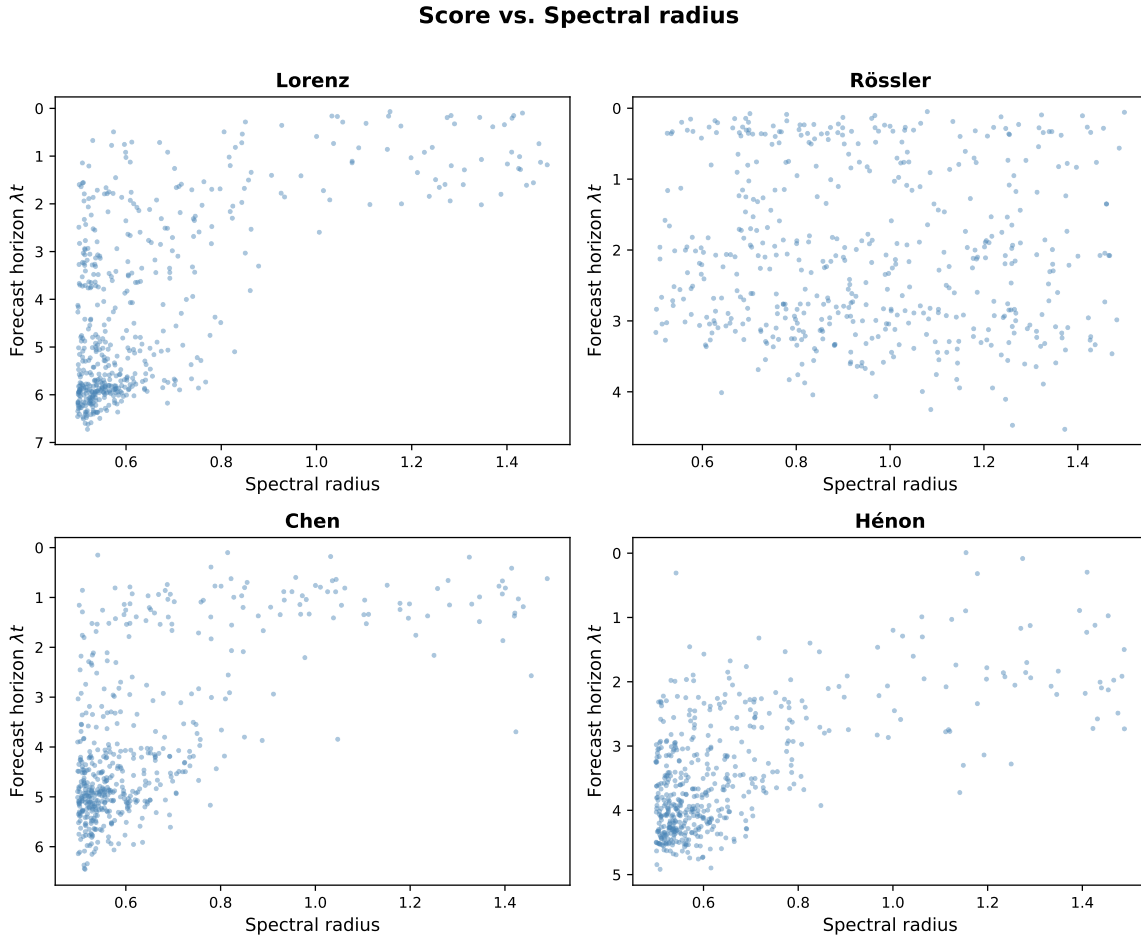


Figure 5.3: Forecast horizon versus spectral radius ρ for all four systems. Each point is a single trial coloured by its EFH value.

The spectral radius (Figure 5.3) exhibits one of the clearest marginal effects in the search space. For the Lorenz, Rössler, and Chen systems, the best-performing trials concentrate at low spectral radii, $\rho \approx 0.4-0.6$, with a marked negative correlation between ρ and forecast horizon: as ρ increases beyond ~ 0.8 , the attainable EFH degrades substantially and the scatter widens, indicating that large spectral radii admit only isolated pockets of good performance contingent on favourable settings of other hyperparameters. This is consistent with the widespread empirical practice of setting $\rho < 1$, though as discussed in Section 1.2, the condition $\rho(\mathbf{W}) < 1$ is neither necessary nor sufficient for the echo state property—what the data reveal is that, across these three continuous flows, the TPE sampler finds the most reliably high-performing configurations well below the spectral unit circle. The Hénon map is the clear exception: the correlation between ρ and forecast horizon is considerably weaker, with competitive trials appearing across the full search range. This reduced sensitivity may reflect the discrete-time nature of the map, where the absence of a continuous-to-discrete timescale mismatch lessens the role of the spectral radius as a dynamical tuning knob.

The input scaling σ_{in} (Figure 5.4) displays a sharp, monotone marginal effect across all four systems: the best-performing trials cluster at small values, $\sigma_{\text{in}} \lesssim 1$, and the forecast horizon deteriorates steadily as the scaling increases toward the upper end of the search range. The effect is most pronounced for the Lorenz and Hénon systems, where nearly all competitive trials lie below $\sigma_{\text{in}} \approx 0.5$; the Rössler system admits a somewhat broader band

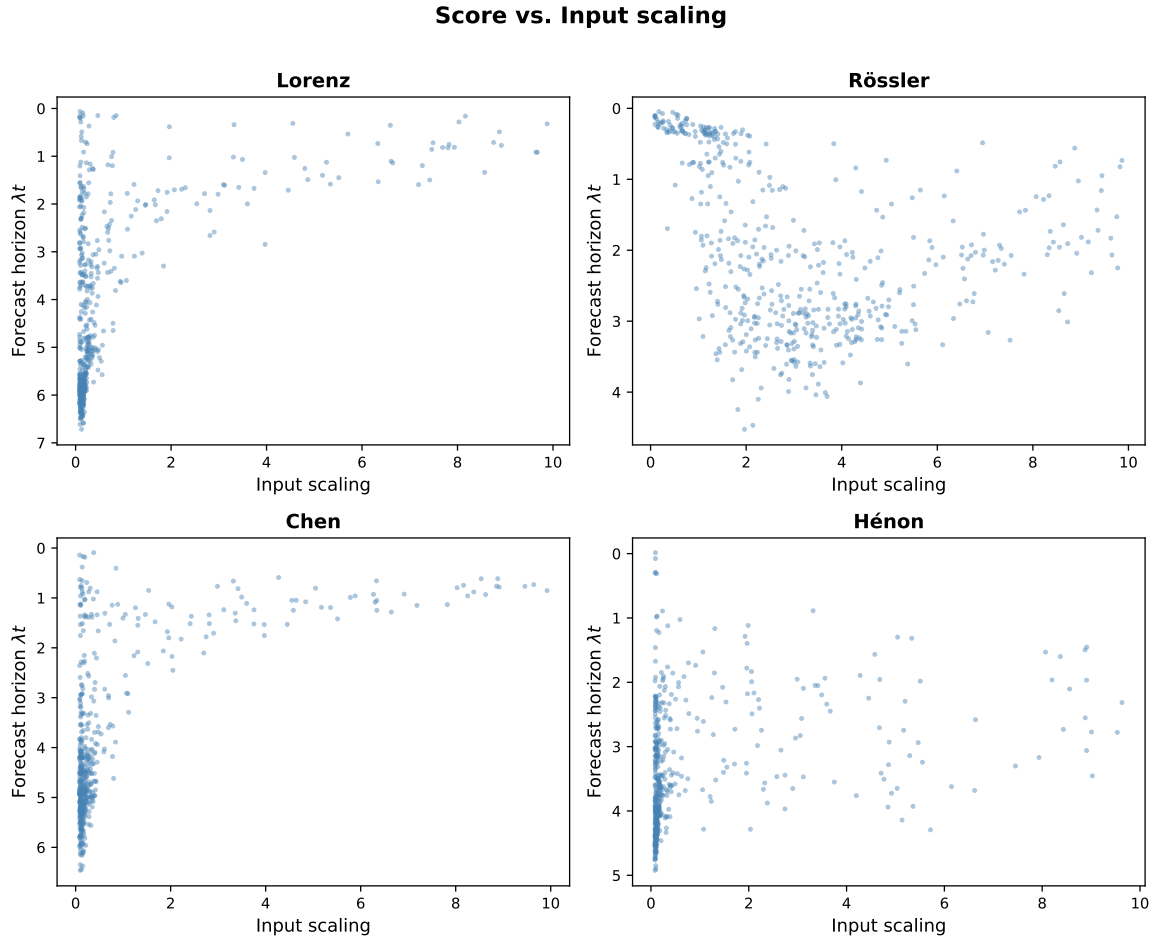


Figure 5.4: Forecast horizon versus input scaling σ_{in} for all four systems.

of viable scalings, with good trials extending to $\sigma_{\text{in}} \approx 2$, consistent with its slower and more regular dynamics requiring less aggressive input attenuation. This observation appears to contradict the finding of Chapter 2, where stronger input scaling was shown to accelerate synchronization by forcing the reservoir toward the driven solution. The tension resolves once one distinguishes the *convergence* objective from the *forecasting* objective. Large σ_{in} accelerates the transient decay because the input term dominates the recurrent dynamics, rapidly washing out initial-condition dependence—but that same dominance drives the $\tanh$ activations into saturation, compressing the reservoir state space and destroying the fine-grained sensitivity to input differences on which long-horizon prediction depends. The echo state property is a *prerequisite* for forecasting, not a proxy for forecasting quality: once ESP is satisfied, further increasing the input scaling yields faster synchronization at the cost of a poorer internal representation. The optimal regime is one where the input is strong enough to guarantee ESP yet weak enough to keep the reservoir in the weakly nonlinear band where its computational capacity is maximal. The TPE sampler, optimizing forecast horizon rather than convergence speed, identifies this regime unambiguously.

The leak rate α (Figure 5.5) shows the most system-dependent marginal structure of all hyperparameters. For the Lorenz and Chen systems, the best trials concentrate decisively at $\alpha \rightarrow 1$, with performance degrading as the leak rate decreases and the reservoir retains more memory of previous states. The effect is more pronounced for Chen, whose large Lyapunov

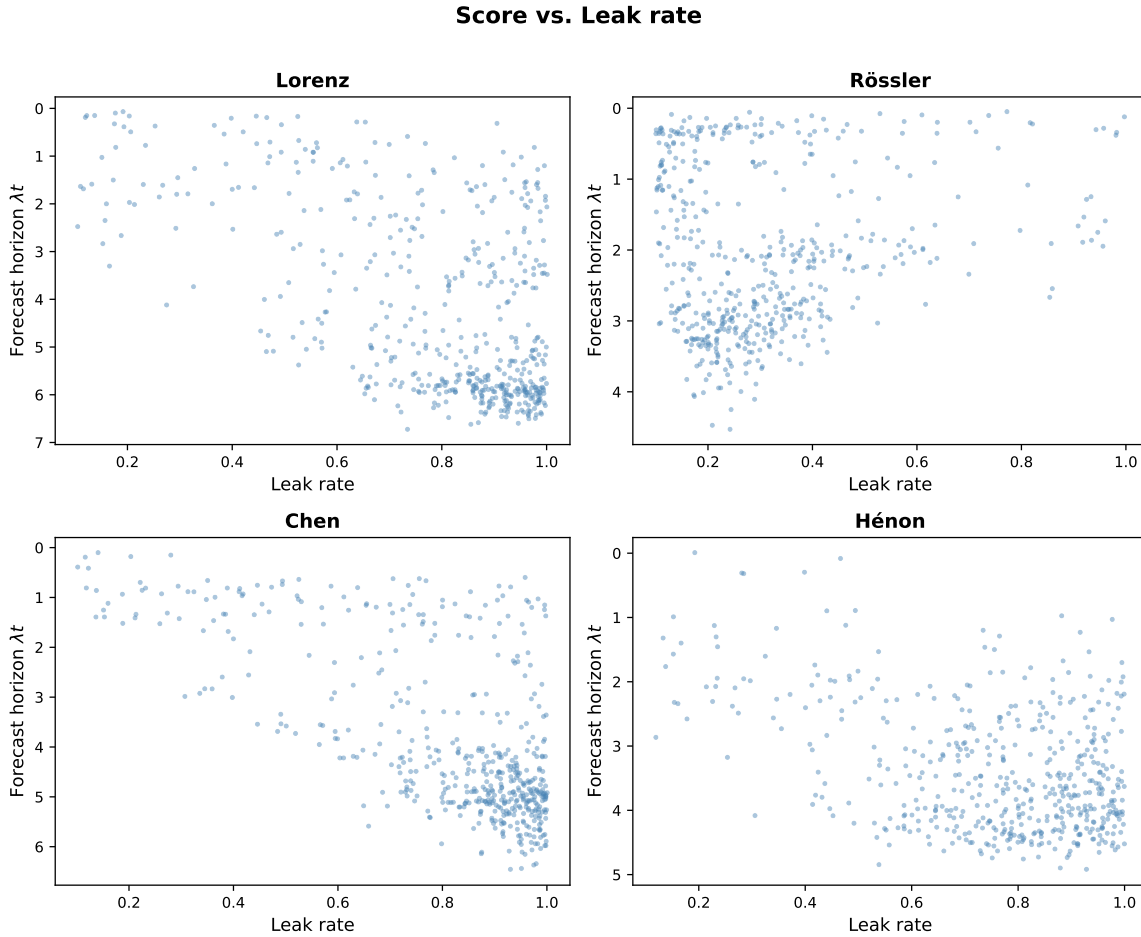


Figure 5.5: Forecast horizon versus leak rate α for all four systems.

exponent ($\lambda_{\max} \approx 2.03$) demands that the reservoir track rapidly diverging trajectories without the inertial smoothing that a small α introduces. The Rössler system presents the opposite pattern: competitive trials cluster at $\alpha \approx 0.1$ – 0.4 , and performance deteriorates markedly for $\alpha > 0.6$. This is entirely consistent with its slow, near-periodic dynamics ($\lambda_{\max} \approx 0.07$), which require the reservoir to integrate information over many time steps rather than react instantaneously—precisely the regime where the leaky integrator acts as a low-pass filter matched to the signal’s intrinsic timescale, as discussed in Section 1.3.1. The ordering Rössler $\rightarrow$ Lorenz $\rightarrow$ Chen thus mirrors the ordering of Lyapunov exponents, confirming the leak rate’s role as a timescale-matching mechanism: faster dynamics demand faster reservoir updates. The Hénon map again stands apart. Although the TPE sampler shows a mild preference for large α , the underlying reason is not timescale matching but redundancy: the Hénon map is already discrete, so the leaky integrator interpolation between successive states is not merely unnecessary but counterproductive— at $\alpha = 1$ the update reduces to the standard ESN equation, and any departure from unity introduces an artificial smoothing that has no counterpart in the generating dynamics.

The topological fractions c and d reveal the most structurally informative marginal relationships in the search space. For the core fraction (Figure 5.6), the Lorenz and Hénon systems show a clear preference for $c \rightarrow 1$: the best-performing trials cluster at the right edge of the panel, with a steady degradation as c decreases. The Rössler system exhibits a milder

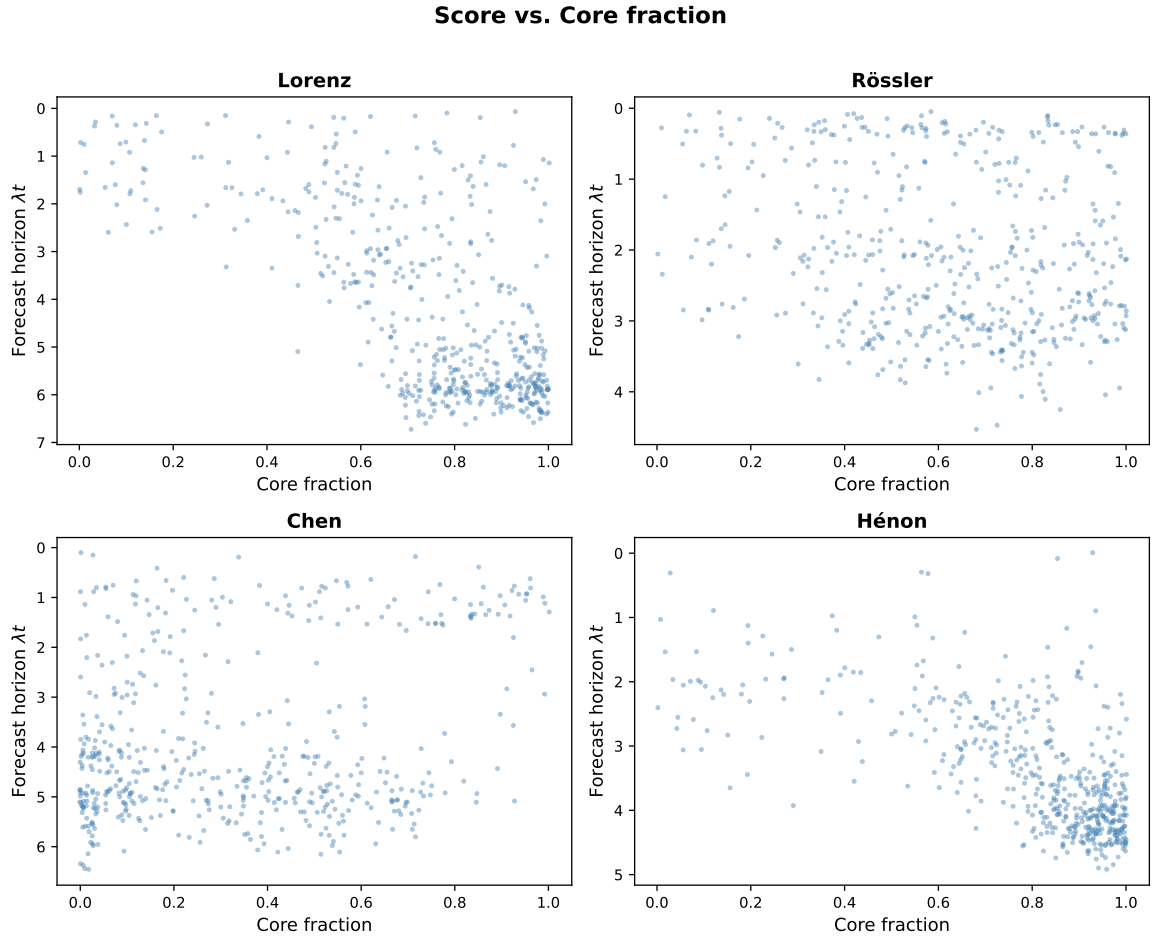


Figure 5.6: Forecast horizon versus core fraction c for all four systems.

version of the same trend, with a slight preference for large c but considerable scatter across the full range. The Chen system, however, displays the opposite pattern: competitive trials appear predominantly at low core fractions, with the densest concentration of high forecast horizon points near $c \approx 0$.

The dendritic fraction (Figure 5.7) complements this picture. For Lorenz and Hénon, d is confined to small values—a direct geometric consequence of the constraint $d \leq 1 - c$ when c is already close to unity. The Rössler system similarly shows a mild preference for small d , again reflecting the slight bias toward cycle-dominated topologies. For Chen, however, the pattern inverts: the best trials extend to large dendritic fractions, and there is a visible positive correlation between d and forecast horizon. Crucially, this tendency is not forced by the constraint— d can take any value in $[0, 1 - c]$, and with c small, the upper bound is not binding. The optimizer is genuinely selecting configurations with extensive dendritic chains.

The divergence between Chen and the other systems admits a mechanistic interpretation rooted in the distinction between core and dendritic nodes established in Section 2.7. Core nodes participate in directed cycles and sustain autonomous recurrent feedback; dendritic nodes lie on feedforward chains downstream of cycles, providing temporal depth without recurrent amplification. One possible explanation is that for systems with moderate Lyapunov exponents (Lorenz, Hénon), the recurrent dynamics of cycle-dominated reservoirs

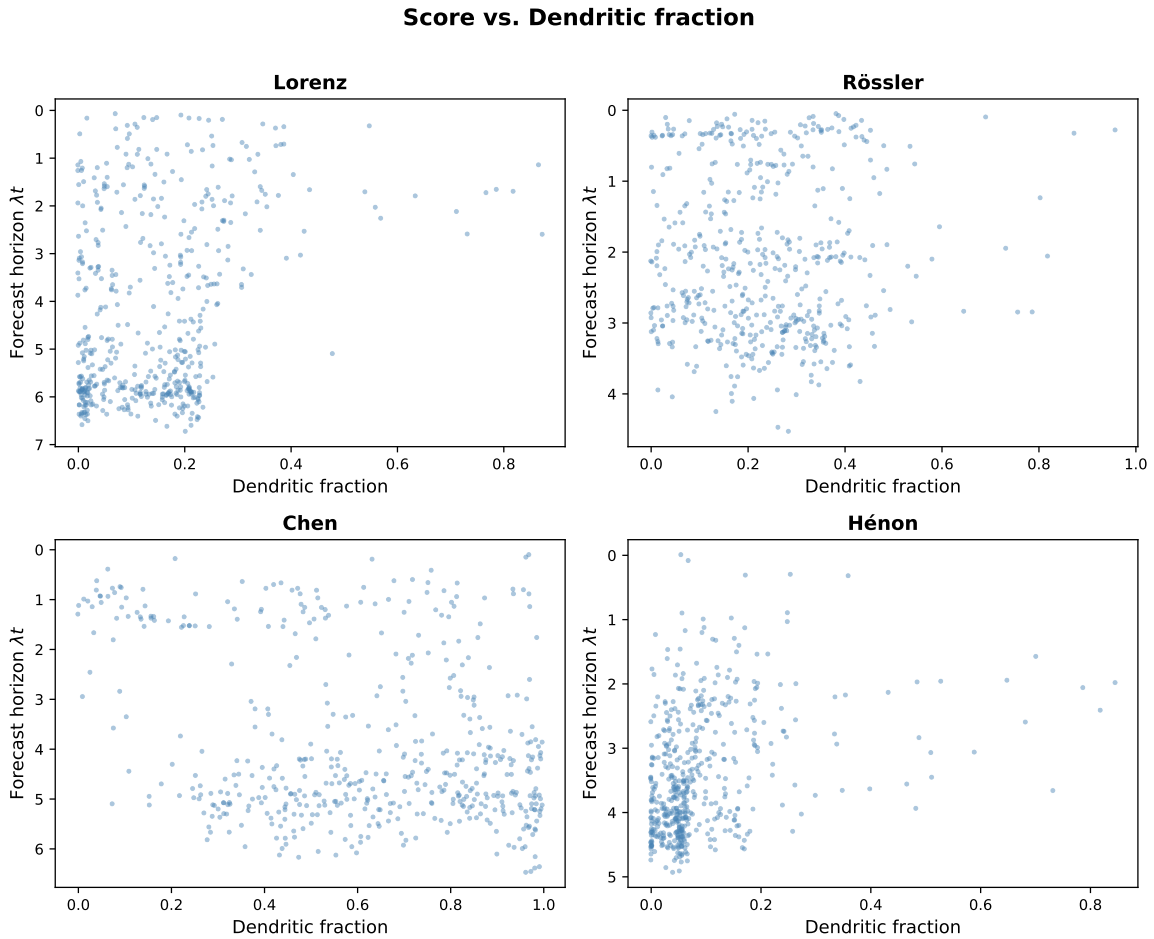


Figure 5.7: Forecast horizon versus dendritic fraction d for all four systems.

enrich the state-space representation without destabilising it— the feedback loops generate diverse nonlinear features that the readout can exploit. For Chen, whose $\lambda_{\max} \approx 2.03$ imposes extremely fast error growth, heavy recurrence becomes a liability: errors injected into the cycle during closed-loop prediction are amplified at each pass, precisely the regime where recurrent feedback is most dangerous. Dendritic chains, being feedforward, provide the temporal memory that forecasting requires— each chain node delays and nonlinearly transforms the signal from the cycle upstream— without feeding prediction errors back into the reservoir’s core dynamics. The optimizer, confronted with Chen’s aggressive divergence rate, learns to shift the active-node budget from cycles to dendrites, trading recurrent richness for feedforward stability.

In all four systems, however, the common thread is that the best trials avoid low values of $c + d$: whether the active fraction is allocated to cycles or dendrites, the elimination of quiescent nodes remains the dominant structural effect. The marginal plots thus foreshadow the joint c – d analysis of Section 5.5, where this observation is confirmed directly. A more detailed analysis is needed to validate this behaviour and observe the joint role of both the fractions c and d .

5.5 Influence of core and dendritic fractions

The most striking structural result of the optimization campaign concerns the topological composition of the reservoir. In Chapter 2, the dendrocycle experiments on the Lorenz system demonstrated that forecast performance improves monotonically as the combined active fraction $c + d$ approaches unity, where every node either belongs to the core cycle or lies on a dendritic chain downstream of it, and the quiescent fraction vanishes. That analysis was conducted on a fixed grid of (c, d) pairs with all other hyperparameters held constant. The question addressed here is whether the same structural preference emerges when the optimizer is free to explore the full eight-dimensional search space simultaneously.

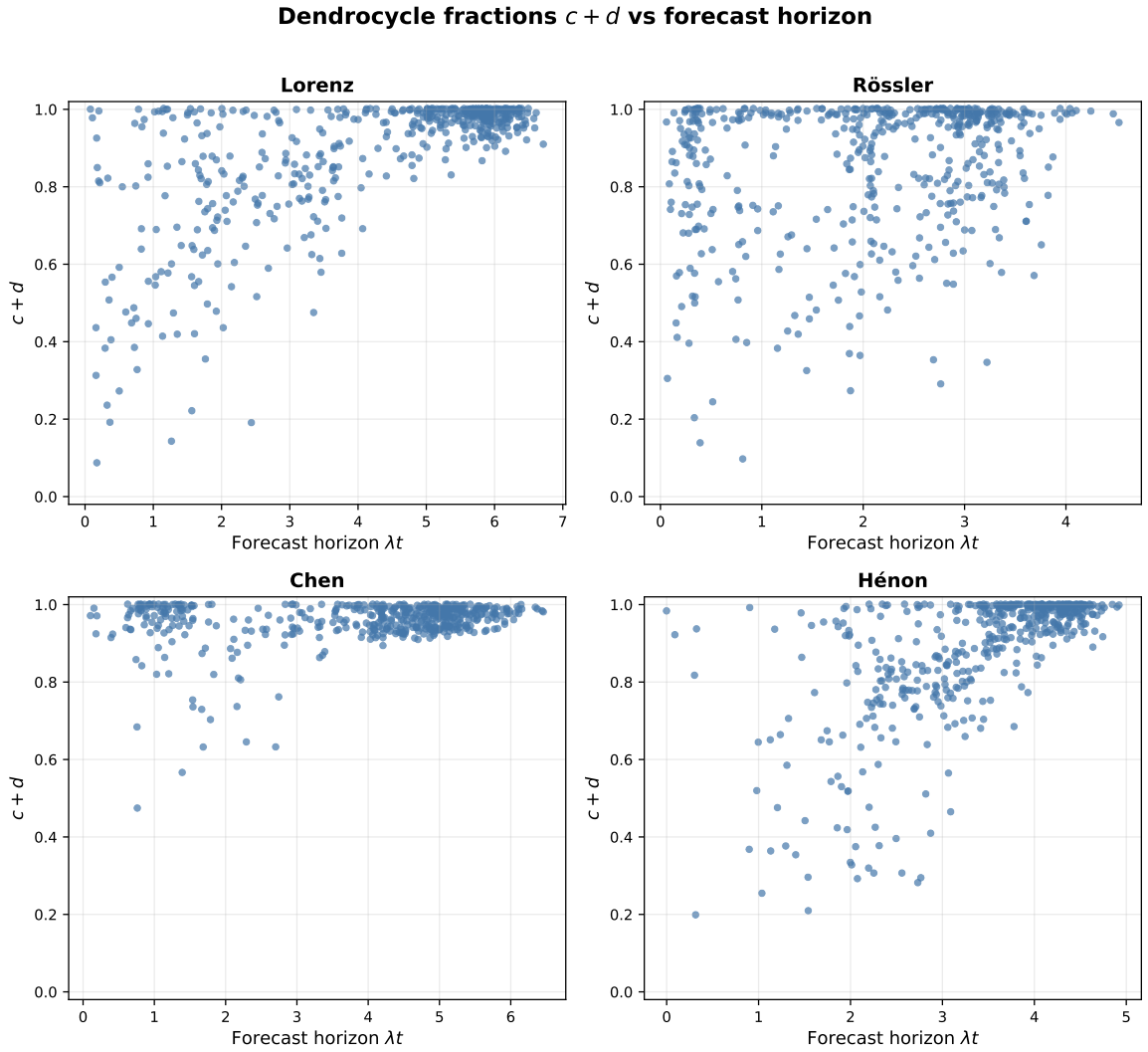


Figure 5.8: Scatter plot of the core fraction c and dendritic fraction d for all trials, coloured by the EFH in Lyapunov times. The diagonal $c + d = 1$ is indicated. Across all four systems, the highest-performing trials concentrate near this diagonal, confirming that the optimizer rediscovers the design principle of Chapter 2 without any structural constraint being imposed.

Figure 5.8 plots the total active fraction $c + d$ against the achieved forecast horizon EFH_λ for all four systems, collapsing the two-dimensional (c, d) plane onto a single axis that captures the quiescence structure of the reservoir. The overall pattern is consistent across systems, high-performing trials are concentrated at $c + d \approx 1$, while low values of $c + d$

are associated with poor forecasts. The relationship is not symmetric, however—the upper band $c + d \in [0.9, 1.0]$ contains both the best and many mediocre trials, confirming that high active fraction is necessary but not sufficient for strong performance. The remaining hyperparameters (spectral radius, input scaling, regularization, and the c/d balance itself) determine which of the high- $c + d$ configurations actually achieves a long horizon.

The four systems differ considerably in how sharply performance degrades as $c + d$ decreases. For Lorenz and Chen the drop is abrupt: trials with $c + d \lesssim 0.8$ are almost uniformly confined to low EFH values, with very few exceptions scattered across the interior. The performance landscape is cliff-like, and the optimizer has learned to avoid the low- $c + d$ region almost entirely—the point density itself is visibly lower there, reflecting the TPE sampler’s concentration of proposals near the active boundary. Hénon shows a similar pattern despite its discrete-time nature and comparatively modest forecast horizons. Rössler is again the outlier: the scatter extends more smoothly downward, and isolated trials with $c + d$ as low as 0.6 still achieve competitive EFH values. This is consistent with the near-periodic, low-dimensional character of the Rössler attractor, which places less demand on the reservoir’s internal memory structure and is therefore more tolerant of dynamically inert nodes.

The horizontal spread within the $c + d \approx 1$ band is itself informative. For Lorenz and Chen, the best trials reach EFH values that are an order of magnitude larger than the median, yet both extremes sit at $c + d \approx 1$ —the active fraction alone cannot discriminate them. This underscores the point that $c + d$ is a necessary condition for high performance, not a sufficient one: it filters out the clearly suboptimal configurations (quiescent reservoirs) while leaving a rich landscape of variability among the active ones. The practical implication is that any hyperparameter search should enforce $c + d \approx 1$ as a soft constraint or prior, concentrating the optimization budget on the subspace where good solutions actually live, while still allowing the remaining degrees of freedom to be tuned to the target system.

The mechanistic interpretation mirrors that of the diagonal analysis: quiescent nodes contribute no independent memory to the reservoir’s internal dynamics, and their activations reduce to a redundant echo of the fed-back prediction signal. The TPE sampler discovers this structure empirically, iteratively shifting probability mass toward $c + d \approx 1$ as it accumulates evidence that trials in the interior of the unit square consistently underperform. That this emerges without any explicit constraint on the topology is itself a validation of the dendrocycle framework: the optimizer recovers, from raw performance data alone, the same structural principle that the theoretical analysis of reservoir capacity predicts.

5.6 Conclusions

This chapter set out to answer a specific question: do the methodological tools developed independently in Chapters 2–4 compose into a coherent, automated pipeline, and if so, what does that pipeline reveal about the structure of the ESN design space for chaotic forecasting? The answer, demonstrated across four dynamical systems spanning nearly two orders of magnitude in Lyapunov exponent, is affirmative on both counts— and the revealed structure is richer than anticipated.

The sagitta discretization of Chapter 3 normalises the geometric difficulty of each system before any reservoir is constructed, ensuring that the sampling step Δt^* reflects the intrinsic curvature of the attractor rather than an arbitrary experimenter choice. The EFH metric of Chapter 4, used as the optimization objective, provides the TPE sampler with a smooth, bounded signal that admits meaningful gradient information near the failure onset—precisely where the brittle VPT would present a discontinuous jump. And the dendrocycle topology of Chapter 2, parameterised by the core and dendritic fractions, exposes a structural axis of the reservoir design space that conventional random sparse topologies leave hidden. None of these components was designed with the others in mind, yet they interlock without friction: sagitta feeds the preprocessing, the dendrocycle defines the search space, and the EFH drives the sampler. The pipeline requires no manual intervention beyond specifying the system equations and the hyperparameter ranges.

What the pipeline reveals is a design space that is simultaneously low-dimensional and system-dependent. The fANOVA analysis shows that, for any given system, at most three hyperparameters account for the majority of the variance in forecast performance—but which three depends on the dynamical regime. For the Lorenz system it is the spectral radius, core fraction, and input scaling; for Rössler, the input scaling and leak rate dominate; for Chen, the input scaling and dendritic weight; for Hénon, the leak rate, spectral radius, and core fraction share the lead. The ridge regularization parameter is negligible everywhere. This low effective dimensionality explains the rapid saturation of the optimization histories: the TPE sampler exhausts the productive dimensions of the search space within $\mathcal{O}(100)$ trials, and the remaining budget yields only incremental gains from exploration of residual interactions.

The marginal analysis exposes three universal tendencies that hold across all four systems despite their dynamical diversity. First, the spectral radius should generally be kept below unity, typically in the range $\rho \approx 0.5\text{--}0.7$, where the reservoir operates near but safely below the edge of stability. Second, the input scaling should be small, $\sigma_{\text{in}} \lesssim 1$, maintaining the reservoir in the weakly nonlinear regime where its activation functions are most sensitive to input variation—a finding that we reconciled with the apparently contradictory role of input scaling in accelerating transient synchronization by distinguishing the convergence prerequisite from the forecasting objective. Third, and most importantly, the combined active fraction $c + d$ should approach unity: quiescent nodes, regardless of the system, are dead weight.

Within this universal envelope, the system-dependent structure is equally informative. The leak rate acts as a timescale-matching dial whose optimal setting mirrors the ordering of Lyapunov exponents: $\alpha \rightarrow 1$ for Chen and Lorenz, $\alpha \approx 0.2$ for Rössler, and $\alpha = 1$ for the Hénon map by a different argument—the discreteness of the map renders any leaky smoothing artificial. The partition of the active fraction between cycles and dendrites is the most striking system-dependent effect: Lorenz, Hénon, and to a lesser extent Rössler prefer cycle-dominated topologies ($c \gg d$), while Chen demands dendrite-dominated architectures ($d \gg c$). The mechanistic interpretation—that Chen’s large Lyapunov exponent penalises recurrent error amplification, driving the optimizer toward feedforward dendritic chains that provide temporal depth without recurrent feedback—represents, to our knowledge, the first empirical evidence that the optimal balance between recurrent and feedforward structure in a reservoir is determined by the divergence rate of the target dynamics.

The centrepiece of the chapter, however, is the unsupervised rediscovery of the $c + d \approx 1$ design principle. In Chapter 2, this principle was established through controlled experiments on a single system with all other hyperparameters held fixed—a setting that left open the possibility that the result was an artifact of the specific operating point or the particular attractor geometry. Here, the TPE sampler—given no structural constraints, no prior on the (c, d) plane, and full freedom to explore an eight-dimensional search space—converges to the $c + d \approx 1$ diagonal in all four systems independently. The convergence is not merely statistical: for Chen and Lorenz, the performance ridge along the diagonal is sharp enough that virtually no high-EFH trial survives at $c + d < 0.8$. This is the strongest form of validation a design principle can receive short of a formal proof: an optimizer that knows nothing about graph theory, strongly connected components, or the distinction between core and quiescent nodes nonetheless learns, from performance data alone, that every node in the reservoir should be dynamically active.

The practical implications are concrete. For a practitioner building an ESN for chaotic forecasting, the pipeline described here reduces the design problem to a modest computational investment—a few hundred TPE trials with the EFH as objective—preceded by a one-time sagitta calibration of the sampling step. The dendrocycle topology, with $c + d \geq 0.9$ and the c/d balance informed by the expected Lyapunov exponent of the target system, provides a principled starting point that outperforms the conventional random sparse reservoir by construction: it eliminates quiescent nodes by design rather than by accident. All components are available in the open-source ResDAG library [43].

6 Conclusions

This thesis set out to place the design of Echo State Networks for chaotic forecasting on principled foundations by closing three methodological gaps that had limited the reproducibility, comparability, and optimality of reservoir-computing-based predictions. The gaps— uncontrolled synchronization transients, arbitrary temporal discretization, and brittle evaluation metrics— were each addressed with a self-contained contribution, and their integration into a single automated pipeline demonstrated that the whole is greater than the sum of its parts.

Transient dynamics and reservoir topology. Chapter 2 transformed the informal notion of “washout” into a quantitative, measurable system property. For reservoirs whose spectral radius satisfies $\rho < 1$, an analytical upper bound on the synchronization time was derived from the linearized dynamics and verified numerically. For $\rho \geq 1$, ensemble statistics over 10^4 reservoir realizations revealed that the transient-time distribution develops heavy tails, with rare realizations requiring an order of magnitude longer than the median. A graph-theoretic decomposition of the recurrent weight matrix into strongly connected components identified three structural classes of nodes— *core* (on cycles), *dendritic* (downstream of cycles), and *quiescent* (on acyclic subgraphs)— whose distinct long-term behaviors under zero input explain the coexistence of oscillatory and decaying units. The forecasting experiments on the Lorenz system demonstrated that performance improves monotonically as the combined active fraction $c + d$ approaches unity, establishing a concrete design principle: every node in the reservoir should be dynamically active. The *dendrocycle* topology, which achieves $c + d = 1$ by construction, was proposed as a principled alternative to conventional random sparse reservoirs.

Geometric time discretization. Chapter 3 developed a derivative-free criterion for selecting the sampling step Δt based on the *sagitta*— the perpendicular distance from the midpoint of a chord to the curve. The key theoretical result is that the sagitta scales as $s(t; h) = \frac{1}{2}|\mathbf{a}_\perp(t)| h^2 + \mathcal{O}(h^3)$, providing a curvature proxy that requires no numerical differentiation. Both absolute and relative selection rules were defined, and their *idempotence* was proved: resampling a trajectory at the selected step and reapplying the rule returns the same step. Applied to the Lorenz, Rössler, and Chen attractors, the sagitta method produces sampling

intervals that are geometrically comparable across systems with widely different natural time scales, thereby enabling the first fair cross-system benchmarks in reservoir computing. The single tolerance parameter ε provides an interpretable, transferable knob controlling the geometric fidelity of the discretization.

The Expected Forecast Horizon. Chapter 4 introduced the EFH as a continuous, differentiable measure of predictive skill that replaces the binary valid prediction time. Constructed through a survival-process analogy—a soft sigmoidal gate converts the pointwise error signal into an acceptability probability, and the EFH is defined as the sum of the resulting survival curve—the metric was shown to be bounded, monotone in the error threshold τ , scale-equivariant, and formally equivalent to a restricted mean survival time. In the limit $\delta \rightarrow 0^+$, the EFH recovers the hard hitting time exactly. A gradient-concentration result demonstrated that the EFH derivative is localized at the failure onset, making it an effective and well-targeted training signal. Noise-robustness experiments confirmed a substantially lower coefficient of variation than the hard VPT, establishing the EFH as a superior benchmarking statistic for chaotic forecasting evaluation.

Hyperparameter optimization and the integrated pipeline. Chapter 5 brought the three contributions together in a systematic optimization campaign using the tree-structured Parzen estimator across four benchmark systems—Hénon, Lorenz, Rössler, and Chen—spanning nearly two orders of magnitude in Lyapunov exponent. The sagitta preprocessing normalized geometric difficulty before any reservoir was constructed; the dendrocycle topology exposed the topological axis of the design space; and the EFH, serving as the smooth optimization objective, provided the sampler with a well-behaved performance signal. The fANOVA analysis revealed that the effective dimensionality of the design space is low: at most three hyperparameters account for the majority of the variance in forecast quality for any given system, though which three depends on the dynamical regime. Three universal tendencies emerged: the spectral radius should be kept below unity ($\rho \approx 0.5\text{--}0.7$), the input scaling should be small ($\sigma_{\text{in}} \lesssim 1$), and the active fraction should satisfy $c + d \approx 1$. The system-dependent structure was equally revealing: the leak rate mirrors the ordering of Lyapunov exponents (fast dynamics demand $\alpha \rightarrow 1$; slow dynamics benefit from small α), and the partition between cyclic and dendritic nodes is governed by the divergence rate of the target attractor—cycle-dominated topologies excel for moderate Lyapunov exponents, while dendrite-dominated architectures are preferred when recurrent error amplification is severe.

The centrepiece result is the *unsupervised rediscovery* of the $c + d \approx 1$ design principle. The TPE sampler, given no structural constraints and no prior on the topological fractions, converged independently to the $c + d \approx 1$ diagonal in all four systems. For the Lorenz and Chen attractors, the performance ridge along this diagonal was sharp enough that virtually no high-EFH trial survived at $c + d < 0.8$. This constitutes the strongest empirical validation a design principle can receive short of a formal proof: an optimizer that knows nothing about graph theory or strongly connected components learns, from performance data alone, that every reservoir node should participate in sustained dynamical activity.

Overall achievement. Taken together, the contributions of this thesis establish a complete, end-to-end methodology for the principled design of Echo State Networks for chaotic forecasting. Where previously the practitioner faced a sequence of ad hoc choices— how long to wash out, at what step to sample, how to measure success, and how to search the hyperparameter space— there now exists a coherent framework in which each decision is grounded in theory and validated across heterogeneous dynamical regimes. The sagitta standardizes the data; the dendrocycle structures the reservoir; the EFH evaluates the forecast; and the Bayesian optimizer navigates the design space, autonomously rediscovering the structural principles that the theoretical analysis predicted. The results are not merely a collection of tools but an integrated scientific instrument for reproducible, comparable, and principled chaotic forecasting with reservoir computers.

Bibliography

- [1] E. Ott, *Chaos in Dynamical Systems*, 2nd ed. Cambridge University Press, Aug. 2002, ISBN: 978-0-521-81196-5 978-0-521-01084-9 978-0-511-80326-0. DOI: [10.1017/CBO9780511803260](https://doi.org/10.1017/CBO9780511803260) Accessed: Aug. 12, 2025.
- [2] H. Kantz and T. Schreiber, *Nonlinear Time Series Analysis*, 2nd ed. Cambridge, UK ; New York: Cambridge University Press, 2004, ISBN: 978-0-521-82150-6 978-0-521-52902-0.
- [3] F. Takens, "Detecting strange attractors in turbulence," in *Dynamical Systems and Turbulence, Warwick 1980*, D. Rand and L.-S. Young, Eds., vol. 898, Berlin, Heidelberg: Springer Berlin Heidelberg, 1981, pp. 366–381, ISBN: 978-3-540-11171-9 978-3-540-38945-3. DOI: [10.1007/BFb0091924](https://doi.org/10.1007/BFb0091924) Accessed: Aug. 12, 2025.
- [4] M. Casdagli, "Chaos and Deterministic Versus Stochastic Non-Linear Modelling," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 54, no. 2, pp. 303–328, Jan. 1992, ISSN: 1369-7412, 1467-9868. DOI: [10.1111/j.2517-6161.1992.tb01884.x](https://doi.org/10.1111/j.2517-6161.1992.tb01884.x) Accessed: Feb. 9, 2026.
- [5] P. J. Werbos, "Backpropagation through time: What it does and how to do it," *Proceedings of the IEEE*, vol. 78, no. 10, pp. 1550–1560, 1990. DOI: [10.1109/5.58337](https://doi.org/10.1109/5.58337)
- [6] Y. Bengio, P. Simard, and P. Frasconi, "Learning long-term dependencies with gradient descent is difficult," *IEEE Transactions on Neural Networks*, vol. 5, no. 2, pp. 157–166, 1994. DOI: [10.1109/72.279181](https://doi.org/10.1109/72.279181)
- [7] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," *Proceedings of the 30th International Conference on Machine Learning (ICML)*, pp. 1310–1318, 2013.
- [8] S. Hochreiter, "Untersuchungen zu dynamischen neuronalen Netzen," *Diploma thesis, Institut für Informatik, Technische Universität München*, 1991.
- [9] H. Jaeger, "The "echo state" approach to analysing and training recurrent neural networks – with an Erratum note," 2001.
- [10] W. Maass, T. Natschläger, and H. Markram, "Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations," *Neural Computation*, vol. 14, no. 11, pp. 2531–2560, Nov. 2002, ISSN: 0899-7667, 1530-888X. DOI: [10.1162/089976602760407955](https://doi.org/10.1162/089976602760407955) Accessed: May 30, 2023.

- [11] A. N. Tikhonov, "Solution of incorrectly formulated problems and the regularization method," *Soviet Mathematics Doklady*, vol. 4, pp. 1035–1038, 1963.
- [12] H. Jaeger, I. B. Yildiz, and S. J. Kiebel, "Re-visiting the echo state property," *Neural Networks*, vol. 35, pp. 1–9, Nov. 2012, ISSN: 08936080. DOI: [10.1016/j.neunet.2012.07.005](https://doi.org/10.1016/j.neunet.2012.07.005) Accessed: May 30, 2023.
- [13] G. Manjunath and H. Jaeger, "Echo State Property Linked to an Input: Exploring a Fundamental Characteristic of Recurrent Neural Networks," *Neural Computation*, vol. 25, no. 3, pp. 671–696, Mar. 2013, ISSN: 0899-7667, 1530-888X. DOI: [10.1162/NECO_a_00411](https://doi.org/10.1162/NECO_a_00411) Accessed: Apr. 21, 2025.
- [14] L. Grigoryeva and J.-P. Ortega, "Echo state networks are universal," *Neural Networks*, vol. 108, pp. 495–508, 2018. DOI: [10.1016/j.neunet.2018.08.025](https://doi.org/10.1016/j.neunet.2018.08.025)
- [15] H. Jaeger and M. Lukoševičius, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review*, vol. 3, no. 3, pp. 127–149, Aug. 2009, ISSN: 15740137. DOI: [10.1016/j.cosrev.2009.03.005](https://doi.org/10.1016/j.cosrev.2009.03.005) Accessed: May 30, 2023.
- [16] G. Tanaka et al., "Recent advances in physical reservoir computing: A review," *Neural Networks*, vol. 115, pp. 100–123, 2019. DOI: [10.1016/j.neunet.2019.03.005](https://doi.org/10.1016/j.neunet.2019.03.005)
- [17] M. Cucchi et al., "Hands-on reservoir computing: A tutorial for practical implementation," *Neuromorphic Computing and Engineering*, vol. 2, no. 3, p. 032 002, 2022. DOI: [10.1088/2634-4386/ac7db7](https://doi.org/10.1088/2634-4386/ac7db7)
- [18] H. Jaeger and H. Haas, "Harnessing Nonlinearity, Predicting Chaotic Systems and Saving Energy in Wireless Communication," *Science*, vol. 304, no. 5667, pp. 78–80, Apr. 2004, ISSN: 0036-8075, 1095-9203. DOI: [10.1126/science.1091277](https://doi.org/10.1126/science.1091277) Accessed: May 30, 2023.
- [19] J. Pathak, B. Hunt, M. Girvan, Z. Lu, and E. Ott, "Model-Free Prediction of Large Spatiotemporally Chaotic Systems from Data: A Reservoir Computing Approach," *Physical Review Letters*, vol. 120, no. 2, p. 024 102, Jan. 2018, ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.120.024102](https://doi.org/10.1103/PhysRevLett.120.024102) Accessed: Mar. 17, 2025.
- [20] Z. Lu, J. Pathak, B. Hunt, M. Girvan, R. Brockett, and E. Ott, "Reservoir observers: Model-free inference of unmeasured variables in chaotic systems," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 27, no. 4, p. 041 102, Apr. 2017, ISSN: 1054-1500, 1089-7682. DOI: [10.1063/1.4979665](https://doi.org/10.1063/1.4979665) Accessed: Mar. 17, 2025.
- [21] P. Vlachas et al., "Backpropagation algorithms and Reservoir Computing in Recurrent Neural Networks for the forecasting of complex spatiotemporal dynamics," *Neural Networks*, vol. 126, pp. 191–217, Jun. 2020, ISSN: 08936080. DOI: [10.1016/j.neunet.2020.02.016](https://doi.org/10.1016/j.neunet.2020.02.016) Accessed: Jan. 10, 2024.
- [22] A. Chattopadhyay, P. Hassanzadeh, and D. Subramanian, "Data-driven predictions of a multiscale Lorenz 96 chaotic system using machine-learning methods: Reservoir computing, artificial neural network, and long short-term memory network," *Nonlinear Processes in Geophysics*, vol. 27, no. 3, pp. 373–389, 2020. DOI: [10.5194/npg-27-373-2020](https://doi.org/10.5194/npg-27-373-2020)
- [23] D. J. Gauthier, E. Bollt, A. Griffith, and W. A. S. Barbosa, "Next generation reservoir computing," *Nature Communications*, vol. 12, no. 1, p. 5564, Sep. 2021, ISSN: 2041-1723. DOI: [10.1038/s41467-021-25801-2](https://doi.org/10.1038/s41467-021-25801-2) Accessed: May 30, 2023.

- [24] M. Lukoševičius, "A Practical Guide to Applying Echo State Networks," in *Neural Networks: Tricks of the Trade*, G. Montavon, G. B. Orr, and K.-R. Müller, Eds., vol. 7700, Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 659–686, ISBN: 978-3-642-35288-1 978-3-642-35289-8. DOI: [10.1007/978-3-642-35289-8_36](https://doi.org/10.1007/978-3-642-35289-8_36) Accessed: May 30, 2023.
- [25] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, 2012.
- [26] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in *Advances in Neural Information Processing Systems*, vol. 24, 2011, pp. 2546–2554.
- [27] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," in *Advances in Neural Information Processing Systems*, vol. 25, 2012, pp. 2951–2959.
- [28] B. Bischl et al., "Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges," *WIREs Data Mining and Knowledge Discovery*, vol. 13, no. 2, e1484, 2023. DOI: [10.1002/widm.1484](https://doi.org/10.1002/widm.1484)
- [29] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, *Optuna: A Next-generation Hyperparameter Optimization Framework*, Jul. 2019. DOI: [10.48550/arXiv.1907.10902](https://doi.org/10.48550/arXiv.1907.10902) arXiv: [1907.10902](https://arxiv.org/abs/1907.10902) [cs]. Accessed: Feb. 23, 2026.
- [30] P. E. Kloeden and M. Rasmussen, *Nonautonomous Dynamical Systems* (Mathematical Surveys and Monographs). American Mathematical Society, 2011, vol. 176.
- [31] M. Buehner and P. Young, "A tighter bound for the echo state property," *IEEE Transactions on Neural Networks*, vol. 17, no. 3, pp. 820–824, May 2006, ISSN: 1045-9227, 1941-0093. DOI: [10.1109/TNN.2006.872357](https://doi.org/10.1109/TNN.2006.872357) Accessed: Aug. 12, 2025.
- [32] L. M. Pecora and T. L. Carroll, "Synchronization in chaotic systems," *Physical Review Letters*, vol. 64, no. 8, pp. 821–824, 1990. DOI: [10.1103/PhysRevLett.64.821](https://doi.org/10.1103/PhysRevLett.64.821)
- [33] J. D. Hart, "Attractor reconstruction with reservoir computers: The effect of the reservoir's conditional Lyapunov exponents on faithful attractor reconstruction," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 34, no. 4, p. 043123, Apr. 2024, ISSN: 1054-1500, 1089-7682. DOI: [10.1063/5.0196257](https://doi.org/10.1063/5.0196257) Accessed: Nov. 23, 2025.
- [34] H. Jaeger, M. Lukoševičius, D. Popovici, and U. Siewert, "Optimization and applications of echo state networks with leaky- integrator neurons," *Neural Networks*, vol. 20, no. 3, pp. 335–352, Apr. 2007, ISSN: 08936080. DOI: [10.1016/j.neunet.2007.04.016](https://doi.org/10.1016/j.neunet.2007.04.016) Accessed: May 30, 2023.
- [35] C. Gallicchio and A. Micheli, *Deep Echo State Network (DeepESN): A Brief Survey*, Sep. 2020. DOI: [10.48550/arXiv.1712.04323](https://doi.org/10.48550/arXiv.1712.04323) arXiv: [1712.04323](https://arxiv.org/abs/1712.04323) [cs]. Accessed: Mar. 7, 2025.
- [36] A. Rodan and P. Tino, "Minimum Complexity Echo State Network," *IEEE Transactions on Neural Networks*, vol. 22, no. 1, pp. 131–144, Jan. 2011, ISSN: 1045-9227, 1941-0093. DOI: [10.1109/TNN.2010.2089641](https://doi.org/10.1109/TNN.2010.2089641) Accessed: Oct. 17, 2025.
- [37] A. Rodan and P. Tiño, "Simple Deterministically Constructed Cycle Reservoirs with Regular Jumps," *Neural Computation*, vol. 24, no. 7, pp. 1822–1852, Jul. 2012, ISSN: 0899-7667, 1530-888X. DOI: [10.1162/NECO_a.00297](https://doi.org/10.1162/NECO_a.00297) Accessed: Oct. 17, 2025.

- [38] E. N. Lorenz, "Deterministic Nonperiodic Flow," *Journal of the Atmospheric Sciences*, vol. 20, no. 2, pp. 130–141, Mar. 1963, ISSN: 0022-4928, 1520-0469. DOI: [10.1175/1520-0469\(1963\)020<0130:DNF>2.0.CO;2](https://doi.org/10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2) Accessed: Jan. 13, 2026.
- [39] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, 1976. DOI: [10.1016/0375-9601\(76\)90101-8](https://doi.org/10.1016/0375-9601(76)90101-8) Accessed: Feb. 9, 2026.
- [40] M. C. Mackey and L. Glass, "Oscillation and chaos in physiological control systems," *Science*, vol. 197, no. 4300, pp. 287–289, 1977. DOI: [10.1126/science.267326](https://doi.org/10.1126/science.267326)
- [41] M. Hénon, "A two-dimensional mapping with a strange attractor," *Communications in Mathematical Physics*, vol. 50, no. 1, pp. 69–77, 1976. DOI: [10.1007/BF01608556](https://doi.org/10.1007/BF01608556)
- [42] G. Chen and T. Ueta, "YET ANOTHER CHAOTIC ATTRACTOR," *International Journal of Bifurcation and Chaos*, vol. 09, no. 07, pp. 1465–1466, Jul. 1999, ISSN: 0218-1274, 1793-6551. DOI: [10.1142/S0218127499001024](https://doi.org/10.1142/S0218127499001024) Accessed: Feb. 9, 2026.
- [43] D. Estevez-Moya, *El3ssar/ResDAG: V0.1.0-alpha*, Zenodo, Jan. 2026. DOI: [10.5281/ZENODO.18236953](https://doi.org/10.5281/ZENODO.18236953) Accessed: Jan. 13, 2026.
- [44] R. J. Williams and D. Zipser, "A learning algorithm for continually running fully recurrent neural networks," *Neural Computation*, vol. 1, no. 2, pp. 270–280, 1989. DOI: [10.1162/neco.1989.1.2.270](https://doi.org/10.1162/neco.1989.1.2.270)
- [45] U. D. Schiller and J. J. Steil, "Analyzing the weight dynamics of recurrent learning algorithms," *Neurocomputing*, vol. 63, pp. 5–23, 2005. DOI: [10.1016/j.neucom.2004.04.009](https://doi.org/10.1016/j.neucom.2004.04.009)
- [46] W. Maass, T. Natschläger, and H. Markram, "Real-time computing without stable states: A new framework for neural computation based on perturbations," *Neural Computation*, vol. 14, no. 11, pp. 2531–2560, 2002. DOI: [10.1162/089976602760407955](https://doi.org/10.1162/089976602760407955)
- [47] D. Verstraeten, B. Schrauwen, M. D'Haene, and D. Stroobandt, "An experimental unification of reservoir computing methods," *Neural Networks*, vol. 20, no. 3, pp. 391–403, 2007. DOI: [10.1016/j.neunet.2007.04.003](https://doi.org/10.1016/j.neunet.2007.04.003)
- [48] J. J. Steil, "Backpropagation-decorrelation: Online recurrent learning with O(n) complexity," in *Proceedings of the IEEE International Joint Conference on Neural Networks (IJCNN)*, vol. 2, 2004, pp. 843–848. DOI: [10.1109/IJCNN.2004.1380039](https://doi.org/10.1109/IJCNN.2004.1380039)
- [49] PLACKETT, "SOME THEOREMS IN LEAST SQUARES," *Biometrika*, 1950. DOI: [10.1093/biomet/37.1-2.149](https://doi.org/10.1093/biomet/37.1-2.149) Accessed: Feb. 11, 2026.
- [50] H. Jaeger, *Echo state network*, Scholarpedia, http://www.scholarpedia.org/article/Echo_state_network, 2007.
- [51] M. C. Ozturk, D. Xu, and J. C. Príncipe, "Analysis and design of echo state networks," *Neural Computation*, vol. 19, no. 1, pp. 111–138, 2007. DOI: [10.1162/neco.2007.19.1.111](https://doi.org/10.1162/neco.2007.19.1.111)
- [52] L. Grigoryeva and J.-P. Ortega, "Echo state networks are universal," *Neural Networks*, vol. 108, pp. 495–508, 2018. DOI: [10.1016/j.neunet.2018.08.025](https://doi.org/10.1016/j.neunet.2018.08.025)
- [53] L. Gonon and J.-P. Ortega, "Fading memory echo state networks are universal," *Neural Networks*, vol. 138, pp. 10–13, 2021. DOI: [10.1016/j.neunet.2021.01.025](https://doi.org/10.1016/j.neunet.2021.01.025)
- [54] L. M. Pecora, T. L. Carroll, G. A. Johnson, D. J. Mar, and J. F. Heagy, "Fundamentals of synchronization in chaotic systems, concepts, and applications," *Chaos*, vol. 7, no. 4, pp. 520–543, 1997. DOI: [10.1063/1.166278](https://doi.org/10.1063/1.166278)

- [55] T. Lymburn, D. M. Walker, M. Small, and T. Jüngling, "The reservoir's perspective on generalized synchronization," *Chaos*, vol. 29, no. 9, p. 093 133, 2019. DOI: [10.1063/1.5120733](https://doi.org/10.1063/1.5120733)
- [56] H. Jaeger, M. Lukoševičius, D. Popovici, and U. Siewert, "Optimization and applications of echo state networks with leaky-integrator neurons," *Neural Networks*, vol. 20, no. 3, pp. 335–352, 2007. DOI: [10.1016/j.neunet.2007.04.016](https://doi.org/10.1016/j.neunet.2007.04.016)
- [57] C. Gallicchio, A. Micheli, and L. Pedrelli, "Deep reservoir computing: A critical experimental analysis," *Neurocomputing*, vol. 268, pp. 87–99, 2017. DOI: [10.1016/j.neucom.2016.12.089](https://doi.org/10.1016/j.neucom.2016.12.089)
- [58] C. Gallicchio and A. Micheli, "Echo state property of deep reservoir computing networks," *Cognitive Computation*, vol. 9, no. 3, pp. 337–350, 2017. DOI: [10.1007/s12559-017-9461-9](https://doi.org/10.1007/s12559-017-9461-9)
- [59] C. Gallicchio, A. Micheli, and L. Pedrelli, "Design of deep echo state networks," *Neural Networks*, vol. 108, pp. 33–47, 2018. DOI: [10.1016/j.neunet.2018.08.002](https://doi.org/10.1016/j.neunet.2018.08.002)
- [60] C. Sheng, J. Wang, Y.-a. Ren, W.-y. Wang, and Z.-y. Qu, "Ensemble of echo state networks with PCA dimensionality reduction for time series forecasting," *Neural Computing and Applications*, vol. 35, pp. 17 825–17 842, 2023. DOI: [10.1007/s00521-023-08646-4](https://doi.org/10.1007/s00521-023-08646-4)
- [61] J. Fu, G. Li, J. Tang, L. Xia, L. Wang, and S. Duan, "A double-cycle echo state network topology for time series prediction," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 33, no. 9, p. 093 113, Sep. 2023, ISSN: 1054-1500, 1089-7682. DOI: [10.1063/5.0159966](https://doi.org/10.1063/5.0159966) Accessed: Aug. 12, 2025.
- [62] A. Haluszczynski and C. Răth, "Good and bad predictions: Assessing and improving the replication of chaotic attractors by means of reservoir computing," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 29, no. 10, p. 103 143, Oct. 2019, ISSN: 1054-1500, 1089-7682. DOI: [10.1063/1.5118725](https://doi.org/10.1063/1.5118725) Accessed: Aug. 12, 2025.
- [63] L. Jaurigue, "Chaotic attractor reconstruction using small reservoirs—the influence of topology," *Machine Learning: Science and Technology*, vol. 5, no. 3, p. 035 058, Sep. 2024, ISSN: 2632-2153. DOI: [10.1088/2632-2153/ad6ee8](https://doi.org/10.1088/2632-2153/ad6ee8) Accessed: Aug. 12, 2025.
- [64] Y. Kawai, J. Park, and M. Asada, "A small-world topology enhances the echo state property and signal propagation in reservoir computing," *Neural Networks*, vol. 112, pp. 15–23, Apr. 2019, ISSN: 08936080. DOI: [10.1016/j.neunet.2019.01.002](https://doi.org/10.1016/j.neunet.2019.01.002) Accessed: Mar. 12, 2025.
- [65] X. Li, F. Bi, X. Yang, and X. Bi, "An Echo State Network With Improved Topology for Time Series Prediction,"
- [66] H. Ma, D. Prosperino, and C. Răth, "A novel approach to minimal reservoir computing," *Scientific Reports*, vol. 13, no. 1, p. 12 970, Aug. 2023, ISSN: 2045-2322. DOI: [10.1038/s41598-023-39886-w](https://doi.org/10.1038/s41598-023-39886-w) Accessed: Aug. 12, 2025.
- [67] P. Erdős and A. Rényi, "On random graphs I," *Publicationes Mathematicae (Debrecen)*, vol. 6, pp. 290–297, 1959.
- [68] A. Rodan and P. Tiño, "Simple deterministically constructed cycle reservoirs with regular jumps," *Neural Computation*, vol. 24, no. 7, pp. 1822–1852, 2012. DOI: [10.1162/NECO_a.00297](https://doi.org/10.1162/NECO_a.00297)

- [69] B. Li, R. Fong, and P. Tiño, “Simple cycle reservoirs are universal,” *arXiv preprint arXiv:2409.14065*, 2024.
- [70] R. Fong, B. Li, and P. Tiño, “Universality of real minimal complexity reservoir,” *arXiv preprint arXiv:2501.13450*, 2025.
- [71] Y. Kawai, J. Park, and M. Asada, “A small-world topology enhances the echo state property and signal propagation in reservoir computing,” *Neural Networks*, vol. 112, pp. 15–23, 2019. DOI: [10.1016/j.neunet.2019.01.002](https://doi.org/10.1016/j.neunet.2019.01.002)
- [72] Z. Deng and Y. Zhang, “Collective behavior of a small-world recurrent neural system with scale-free distribution,” *IEEE Transactions on Neural Networks*, vol. 18, no. 5, pp. 1364–1375, 2007. DOI: [10.1109/TNN.2007.894082](https://doi.org/10.1109/TNN.2007.894082)
- [73] M. Lukoševičius, “A practical guide to applying echo state networks,” *Lecture Notes in Computer Science*, vol. 7700, pp. 659–686, 2012. DOI: [10.1007/978-3-642-35289-8_36](https://doi.org/10.1007/978-3-642-35289-8_36)
- [74] L. Grigoryeva, A. Hart, and J.-P. Ortega, “Learning strange attractors with reservoir systems,” *Nonlinearity*, vol. 36, no. 9, pp. 4674–4708, Sep. 2023, ISSN: 0951-7715, 1361-6544. DOI: [10.1088/1361-6544/ace492](https://doi.org/10.1088/1361-6544/ace492) Accessed: Jul. 28, 2025.
- [75] S. Basterrech, *Empirical Analysis of the Necessary and Sufficient Conditions of the Echo State Property*, Mar. 2017. arXiv: [1703.06664 \[cs\]](https://arxiv.org/abs/1703.06664). Accessed: Oct. 21, 2024.
- [76] C. Gallicchio, *Chasing the Echo State Property*, Sep. 2019. arXiv: [1811.10892 \[cs, stat\]](https://arxiv.org/abs/1811.10892). Accessed: May 30, 2023.
- [77] F. Martinuzzi, “Minimal Deterministic Echo State Networks Outperform Random Reservoirs in Learning Chaotic Dynamics,” vol. 35, no. 9, p. 091 108, Jul. 2025. DOI: [10.1063/5.0288751](https://doi.org/10.1063/5.0288751) Accessed: Aug. 12, 2025.
- [78] Y. Sakemi, S. Nobukawa, T. Matsuki, T. Morie, and K. Aihara, “Learning reservoir dynamics with temporal self-modulation,” *Communications Physics*, vol. 7, no. 1, p. 29, Jan. 2024, ISSN: 2399-3650. DOI: [10.1038/s42005-023-01500-w](https://doi.org/10.1038/s42005-023-01500-w) Accessed: Aug. 12, 2025.
- [79] A. Pikovsky, M. Rosenblum, and J. Kurths, “Synchronization: A universal concept in nonlinear sciences,”
- [80] A. Nazerian, J. D. Hart, M. Lodi, and F. Sorrentino, “The efficiency of synchronization dynamics and the role of network syncreactivity,” *Nature Communications*, vol. 15, no. 1, p. 9003, Oct. 2024, ISSN: 2041-1723. DOI: [10.1038/s41467-024-52486-0](https://doi.org/10.1038/s41467-024-52486-0) Accessed: Dec. 3, 2025.
- [81] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, Second edition. Boulder, CO: Westview Press, a member of the Perseus Books Group, 2015, ISBN: 978-0-8133-4910-7.
- [82] L. Arnold, *Random Dynamical Systems* (Springer Monographs in Mathematics). Berlin, Heidelberg: Springer Berlin Heidelberg, 1998, ISBN: 978-3-642-08355-6 978-3-662-12878-7. DOI: [10.1007/978-3-662-12878-7](https://doi.org/10.1007/978-3-662-12878-7) Accessed: Jul. 28, 2025.
- [83] H. Crauel, A. Debussche, and F. Flandoli, “Random attractors,”
- [84] L. Jaurigue, “Chaotic attractor reconstruction using small reservoirs—the influence of topology,” *Machine Learning: Science and Technology*, vol. 5, no. 3, p. 035 058, Sep. 2024, ISSN: 2632-2153. DOI: [10.1088/2632-2153/ad6ee8](https://doi.org/10.1088/2632-2153/ad6ee8) Accessed: Nov. 23, 2025.

- [85] Y. Kuramoto and D. Battogtokh, "Coexistence of Coherence and Incoherence in Nonlocally Coupled Phase Oscillators," vol. 5, no. 4, 2002.
- [86] D. M. Abrams and S. H. Strogatz, "Chimera States for Coupled Oscillators," *Physical Review Letters*, vol. 93, no. 17, p. 174 102, Oct. 2004, ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.93.174102](https://doi.org/10.1103/PhysRevLett.93.174102) Accessed: Jul. 28, 2025.
- [87] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*. Englewood Cliffs, N.J: Prentice-Hall, 1976, ISBN: 978-0-13-212589-5.
- [88] "Solving ordinary differential equations I. nonstiff problems," *Mathematics and Computers in Simulation*, vol. 29, no. 5, p. 447, Oct. 1987, ISSN: 03784754. DOI: [10.1016/0378-4754\(87\)90083-8](https://doi.org/10.1016/0378-4754(87)90083-8) Accessed: Feb. 9, 2026.
- [89] J. Hoschek, "Spline approximation of offset curves," *Computer Aided Geometric Design*, 1988. DOI: [10.1016/0167-8396\(88\)90018-0](https://doi.org/10.1016/0167-8396(88)90018-0) Accessed: Feb. 9, 2026.
- [90] Y. Nievergelt, "Computing circles and spheres of arithmetic least squares," *Computer Physics Communications*, 1994. DOI: [10.1016/0010-4655\(94\)90082-5](https://doi.org/10.1016/0010-4655(94)90082-5) Accessed: Feb. 9, 2026.
- [91] T. Sauer, J. A. Yorke, and M. Casdagli, "Embedology," *Journal of Statistical Physics*, 1991. DOI: [10.1007/bf01053745](https://doi.org/10.1007/bf01053745) Accessed: Jan. 29, 2026.
- [92] A. M. Fraser, "Using Mutual Information to Estimate Metric Entropy," in *Dimensions and Entropies in Chaotic Systems*, G. Mayer-Kress, Ed., Berlin, Heidelberg: Springer, 1986, pp. 82–91, ISBN: 978-3-642-71001-8. DOI: [10.1007/978-3-642-71001-8_11](https://doi.org/10.1007/978-3-642-71001-8_11)
- [93] M. B. Kennel, R. Brown, and H. D. I. Abarbanel, "Determining embedding dimension for phase-space reconstruction using a geometrical construction," *Physical Review A*, vol. 45, no. 6, pp. 3403–3411, Mar. 1992. DOI: [10.1103/PhysRevA.45.3403](https://doi.org/10.1103/PhysRevA.45.3403) Accessed: Jan. 29, 2026.
- [94] E. Ott, *Chaos in Dynamical Systems*, 2nd ed. Cambridge University Press, Aug. 2002, ISBN: 978-0-521-81196-5 978-0-521-01084-9 978-0-511-80326-0. DOI: [10.1017/CBO9780511803260](https://doi.org/10.1017/CBO9780511803260) Accessed: Feb. 8, 2026.
- [95] A. Wolf, J. Swift, H. L. Swinney, and J. Vastano, "Determining Lyapunov exponents from a time series," *Physica D: Nonlinear Phenomena*, vol. 16, no. 3, pp. 285–317, Jul. 1985. DOI: [10.1016/0167-2789\(85\)90011-9](https://doi.org/10.1016/0167-2789(85)90011-9) Accessed: Feb. 8, 2026.
- [96] P. Royston and M. K. B. Parmar, "Restricted mean survival time: An alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome," *BMC medical research methodology*, vol. 13, p. 152, Dec. 2013, ISSN: 1471-2288. DOI: [10.1186/1471-2288-13-152](https://doi.org/10.1186/1471-2288-13-152)
- [97] X. Chen and M. Baron, "Change-Point Analysis of Survival Data with Application in Clinical Trials," *Open Journal of Statistics*, vol. 04, no. 09, pp. 663–677, 2014, ISSN: 2161-718X, 2161-7198. DOI: [10.4236/ojs.2014.49062](https://doi.org/10.4236/ojs.2014.49062) Accessed: Feb. 8, 2026.
- [98] Y. Wei, C. Ji, F. Galvan, S. Couvillon, G. Orellana, and J. Momoh, "Non-Stationary Random Process for Large-Scale Failure and Recovery of Power Distribution," *Applied Mathematics*, vol. 7, no. 3, pp. 233–249, Feb. 2016. DOI: [10.4236/am.2016.73022](https://doi.org/10.4236/am.2016.73022) Accessed: Feb. 8, 2026.
- [99] F. Hutter, H. Hoos, and K. Leyton-Brown, "An Efficient Approach for Assessing Hyperparameter Importance,"

- [100] A. M. Fraser and H. L. Swinney, "Independent coordinates for strange attractors from mutual information," *Physical Review A*, 1986. DOI: [10.1103/PhysRevA.33.1134](https://doi.org/10.1103/PhysRevA.33.1134) Accessed: Jan. 29, 2026.

List of Figures

1.1	Schematic of the reservoir computing framework. The input signal $\mathbf{u}(n)$ is fed into a reservoir with fixed input weights $\mathbf{W}_{\text{in}}$ and recurrent weights $\mathbf{W}$. The high-dimensional state $\mathbf{x}(n)$ is read out by a trainable linear layer $\mathbf{W}_{\text{out}}$.	8
1.2	Comparison of common reservoir topologies. (a) Random sparse with Erdős-Rényi connectivity. (b) Simple cycle reservoir (SCR): a single directed ring. (c) Cycle with jumps (CRJ): cycle plus long-range skip connections. (d) Delay line reservoir (DLR): a unidirectional chain.	17
1.3	The ESN training and inference pipeline. During the open-loop phase, the true input drives the reservoir and states are collected for readout training. During the closed-loop phase, the network's own output is fed back as input for autonomous forecasting.	19
2.1	Synchronization of reservoir states from different initial conditions. (a) $D(t) = \ \mathbf{x}_1(t) - \mathbf{x}_2(t)\ $ for a single input sequence across several random initializations. (b) $D(t)$ averaged over random initializations for each of several input sequences (blue), and averaged over both initializations and input sequences (green).	26
2.2	The convergence of pairs of trajectories of internal states $\ \mathbf{x}_1(t) - \mathbf{x}_2(t)\ $ where one starts from random initial conditions, the other from the zero-state, for four randomly picked reservoirs each. While under the purely linear evolution Eq. (2.3) all distances decay with ρ^t , a faster decay is observed for the non-linear system. For $\rho = 0.5$ (a), the two types of decay are very similar. For $\rho = 0.98$ (b), the damping due to the activation function becomes much more relevant, and the non-linear system's decay depends more strongly on the individual reservoir.	28
2.3	The convergence of pairs of trajectories of internal states $ \mathbf{x}_1(t) - \mathbf{x}_2(t) $ for a single reservoir with spectral radius $\rho = 1.1$. While under the purely linear dynamics the distances grow asymptotically like ρ^t , the distances in the non-linear system without input remain constant, and with driving (here: logistic equation), this system exhibits the ESP and the internal states converge faster as the input scaling $a \rightarrow 1$.	29

2.4	A histogram of transient times for four different reservoirs (each color represents one reservoir) with identical hyperparameters but different random weight initializations. For the upper panel, the driving signal is the logistic map; for the lower panel, the driving signal is white noise.	30
2.5	The number of exceedances of mean transient times obtained for 10000 reservoirs, for different spectral radii ρ ($N = 100$ nodes, $n_{\text{links}} = 200$, $\alpha = 0.1$). The black vertical markers indicate the benchmarks derived for the linear update Eq. (2.3) for $\rho < 1$. Driving signal: logistic equation. See also Table 2.1.	31
2.6	The number of exceedances of mean transient times obtained for 10000 reservoirs for different link numbers n_{links} , $N = 100$, $\rho = 1.5$, driving signal is logistic map.	33
2.7	The number of exceedances of mean transient times obtained for 10000 reservoirs with different numbers of nodes N but fixed link density of 1%, spectral radius $\rho = 1.5$, driving signal logistic map.	34
2.8	The number of exceedances of mean transient times obtained for 10000 reservoirs with different numbers of nodes N but fixed number of links $n_{\text{links}} = 200$, spectral radius $\rho = 1.5$, driving signal logistic map.	35
2.9	Number of exceedances of mean transient times obtained for 10^4 reservoirs for different leak rates α ($N = 100$, $n_{\text{links}} = 100$ (1%), $\rho = 1.5$). Driving signal: logistic map.	36
2.10	Classification of nodes according to their graph-theoretic role. Core nodes (green) belong to cycles and sustain autonomous activity; driven nodes (blue) inherit oscillations from cores; quiescent nodes (gray) collapse to zero. Parameters: $\rho = 1.5$, $N = 50$, $n_{\text{links}} = 50$	37
2.11	Schematic of the dendrocycle topology. Core nodes (green) form a directed cycle providing recurrent feedback. Dendritic nodes (blue) are arranged in chains emanating from the core, receiving but not returning activity. Quiescent nodes (gray) form an acyclic subgraph disconnected from any cycle. Arrows indicate edge directions.	39
2.12	Mean valid prediction time (in Lyapunov times λt) for Lorenz forecasting as a function of core fraction c and dendritic fraction d . Performance improves along the diagonal $c + d \rightarrow 1$, where quiescent nodes are minimized. Each cell averages over 20 reservoir realizations and 20 independent trajectories.	40
3.1	Geometric definition of the sagitta. (a) A smooth arc with triplet (A, B, C) spaced by half-width h . The sagitta $s(t; h)$ is the orthogonal distance from B to the chord $\overline{AC}$; the shaded area shows the region where the trajectory deviates from linear interpolation. (b) Pointwise sagitta $s_i(m)$ computed along a Lorenz trajectory (projected onto the x - y plane), with $m = 4$ and $\Delta t_0 = 0.005$. Red segments mark the sagitta at selected high-curvature triplets.	48
3.2	Verification of the quadratic scaling $S(m) \propto h^2$ (Proposition 3.1) for (a) the Lorenz system and (b) the Rössler system. The base sampling step is $\Delta t_0 = 0.005$ with 95th-percentile aggregation. Dashed red lines show the h^2 power law fitted to the first 20 spans; the fitted slopes are indicated in the legend. The horizontal dotted line marks $\varepsilon = 0.1$, and the dash-dotted vertical line marks the selected Δt^*	50

3.3	Scaling of the absolute and relative sagitta for scalar observations. The x -component of the Lorenz system is delay-embedded ($\ell = 17$, $d_e = 3$) and standardized. (a) $S(m) \propto h^2$ with fitted slope 1.91. (b) $R(m) \propto h$ with fitted slope 0.96, confirming the linear scaling of Proposition 3.1. The horizontal line marks $\varepsilon = 0.1$	51
3.4	Idempotence of the absolute selection rule on the Lorenz system. (a) $S(m)$ on the original series ($\Delta t_0 = 0.002$); the shaded region indicates admissible spans, with $m^* = 11$ and $\Delta t^* = 0.022$. (b) $\hat{S}(k)$ on the decimated series: $k = 1$ (square) satisfies $\hat{S}(1) = 0.0896 \leq \varepsilon$, while $k = 2$ (triangle) exceeds the tolerance at $\hat{S}(2) = 0.3387$, so $\hat{m}^* = 1$ and the rule returns Δt^* unchanged.	53
3.5	Selected time step Δt^* as a function of tolerance ε for the Lorenz, Rössler, and Chen systems ($\Delta t_0 = 0.002$, 95th-percentile aggregation). The vertical dotted line marks $\varepsilon = 0.1$. Each system occupies a distinct band reflecting its characteristic curvature; the staircase structure arises from the integer constraint on the span m^*	56
3.6	Lorenz attractor colored by pointwise sagitta $s_i(m)$ with $m = 3$ and $\Delta t_0 = 0.01$. High curvature (bright) concentrates at the outer rims of the lobes and at lobe-switching transitions near the origin; the lobe interiors (dark) are nearly linear at this time scale. Color scale clipped at the 97th percentile.	57
4.1	Synthetic error signal e_t defined in Eq. (4.4), used as the reference test case for the sensitivity analysis of EFH in Sections 4.1.4–4.2.7.	62
4.2	Gate function $g_t(\tau, \delta)$ for different softness ratios $R = \delta/\tau$. Dashed vertical lines indicate the corresponding EFH values. As R increases, the transition region broadens and the effective horizon decreases.	64
4.3	Survival process $s_t(\tau, \delta)$ for different softness ratios. The survival curve collapses once the cumulative effect of below-unity gate values accumulates sufficiently. Dashed lines mark the EFH values.	65
4.4	Cumulative sum $\sum_{k=1}^t s_k$ as a function of time. The plateau value at $t = T$ equals $\text{EFH}(\tau, \delta)$. Curves are shown for different softness ratios R	66
4.5	EFH as a function of the softness ratio $R = \delta/\tau$ (log scale). The horizon is stable for $R \leq 0.1$ and degrades rapidly for $R \geq 0.3$, motivating the practical recommendation $R \in [0.01, 0.1]$	67
4.6	Monotonicity of $\text{EFH}(\tau, \delta)$ with respect to τ , for several values of δ . The black dashed curve shows the hard hitting time $t^* - 1$. All EFH curves are monotonically nondecreasing in τ , as guaranteed by Lemma 4.2 (1), and converge to the hard hitting time as δ decreases.	69
4.7	Convergence of $\text{EFH}(\tau, \delta)$ to the hard hitting time $t^* - 1$ as $\delta \rightarrow 0^+$. The EFH (blue curve) approaches $t^* - 1 = 39$ for small δ and tends to $1 - 2^{-T} \approx 1$ for large δ , in agreement with Lemmas 4.3 and 4.4.	70
4.8	Survival-analysis decomposition of EFH for the reference error signal ($\tau = 0.2$, $R = 0.1$). Top: Gate g_t . Middle: Stepwise hazard $h_t = -\log g_t$. Bottom: Cumulative hazard H_t (green, left axis) and survival $s_t = \exp(-H_t)$ (red dashed, right axis). The rapid accumulation of hazard near the threshold crossing drives the exponential collapse of s_t	72

4.9	Gradient concentration of the EFH loss. The blue curve shows the error signal e_t ; the red shaded region shows $-\partial \text{EFH} / \partial e_t$ (right axis). The gradient is sharply localised near the threshold crossing ($t \approx 39$), confirming Proposition 4.5: optimization pressure concentrates at the onset of failure.	73
4.10	Top: Mean $\pm$ standard deviation of EFH and hard hitting time over 500 noise realizations at each noise level σ . Bottom: Coefficient of variation (CV). EFH maintains a substantially lower CV than the hard hitting time across all noise levels, demonstrating its robustness as a benchmarking statistic.	74
4.11	EFH evaluated on four qualitatively different error profiles: exponential growth, linear ramp, sigmoidal transition, and step-like failure. In each panel, the coloured curve is e_t , the shaded region shows s_t , the dot-dashed line marks the EFH, and the dotted line marks the hard hitting time $t^* - 1$. EFH produces sensible horizons across all divergence patterns.	75
5.1	Optimization history for the four benchmark systems. Each point represents a single TPE trial; the step curve tracks the cumulative best EFH in Lyapunov times. All four systems exhibit rapid initial improvement followed by saturation within the first 100–150 trials.	80
5.2	Relative hyperparameter importance computed via the fANOVA decomposition for each benchmark system. The ranking varies across systems, reflecting differences in the dynamical regime and the sensitivity of the ESN to each design choice.	82
5.3	Forecast horizon versus spectral radius ρ for all four systems. Each point is a single trial coloured by its EFH value.	84
5.4	Forecast horizon versus input scaling σ_{in} for all four systems.	85
5.5	Forecast horizon versus leak rate α for all four systems.	86
5.6	Forecast horizon versus core fraction c for all four systems.	87
5.7	Forecast horizon versus dendritic fraction d for all four systems.	88
5.8	Scatter plot of the core fraction c and dendritic fraction d for all trials, coloured by the EFH in Lyapunov times. The diagonal $c + d = 1$ is indicated. Across all four systems, the highest-performing trials concentrate near this diagonal, confirming that the optimizer rediscovers the design principle of Chapter 2 without any structural constraint being imposed.	89

List of Tables

2.1	The mean transient time $\bar{T}$ (average over 10000 reservoirs) in dependence on spectral radius ρ . Input data is the logistic map, $N = 100$, $n_{\text{links}} = 200$. . .	31
2.2	Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the number of links n_{links} , for reservoirs of size $N = 100$ with spectral radius $\rho = 1.5$. The driving signal is the logistic map. The fraction of reservoirs that converged within 10^4 steps is also reported.	32
2.3	Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the number of nodes N , with link density fixed at 1% ($n_{\text{links}} = 0.01N^2$). Spectral radius is $\rho = 1.5$ and the driving signal is the logistic map. The fraction of reservoirs that converged within 10^4 steps is reported, along with the mean transient time $\bar{T}$ of successful cases.	33
2.4	Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the number of nodes N , with the total number of links fixed at $n_{\text{links}} = 200$. Spectral radius is $\rho = 1.5$ and the driving signal is the logistic map. Reported are the fraction of reservoirs that converged within 10^4 steps and the mean transient time $\bar{T}$ of successful cases.	34
2.5	Mean transient time $\bar{T}$ (averaged over 10^4 reservoirs) as a function of the leak rate α , for reservoirs of size $N = 100$ with link density 1% ($n_{\text{links}} = 100$) and spectral radius $\rho = 1.5$. The driving signal is the logistic map. Reported are the fraction of reservoirs that converged within 10^4 steps and the mean transient time $\bar{T}$ of successful cases.	35
5.1	Hyperparameter search space for the TPE optimization. Log-scale sampling is used for parameters spanning multiple orders of magnitude.	80

A Appendix: Transient Times

A.1 Bounds on Initial Separation

This appendix provides a rigorous derivation of the estimate used in Eq. (2.4) and clarifies the difference between the typical separation of random initial states and the strict maximal separation.

Consider two independent random initial states $\vec{s}_1(0), \vec{s}_2(0) \in [-1, 1]^N$, each drawn component-wise from a uniform distribution $U(-1, 1)$. Define the initial difference

$$D(0) = \|\vec{s}_1(0) - \vec{s}_2(0)\|. \quad (\text{A.1})$$

For each coordinate $d_i = s_{1,i}(0) - s_{2,i}(0)$, with $s_{1,i}(0), s_{2,i}(0) \sim U(-1, 1)$ independently, one computes

$$\mathbb{E}[d_i] = 0, \quad \mathbb{E}[d_i^2] = \mathbb{E}[s_{1,i}(0)^2] + \mathbb{E}[s_{2,i}(0)^2] = \frac{2}{3}. \quad (\text{A.2})$$

Thus, the expected squared distance between the two initial states is

$$\mathbb{E}[D(0)^2] = N \cdot \frac{2}{3}, \quad (\text{A.3})$$

so that the root-mean-square (RMS) value is

$$D_{\text{RMS}} = \sqrt{\frac{2N}{3}}. \quad (\text{A.4})$$

By concentration of measure for sums of i.i.d. bounded random variables (Hoeffding inequality), $D(0)$ is tightly concentrated around D_{RMS} with high probability as N grows.

Hence, for typical random initializations, the linearized dynamics give

$$D(t) = \|\vec{s}_1(t) - \vec{s}_2(t)\| \leq \rho^t \|\vec{s}_1(0) - \vec{s}_2(0)\|, \quad (\text{A.5})$$

$$D(t) < \epsilon \quad \Rightarrow \quad \rho^t D(0) < \epsilon, \quad (\text{A.6})$$

$$t < \frac{\ln(\epsilon) - \ln(D(0))}{\ln \rho}, \quad (\text{A.7})$$

$$D(0) \approx \sqrt{2N/3}, \quad (\text{A.8})$$

$$T < \frac{\ln(10^{-7}) - \ln(\sqrt{2N/3})}{\ln \rho}. \quad (\text{A.9})$$

For a strict upper bound, since each coordinate difference satisfies $|d_i| \leq 2$, one has

$$D(0)^2 = \sum_{i=1}^N d_i^2 \leq 4N, \quad (\text{A.10})$$

so that

$$D(0) \leq 2\sqrt{N}. \quad (\text{A.11})$$

Using this maximal separation, the worst-case transient time bound is obtained:

$$T < \frac{\ln(10^{-7}) - \ln(2\sqrt{N})}{\ln \rho}. \quad (\text{A.12})$$

Equation (A.9) provides an RMS estimate of the transient time, valid with high probability for random initializations due to concentration of measure. In contrast, Eq. (A.12) offers a rigorous worst-case bound. Since the transient time depends only logarithmically on the initial separation $D(0)$, the difference between these estimates is negligible in practice. Therefore, it is preferable to employ the RMS-based expression

$$T \approx \frac{\ln(10^{-7}/\sqrt{2N/3})}{\ln \rho}, \quad (\text{A.13})$$

as reported in the main text, rather than the more pessimistic worst-case bound. This completes the proof of Eq. (2.4).

B Appendix: Geometric time discretization

This appendix provides complete derivations and proofs for the results stated in Chapter 3. We maintain the notation established there: $y : [0, T] \rightarrow \mathbb{R}^d$ denotes a smooth trajectory, $v = \dot{y}$ and $a = \ddot{y}$ are the velocity and acceleration, $|\cdot|$ is the Euclidean norm, and $P_v = vv^\top / |v|^2$ is the orthogonal projector onto the span of v .

B.1 Derivation of the sagitta-curvature relation

We derive the fundamental relation between the sagitta, the normal acceleration, and the curvature. The argument proceeds in three steps: we first establish the connection between linearization error and normal acceleration, then define the sagitta and show it captures the same quantity, and finally express the result in terms of curvature.

Linearization error and normal acceleration. Fix a time t and a half-width $h > 0$. Consider the triplet of points $A = y(t - h)$, $B = y(t)$, and $C = y(t + h)$. Linear interpolation between A and C approximates the trajectory by the chord joining these endpoints. The interpolant at the midpoint is

$$y_{\text{lin}}(t) = \frac{A + C}{2}. \quad (\text{B.1})$$

The error of this approximation is $e(t; h) = B - y_{\text{lin}}(t) = B - (A + C)/2$.

To evaluate this error, expand A and C in Taylor series around t . Writing $v = \dot{y}(t)$, $a = \ddot{y}(t)$, and $j = y^{(3)}(t)$ for the velocity, acceleration, and jerk, respectively, we have

$$A = y(t - h) = y(t) - vh + \frac{1}{2}ah^2 - \frac{1}{6}jh^3 + \mathcal{O}(h^4), \quad (\text{B.2})$$

$$C = y(t + h) = y(t) + vh + \frac{1}{2}ah^2 + \frac{1}{6}jh^3 + \mathcal{O}(h^4). \quad (\text{B.3})$$

Adding these expressions,

$$A + C = 2y(t) + ah^2 + \mathcal{O}(h^4), \quad (\text{B.4})$$

where the odd powers of h cancel by symmetry. The linear interpolant at the midpoint is therefore

$$y_{\text{lin}}(t) = \frac{A+C}{2} = y(t) + \frac{1}{2}ah^2 + \mathcal{O}(h^4), \quad (\text{B.5})$$

and the interpolation error is

$$e(t; h) = B - y_{\text{lin}}(t) = y(t) - y(t) - \frac{1}{2}ah^2 + \mathcal{O}(h^4) = -\frac{1}{2}ah^2 + \mathcal{O}(h^4). \quad (\text{B.6})$$

The leading-order error is proportional to the acceleration and scales quadratically with h .

However, the chord from A to C itself has a definite direction, and only the component of e orthogonal to this chord represents geometric deviation that linear interpolation cannot capture. To isolate this component, we compute the chord vector:

$$C - A = 2vh + \frac{1}{3}jh^3 + \mathcal{O}(h^5). \quad (\text{B.7})$$

The magnitude of the chord is

$$|C - A| = 2|v|h \left(1 + \frac{v \cdot j}{6|v|^2}h^2 + \mathcal{O}(h^4) \right) = 2|v|h + \mathcal{O}(h^3), \quad (\text{B.8})$$

and the unit chord direction is

$$u = \frac{C - A}{|C - A|} = \frac{v}{|v|} + \mathcal{O}(h^2) = \hat{v} + \mathcal{O}(h^2), \quad (\text{B.9})$$

where $\hat{v} = v/|v|$ is the unit tangent vector. To leading order, the chord aligns with the instantaneous velocity.

The orthogonal projector onto the complement of the chord direction is $I - uu^\top$. Applying it to the error vector (B.6) and using (B.9),

$$\begin{aligned} (I - uu^\top)e(t; h) &= (I - \hat{v}\hat{v}^\top + \mathcal{O}(h^2)) \left(-\frac{1}{2}ah^2 + \mathcal{O}(h^4) \right) \\ &= -\frac{1}{2}(I - \hat{v}\hat{v}^\top)a h^2 + \mathcal{O}(h^4) \\ &= -\frac{1}{2}a_\perp h^2 + \mathcal{O}(h^4), \end{aligned} \quad (\text{B.10})$$

where $a_\perp = (I - P_v)a$ is the normal component of acceleration. The tangential component $a_\parallel = P_v a$ is annihilated by the projector. Taking norms,

$$|e_\perp(t; h)| = \frac{1}{2}|a_\perp(t)|h^2 + \mathcal{O}(h^4). \quad (\text{B.11})$$

This establishes equation (3.6) of the main text: the orthogonal interpolation error is controlled by the normal acceleration and scales as h^2 .

The sagitta. The sagitta measures the same geometric deviation but is defined without reference to the error of an interpolant. Recall the definition (3.12):

$$s(t; h) = |(I - uu^\top)(B - A)|. \quad (\text{B.12})$$

This is the orthogonal distance from the midpoint B to the line through A and C . We now show that $s(t; h)$ equals $|e_\perp(t; h)|$ to leading order.

From the Taylor expansion (B.2), the displacement from A to B is

$$B - A = y(t) - y(t - h) = vh - \frac{1}{2}ah^2 + \frac{1}{6}jh^3 + \mathcal{O}(h^4). \quad (\text{B.13})$$

Applying the orthogonal projector $(I - uu^\top)$ with $u = \hat{v} + \mathcal{O}(h^2)$:

$$\begin{aligned} (I - uu^\top)(B - A) &= (I - \hat{v}\hat{v}^\top) \left(vh - \frac{1}{2}ah^2 + \frac{1}{6}jh^3 \right) + \mathcal{O}(h^4) \\ &= (I - \hat{v}\hat{v}^\top)(vh) - \frac{1}{2}(I - \hat{v}\hat{v}^\top)(a)h^2 + \mathcal{O}(h^3). \end{aligned} \quad (\text{B.14})$$

The first term vanishes because $(I - \hat{v}\hat{v}^\top)v = v - \hat{v}(\hat{v} \cdot v) = v - \hat{v}|v| = v - v = 0$. Thus,

$$(I - uu^\top)(B - A) = -\frac{1}{2}a_\perp h^2 + \mathcal{O}(h^3), \quad (\text{B.15})$$

and taking norms,

$$s(t; h) = \frac{1}{2}|a_\perp(t)|h^2 + \mathcal{O}(h^3). \quad (\text{B.16})$$

This is equation (3.20) of the main text. The sagitta and the orthogonal interpolation error coincide to leading order; they differ only in higher-order terms arising from the slight misalignment between the chord direction u and the tangent $\hat{v}$.

Connection to curvature. The curvature $\kappa(t)$ of a curve parametrized by time is defined as the rate of rotation of the unit tangent per unit arc length. For a trajectory $y(t)$ with velocity $v = \dot{y}$ and acceleration $a = \ddot{y}$, the curvature is given by

$$\kappa(t) = \frac{|v \times a|}{|v|^3} \quad (\text{in } \mathbb{R}^3), \quad \kappa(t) = \frac{\sqrt{|v|^2|a|^2 - (v \cdot a)^2}}{|v|^3} \quad (\text{in } \mathbb{R}^d). \quad (\text{B.17})$$

The numerator in the general formula is $|v| \cdot |a_\perp|$, since

$$|a_\perp|^2 = |a|^2 - |a_\parallel|^2 = |a|^2 - \frac{(v \cdot a)^2}{|v|^2}, \quad (\text{B.18})$$

which gives $|v|^2|a_\perp|^2 = |v|^2|a|^2 - (v \cdot a)^2$. Therefore,

$$\kappa(t) = \frac{|a_\perp(t)|}{|v(t)|^2}, \quad (\text{B.19})$$

or equivalently,

$$|a_\perp(t)| = \kappa(t)|v(t)|^2. \quad (\text{B.20})$$

This is equation (3.7) of the main text.

Substituting (B.20) into the sagitta formula (B.16),

$$s(t; h) = \frac{1}{2}\kappa(t)|v(t)|^2h^2 + \mathcal{O}(h^3). \quad (\text{B.21})$$

To express this in terms of the chord length $L = |C - A|$, use (B.8):

$$L = 2|v|h + \mathcal{O}(h^3), \quad \text{so} \quad |v|^2h^2 = \frac{L^2}{4} + \mathcal{O}(h^4). \quad (\text{B.22})$$

Substituting into (B.21),

$$s(t; h) = \frac{1}{2}\kappa(t) \cdot \frac{L^2}{4} + \mathcal{O}(h^3) = \frac{\kappa(t)L^2}{8} + \mathcal{O}(h^3). \quad (\text{B.23})$$

This is the classical sagitta-curvature relation stated in equation (3.21) of the main text. It expresses the sagitta purely in terms of the local curvature and the chord length, independent of the time parametrization.

Summary. The derivations establish a chain of equalities linking the orthogonal interpolation error, the sagitta, the normal acceleration, and the curvature:

$$|e_\perp(t; h)| = s(t; h) + \mathcal{O}(h^4) = \frac{1}{2}|a_\perp(t)|h^2 + \mathcal{O}(h^3) = \frac{\kappa(t)L^2}{8} + \mathcal{O}(h^3). \quad (\text{B.24})$$

All four quantities capture the same geometric information—the deviation of the trajectory from local linearity—but the sagitta has the practical advantage of being computable from point coordinates alone, without differentiation.

B.2 Proof of Proposition 3.1

Proposition 3.1 asserts that the discrete aggregates $S(m)$ and $R(m)$ inherit the scaling behavior of the continuous sagitta. We now provide a complete proof.

Proposition B.1 (Restatement of Proposition 3.1). *Let $y \in C^3([0, T], \mathbb{R}^d)$ with $\dot{y}(t) \neq 0$ for almost every $t \in [0, T]$, and suppose Δt_0 is small enough that the expansion (B.16) holds for $h = m\Delta t_0$ over the range of spans $m \in \{1, 2, \dots, M\}$ under consideration. Then*

$$S(m) = c(m\Delta t_0)^2 + \mathcal{O}((m\Delta t_0)^3), \quad c = \frac{1}{2}\mathcal{A}(\{|a_\perp(t_i)|\}), \quad (\text{3.26})$$

$$R(m) = \tilde{c}(m\Delta t_0) + \mathcal{O}((m\Delta t_0)^2), \quad \tilde{c} = \frac{1}{4}\mathcal{A}\left(\left\{\frac{|a_\perp(t_i)|}{|v(t_i)|}\right\}\right), \quad (\text{3.27})$$

where $\mathcal{A}$ is a robust, order-preserving aggregate operator applied over the index set $\{i : m \leq i \leq N - 1 - m\}$.

Proof. We prove the two claims in sequence.

Part 1: Asymptotics of $S(m)$. Fix a span $m \geq 1$ and consider the discrete sagitta at index i , defined by equation (3.22):

$$s_i(m) = |(I - u_i u_i^\top)(y_i - y_{i-m})|, \quad (\text{B.25})$$

where $u_i = (y_{i+m} - y_{i-m})/|y_{i+m} - y_{i-m}|$. The triplet (y_{i-m}, y_i, y_{i+m}) corresponds to the continuous triplet $(A, B, C) = (y(t_i - h), y(t_i), y(t_i + h))$ with $h = m\Delta t_0$ and $t_i = i\Delta t_0$.

By the continuous sagitta expansion (B.16), applied at time t_i with half-width $h = m\Delta t_0$,

$$s(t_i; m\Delta t_0) = \frac{1}{2} |a_\perp(t_i)| (m\Delta t_0)^2 + r_i(m), \quad (\text{B.26})$$

where the remainder satisfies $|r_i(m)| \leq C_3(m\Delta t_0)^3$ for some constant C_3 depending on bounds for $|y^{(3)}|$ and $|v|^{-1}$ over $[0, T]$.

The discrete sagitta $s_i(m)$ evaluates the same geometric construction on the sampled points. Since $y_{i\pm m} = y(t_i \pm m\Delta t_0)$ exactly (no interpolation or approximation is involved in the sampling), and since the sagitta formula (B.25) is the discrete instantiation of (3.12), we have

$$s_i(m) = s(t_i; m\Delta t_0) = \frac{1}{2} |a_\perp(t_i)| (m\Delta t_0)^2 + r_i(m). \quad (\text{B.27})$$

The discrete and continuous sagittae coincide pointwise; the expansion (B.26) applies without modification.

The aggregated sagitta is

$$S(m) = \mathcal{A}(\{s_i(m) : i \in I_m\}), \quad (\text{B.28})$$

where $I_m = \{m, m+1, \dots, N-1-m\}$ is the set of admissible indices. Substituting the expansion (B.27),

$$S(m) = \mathcal{A}\left(\left\{\frac{1}{2} |a_\perp(t_i)| (m\Delta t_0)^2 + r_i(m)\right\}_{i \in I_m}\right). \quad (\text{B.29})$$

We now use two properties of $\mathcal{A}$. First, $\mathcal{A}$ is *positive homogeneous of degree one*: for any $\lambda > 0$ and any finite set $\{x_i\}$,

$$\mathcal{A}(\{\lambda x_i\}) = \lambda \mathcal{A}(\{x_i\}). \quad (\text{B.30})$$

This holds for percentiles, trimmed means, and most robust location estimators. Second, $\mathcal{A}$ is *Lipschitz* in the sense that if $|x_i - \tilde{x}_i| \leq \eta$ for all i , then

$$|\mathcal{A}(\{x_i\}) - \mathcal{A}(\{\tilde{x}_i\})| \leq L\eta, \quad (\text{B.31})$$

for some constant $L \geq 1$ (for quantiles and trimmed means, $L = 1$).

Applying homogeneity to the leading term in (B.29),

$$\begin{aligned} S(m) &= \mathcal{A}\left(\left\{\frac{1}{2} |a_\perp(t_i)| (m\Delta t_0)^2\right\}_{i \in I_m}\right) + \mathcal{O}(\mathcal{A}(\{|r_i(m)|\})) \\ &= (m\Delta t_0)^2 \cdot \mathcal{A}\left(\left\{\frac{1}{2} |a_\perp(t_i)|\right\}_{i \in I_m}\right) + \mathcal{O}((m\Delta t_0)^3) \\ &= c(m\Delta t_0)^2 + \mathcal{O}((m\Delta t_0)^3), \end{aligned} \quad (\text{B.32})$$

where

$$c = \frac{1}{2} \mathcal{A}(\{|a_\perp(t_i)|\}_{i \in I_m}). \quad (\text{B.33})$$

The remainder estimate uses $|r_i(m)| \leq C_3(m\Delta t_0)^3$ and the boundedness of $\mathcal{A}$ on uniformly bounded inputs.

Strictly speaking, the index set I_m depends on m , so c could vary with m . However, for m small relative to N , the set I_m covers most of the trajectory, and as $\Delta t_0 \rightarrow 0$ with T fixed, the aggregate $\mathcal{A}(\{|a_\perp(t_i)|\})$ converges to the corresponding population quantity (e.g., the p th percentile of $|a_\perp|$ over $[0, T]$). For practical purposes, c is effectively constant over the range of small spans, and (B.32) establishes the claimed quadratic scaling.

Part 2: Asymptotics of $R(m)$. The relative sagitta is $R(m) = S(m)/L_{\text{med}}(m)$, where

$$L_{\text{med}}(m) = \text{median}(\{|y_{i+m} - y_{i-m}| : i \in I_m\}). \quad (\text{B.34})$$

We first establish the asymptotics of $L_{\text{med}}(m)$.

The chord length at index i is $L_i(m) = |y_{i+m} - y_{i-m}|$. Using the expansion (B.8) at time t_i with $h = m\Delta t_0$,

$$L_i(m) = 2|v(t_i)|(m\Delta t_0) + \tilde{r}_i(m), \quad (\text{B.35})$$

where $|\tilde{r}_i(m)| \leq \tilde{C}_3(m\Delta t_0)^3$ for some constant $\tilde{C}_3$.

Applying the median (a special case of the aggregate $\mathcal{A}$ with homogeneity and Lipschitz properties),

$$\begin{aligned} L_{\text{med}}(m) &= \text{median}(\{2|v(t_i)|(m\Delta t_0) + \tilde{r}_i(m)\}) \\ &= 2(m\Delta t_0) \text{median}(\{|v(t_i)|\}) + \mathcal{O}((m\Delta t_0)^3) \\ &= 2\bar{v}(m\Delta t_0) + \mathcal{O}((m\Delta t_0)^3), \end{aligned} \quad (\text{B.36})$$

where $\bar{v} = \text{median}(\{|v(t_i)|\})$.

For the quotient, using (B.32) and (B.36),

$$\begin{aligned} R(m) &= \frac{S(m)}{L_{\text{med}}(m)} \\ &= \frac{c(m\Delta t_0)^2 + \mathcal{O}((m\Delta t_0)^3)}{2\bar{v}(m\Delta t_0) + \mathcal{O}((m\Delta t_0)^3)} \\ &= \frac{c(m\Delta t_0)^2}{2\bar{v}(m\Delta t_0)} \cdot \frac{1 + \mathcal{O}(m\Delta t_0)}{1 + \mathcal{O}((m\Delta t_0)^2)} \\ &= \frac{c}{2\bar{v}}(m\Delta t_0) \cdot (1 + \mathcal{O}(m\Delta t_0)) \\ &= \tilde{c}(m\Delta t_0) + \mathcal{O}((m\Delta t_0)^2), \end{aligned} \quad (\text{B.37})$$

where

$$\tilde{c} = \frac{c}{2\bar{v}} = \frac{\frac{1}{2}\mathcal{A}(\{|a_{\perp}(t_i)|\})}{2\text{median}(\{|v(t_i)|\})} = \frac{\mathcal{A}(\{|a_{\perp}(t_i)|\})}{4\text{median}(\{|v(t_i)|\})}. \quad (\text{B.38})$$

To express $\tilde{c}$ more directly, note that if $\mathcal{A}$ and the median were both linear operators (e.g., both arithmetic means), we would have

$$\tilde{c} = \frac{1}{4}\mathcal{A}\left(\left\{\frac{|a_{\perp}(t_i)|}{|v(t_i)|}\right\}\right). \quad (\text{B.39})$$

In general, for nonlinear aggregates, equation (B.38) gives the exact form while (B.39) holds approximately when the distributions of $|a_{\perp}|$ and $|v|$ are not strongly correlated. For the purposes of the main text, we write $\tilde{c} \propto \mathcal{A}(\{|a_{\perp}(t_i)|/|v(t_i)|\})$ to indicate the proportionality without committing to exact equality.

Monotonicity. The quadratic scaling $S(m) \sim c(m\Delta t_0)^2$ with $c > 0$ implies that $S(m)$ is strictly increasing in m for small spans, provided Δt_0 is small enough that the $\mathcal{O}((m\Delta t_0)^3)$ remainder does not dominate. Specifically, for $m_2 > m_1$ both in the small-span regime,

$$\begin{aligned} S(m_2) - S(m_1) &= c((m_2\Delta t_0)^2 - (m_1\Delta t_0)^2) + \mathcal{O}((m_2\Delta t_0)^3) \\ &= c\Delta t_0^2(m_2^2 - m_1^2) + \mathcal{O}(\Delta t_0^3) \\ &> 0, \end{aligned} \tag{B.40}$$

for Δt_0 sufficiently small. The same reasoning applies to $R(m)$ with linear scaling. This monotonicity underpins the well-posedness of the selection rules (3.28) and (3.30). $\square$

Remark on the constant c . The constant $c = \frac{1}{2}\mathcal{A}(\{|a_\perp(t_i)|\})$ encodes a global characteristic of the trajectory's curvature distribution. When $\mathcal{A}$ is a high percentile (e.g., the 95th), c reflects near-worst-case curvature along the trajectory, providing a conservative bound. When $\mathcal{A}$ is a trimmed mean, c reflects typical curvature with outliers excluded. The choice of $\mathcal{A}$ thus controls the sensitivity of the procedure to rare high-curvature events, and should be selected based on whether such events are physically meaningful or represent noise and transients to be discounted.

B.3 Proofs of Theorems 3.3 and 3.4

We prove the idempotence theorems stated in Section 3.4. Both proofs share a common structure: the span mapping relation (3.2) shows that span k on the decimated series probes physical scale $k\Delta t^*$, which corresponds to span km^* on the original series. Since $2m^* \geq m^* + 1$ for any $m^* \geq 1$, the gap between the threshold and $S(m^* + 1)$ ensures that span 2 on the decimated series exceeds the tolerance, forcing $\hat{m}^* = 1$.

B.3.1 Idempotence of the absolute rule

Theorem B.2 (Restatement of Theorem 3.3). *Let $y \in C^3([0, T], \mathbb{R}^d)$ with $\dot{y}(t) \neq 0$ for almost every $t \in [0, T]$. Suppose $m \mapsto S(m)$ is strictly increasing on $\{1, \dots, M\}$ for some $M \geq m^* + 1$, and let m^* be the maximal span satisfying $S(m^*) \leq \varepsilon < S(m^* + 1)$. If Δt_0 is sufficiently small, then reapplying the selection rule to the decimated series yields $\hat{m}^* = 1$, so that $\hat{\Delta t}^* = \Delta t^*$.*

Proof. **Step 1: Quantify the gap.** By hypothesis, m^* is the largest integer with $S(m^*) \leq \varepsilon$, and the next span exceeds the threshold:

$$\delta := S(m^* + 1) - \varepsilon > 0. \tag{B.41}$$

This gap controls the robustness of the selection.

Step 2: Apply the span mapping. After decimation by stride m^* , the base step becomes $\hat{\Delta}t_0 = m^* \Delta t_0 = \Delta t^*$. By the span mapping relation (3.2), the aggregated sagittae on the decimated series satisfy

$$\hat{S}(1) = S(m^*) + \mathcal{O}((\Delta t^*)^3), \quad (\text{B.42})$$

$$\hat{S}(2) = S(2m^*) + \mathcal{O}((\Delta t^*)^3). \quad (\text{B.43})$$

Since $\Delta t^* = m^* \Delta t_0$, the remainders can be made arbitrarily small by reducing Δt_0 . Choose Δt_0 small enough that both remainder magnitudes are bounded by $\delta/4$.

Step 3: Span 1 is admissible. From (B.42) and the admissibility of m^* ,

$$\hat{S}(1) \leq S(m^*) + \frac{\delta}{4} \leq \varepsilon + \frac{\delta}{4}. \quad (\text{B.44})$$

For sufficiently small Δt_0 , the correction $\delta/4$ can be made smaller than any prescribed margin, so $\hat{S}(1) \leq \varepsilon$ up to $\mathcal{O}((\Delta t^*)^3)$ corrections.

Step 4: Span 2 is inadmissible. By the strict monotonicity of S and the inequality $2m^* \geq m^* + 1$ (valid for all $m^* \geq 1$),

$$S(2m^*) \geq S(m^* + 1) = \varepsilon + \delta. \quad (\text{B.45})$$

Combining with (B.43) and the remainder bound,

$$\hat{S}(2) \geq S(2m^*) - \frac{\delta}{4} \geq \varepsilon + \delta - \frac{\delta}{4} = \varepsilon + \frac{3\delta}{4} > \varepsilon. \quad (\text{B.46})$$

Thus span 2 violates the tolerance on the decimated series.

Step 5: Conclusion. By monotonicity in the small-span regime (Proposition 3.1), $\hat{S}(k) > \hat{S}(2) > \varepsilon$ for all $k \geq 2$. Therefore the maximal admissible span is $\hat{m}^* = 1$, and

$$\hat{\Delta}t^* = \hat{m}^* \cdot \hat{\Delta}t_0 = 1 \cdot \Delta t^* = \Delta t^*. \quad (\text{B.47})$$

□

B.3.2 Idempotence of the relative rule

Theorem B.3 (Restatement of Theorem 3.4). *Under the same hypotheses as Theorem 3.3, let m^* be the maximal span satisfying $R(m^*) \leq \varepsilon < R(m^* + 1)$. If Δt_0 is sufficiently small, then reapplying the relative selection rule to the decimated series yields $\hat{m}^* = 1$, so that $\hat{\Delta}t^* = \Delta t^*$.*

Proof. The argument parallels the absolute case, with R replacing S and the remainder order adjusted.

Step 1: Quantify the gap. Define

$$\delta := R(m^* + 1) - \varepsilon > 0. \quad (\text{B.48})$$

Step 2: Apply the span mapping. By the relative span mapping (3.33),

$$\hat{R}(1) = R(m^*) + \mathcal{O}((\Delta t^*)^2), \quad (\text{B.49})$$

$$\hat{R}(2) = R(2m^*) + \mathcal{O}((\Delta t^*)^2). \quad (\text{B.50})$$

Choose Δt_0 small enough that both remainder magnitudes are bounded by $\delta/4$.

Step 3: Span 1 is admissible. From (B.49) and $R(m^*) \leq \varepsilon$,

$$\hat{R}(1) \leq R(m^*) + \frac{\delta}{4} \leq \varepsilon + \frac{\delta}{4}. \quad (\text{B.51})$$

Step 4: Span 2 is inadmissible. By monotonicity and $2m^* \geq m^* + 1$,

$$\hat{R}(2) \geq R(2m^*) - \frac{\delta}{4} \geq R(m^* + 1) - \frac{\delta}{4} = \varepsilon + \frac{3\delta}{4} > \varepsilon. \quad (\text{B.52})$$

Step 5: Conclusion. The maximal admissible span is $\hat{m}^* = 1$, giving $\hat{\Delta t}^* = \Delta t^*$. □

B.3.3 Remarks

Role of the gap. The gap δ quantifies how decisively the optimal span separates from its neighbors. A larger gap permits coarser Δt_0 while still satisfying the remainder bounds; a smaller gap requires finer sampling. In the degenerate case $S(m^*) = \varepsilon$ exactly, the gap depends on the discretization and idempotence becomes approximate. For generic trajectories and tolerances, δ is bounded away from zero.

Remainder orders. The absolute rule achieves idempotence with $\mathcal{O}((\Delta t^*)^3)$ corrections; the relative rule with $\mathcal{O}((\Delta t^*)^2)$. This difference reflects the scaling laws: $S(m) \sim (m\Delta t_0)^2$ versus $R(m) \sim m\Delta t_0$. The quotient in $R = S/L_{\text{med}}$ introduces a factor $(m\Delta t_0)^{-1}$ that degrades the remainder by one power.

Higher spans. The proofs verify only spans 1 and 2 explicitly. Inadmissibility of $k \geq 3$ follows from monotonicity: $\hat{S}(k) > \hat{S}(2) > \varepsilon$ for $k > 2$, which holds in the small-span regime by Proposition 3.1.

B.4 Extension to scalar time series

For scalar time series $\{x_i\}_{i=0}^{N-1}$, the sagitta rule operates on delay-coordinate embeddings rather than on the raw data. This section specifies the procedures for selecting the embedding lag ℓ and dimension d_e , and how these parameters integrate with the sagitta-based decimation. For theoretical foundations of delay embedding, see Takens [3] and Sauer

et al. [91]; for the lag and dimension selection criteria, see Fraser and Swinney [100] and Kennel et al. [93].

B.4.1 Embedding lag via average mutual information

The embedding lag ℓ is chosen as the first local minimum of the average mutual information (AMI) between x_i and $x_{i+\ell}$. The AMI at lag ℓ is

$$I(\ell) = \sum_{a,b} p_\ell(a,b) \log \frac{p_\ell(a,b)}{p(a)p(b)}, \quad (\text{B.53})$$

where $p(a)$ denotes the marginal histogram of $\{x_i\}$ and $p_\ell(a,b)$ denotes the joint histogram of pairs $(x_i, x_{i+\ell})$. Histograms are constructed with $B = \lfloor \sqrt{N} \rfloor$ bins of uniform width spanning $[\min_i x_i, \max_i x_i]$.

The selected lag is

$$\ell^* = \min\{\ell \geq 1 : I(\ell-1) > I(\ell) < I(\ell+1)\}, \quad (\text{B.54})$$

i.e., the first lag at which I attains a strict local minimum. If no local minimum exists for $\ell \leq \ell_{\max}$, a fallback value $\ell^* = 1$ is used. In the implementation, we set $\ell_{\max} = \min(N/10, 2000)$ to balance thoroughness against computational cost.

B.4.2 Embedding dimension via false nearest neighbors

The embedding dimension d_e is determined by the false nearest neighbor (FNN) criterion, which detects when apparent neighbors in a low-dimensional embedding are artifacts of projection rather than true dynamical proximity.

For each candidate dimension d , construct the embedded point

$$X_i^{(d)} = (x_i, x_{i+\ell^*}, \dots, x_{i+(d-1)\ell^*}) \in \mathbb{R}^d. \quad (\text{B.55})$$

Let $X_j^{(d)}$ be the nearest neighbor of $X_i^{(d)}$ in Euclidean distance, excluding temporal neighbors satisfying $|i-j| \leq w$ for a Theiler window $w \geq \ell^*$. The neighbor is declared *false* if either of the following conditions holds:

$$\frac{|x_{i+d\ell^*} - x_{j+d\ell^*}|}{\|X_i^{(d)} - X_j^{(d)}\|} > r_{\text{tol}}, \quad (\text{B.56})$$

$$\frac{\|X_i^{(d+1)} - X_j^{(d+1)}\|}{R_A} > a_{\text{tol}}, \quad (\text{B.57})$$

where R_A is the standard deviation of the embedded data and serves as a scale reference. The first criterion (B.56) tests whether the neighbor relationship is destroyed upon adding the $(d+1)$ th coordinate; the second (B.57) guards against spurious neighbors arising from attractor size effects.

The FNN fraction at dimension d is

$$F(d) = \frac{\#\{\text{false neighbors at dimension } d\}}{\#\{\text{tested pairs}\}}. \quad (\text{B.58})$$

The selected dimension is

$$d_e = \min\{d \geq 1 : F(d) < F_{\text{tol}}\}, \quad (\text{B.59})$$

with default thresholds $r_{\text{tol}} = 15$, $a_{\text{tol}} = 2$, and $F_{\text{tol}} = 0.01$. If no dimension satisfies the criterion up to $d_{\text{max}} = 10$, the algorithm returns $d_e = d_{\text{max}}$ as a conservative upper bound.

B.4.3 Integration with the sagitta rule

Given ℓ^* and d_e , the embedded trajectory is

$$X_i = (x_i, x_{i+\ell^*}, \dots, x_{i+(d_e-1)\ell^*}) \in \mathbb{R}^{d_e}, \quad i = 0, \dots, N_e - 1, \quad (\text{B.60})$$

where $N_e = N - (d_e - 1)\ell^*$ is the number of valid embedded points.

The relative sagitta rule (Section 3.3) is then applied to $\{X_i\}_{i=0}^{N_e-1}$. Prior to computing sagittae, the embedded vectors are standardized coordinate-wise: each coordinate is divided by its standard deviation across the trajectory. This ensures that all embedding dimensions contribute equally to the Euclidean distances used in the sagitta computation, preventing coordinates with larger variance from dominating the geometric analysis.

Remark. The relative rule is preferred over the absolute rule for embedded scalar series because the embedding dimension d_e is data-dependent and may vary across datasets. The relative sagitta $R(m) = S(m)/L_{\text{med}}(m)$ normalizes by the local arc length, yielding a dimensionless quantity that is more comparable across embeddings of different dimension. The absolute sagitta $S(m)$, by contrast, scales with the ambient dimension and would require dimension-dependent tolerance calibration.